SAMSUNG

Mobile Device

SM-G8870 Common

SERVICE Manual

Mobile Device

CONTENTS

1. Safety Precautions

1-1. Repair Precaution

Before attempting any repair or detailed tuning, shield the device from RF noise or static electricity discharges.

Use only demagnetized tools that are specifically designed for small electronic repairs, as most electronic parts are sensitive to electromagnetic forces.

Use only high quality screwdrivers when servicing products. Low quality screwdrivers can easily damage the heads of screws.

Use only conductor wire of the properly gauge and insulation for low resistance, because of the low margin of error of most testing equipment.
We recommend 22-gauge twisted copper wire.

Hand-soldering is not recommended, because printed circuit boards (PCBs) can be easily damaged, even with relatively low heat. Never use a soldering iron with a power rating of more than 100 watts and use only lead-free solder with a melting point below 250°C (482°F).

Prior to disassembling the battery charger for repair, ensure that the AC power is disconnected.
Always use the replacement parts that are registered in the SEC system. Third-party replacement parts may not function properly.

1. Safety Precautions

1-2. ESD(Electrostatically Sensitive Devices) Precaution

Many semiconductors and ESDs in electronic devices are particularly sensitive to static discharge and can be easily damaged by it. We recommend protecting these components with conductive anti-static bags when you store or transport them.

Always use an anti-static strap or wristband and remove electrostatic buildup or dissipate static electricity from your body before repairing ESDs.

Ensure that soldering irons have AC adapter with ground wires and that the ground wires are properly connected.

Use only desoldering tools with plastic tips to prevent static discharge.

Properly shield the work environment from accidental electrostatic discharge before opening packages containing ESDs.

The potential for static electricity discharge may be increased in low humidity environments, such as air-conditioned rooms. Increase the airflow to the working area to decrease the chance of accidental static electricity discharges.

2. Specification

2-1. CDMA & GPS General Specification

ITEM	CDMA	GPS
Tx Freq. range	824.025~848.985MHz	-
Rx Freq. range	869.025~893.985MHz	1575.42MHz
Channel Bandwidth	1.23MHz	2MHz
Channel Spacing	30KHz	Not Used
Number of Channel	832	1
Duplex Separation	45MHz	-
In/Output Impedance	50Ohm	50Ohm
Tx Local Frequency	F_{Tx} * 1.6666	-
Rx Local Frequency	F_{Rx} * 2	F_{Rx} * 2
TCXO Frequency	38.4MHz	38.4MHz
Freq. Stability	(FRX-45MHz)±300Hz	-
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

2-2. GSM General Specification

Item		GSM 850	EGSM 900	DCS1800	PCS1900
Freq. Band[MHz] Uplink/Downlink		824~849 869~894	880~915 925~960	1710~1785 1805~1880	1850~1910 1930~1990
ARFCN range		128~251	0~124 & 975~1023	512~885	512~810
Tx/Rx spacing		45MHz	45MHz	95MHz	80MHz
Mod. Bit rate/ Bit Period		270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us	270.833kbps 3.692us
Time Slot Period/ Frame Period		576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms	576.9us 4.615ms
Modulation	GSM/ GPRS	0.3GMSK	0.3GMSK	0.3GMSK	0.3GMSK
MS Power		33dBm ~5dBm	33dBm ~5dBm	30dBm ~0dBm	30dBm ~0dBm
Power Class		5pcl ~ 19pcl	5pcl ~ 19pcl	0pcl ~ 15pcl	0pcl ~ 15pcl
Sensitivity		-102dBm	-102dBm	-100dBm	-100dBm
TDMA Mux		8	8	8	8
Cell Radius		35Km	35Km	2Km	2Km

2-3. GSM Tx Power Class

TX Power control level	GSM850	TX Power control level	EGSM900	TX Power control level	DCS1800	TX Power control level	PCS1900
5	33±2 dBm	5	33±2 dBm	0	30±3 dBm	0	30±3 dBm
6	31±2 dBm	6	31±2 dBm	1	28±3 dBm	1	28±3 dBm
7	29±2 dBm	7	29±2 dBm	2	26±3 dBm	2	26±3 dBm
8	27±2 dBm	8	27±2 dBm	3	24±3 dBm	3	24±3 dBm
9	25±2 dBm	9	25±2 dBm	4	22±3 dBm	4	22±3 dBm
10	23±2 dBm	10	23±2 dBm	5	20±3 dBm	5	20±3 dBm
11	21±2 dBm	11	21±2 dBm	6	18±3 dBm	6	18±3 dBm
12	19±2 dBm	12	19±2 dBm	7	16±3 dBm	7	16±3 dBm
13	17±2 dBm	13	17±2 dBm	8	14±3 dBm	8	14±3 dBm
14	15±2 dBm	14	15±2 dBm	9	12±4 dBm	9	12±4 dBm
15	13±2 dBm	15	13±2 dBm	10	10±4 dBm	10	10±4 dBm
16	11±3 dBm	16	11±3 dBm	11	8±4 dBm	11	8±4 dBm
17	9±3 dBm	17	9±3 dBm	12	6±4 dBm	12	6±4 dBm
18	7±3 dBm	18	7±3 dBm	13	4±4 dBm	13	4±4 dBm
19	5±3 dBm	19	5±3 dBm	14	2±5 dBm	14	2±5 dBm
-	-	-	-	15	0±5 dBm	15	0±5 dBm

2. Specification

2-4. WCDMA General Specification

Item	WCDMA2100	WCDMA1900	WCDMA850	WCDMA900
Freq. Band[MHz] Uplink/Downlink	1922~1977 2112~2167	1852~1907 1932~1987	824~849 869~894	880~915 925~960
ARFCN range	UL：9612~9888 DL：10562~10838	UL：9262~9538 DL：9662~9938	UL：4132~4233 DL：4357~4458	UL：2712~2863 DL：2937~3088
Tx/Rx spacing	190MHz	80MHz	45MHz	45MHz
Mod. Bit rate/ Bit Period	3.84 Mcps	3.84 Mcps	3.84 Mcps	3.84 Mcps
Time Slot Period/Frame Period	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms	FrameLength:10ms Slotlength:0.667ms
Modulation	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK	QPSK/HQPSK
MS Power	24dBm~-50dBm	21dBm~-50dBm	24dBm~-50dBm	24dBm~-50dBm
Power Class	3(max+24dBm)	4(max+21dBm)	3(max+24dBm)	3(max+24dBm)
Sensitivity	-106.7dBm	-106.7dBm	-106.7dBm	-106.7dBm
TDMA Mux	8	8	8	8
Cell Radius	2Km	2Km	2Km	2Km

2. Specification

2-5-1. LTE General Specification (FDD)

Item	Band1 (FDD)	Band3 (FDD)	Band5 (FDD)	Band7 (FDD)	Band8 (FDD)
Freq. Band[MHz] UL/DL	1920~1980 2110~2170	1710~1785 1805~1880	824~849 869~894	2500~2570 2620~2690	880~915 925~960
ARFCN range UL/DL	18000~18599 0~599	19200~19949 1200~1949	20400~20649 2400~2649	20750~20449 2750~3449	21450~21799 3450~3799
Tx/Rx spacing	190MHz	95MHz	45MHz	120MHz	45MHz
TCXO Frequency	38.4MHz	38.4MHz	38.4MHz	38.4MHz	38.4MHz
Modulation	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-99.3（BW：5 MHz） -96.3（BW：10 MHz） -94.5（BW：15MHz） -93.3（BW：20MHz）	-101（BW：1.4MHz） -98(BW：3MHz) -96.3（BW：5 MHz） -93.3（BW：10 MHz） -91.5（BW：15MHz） -90.3（BW：20MHz）	-102.5(BW:1.4MHz） -99.5(BW：3MHz) -97.3（BW：5 MHz） -94.3（BW：10 MHz）	-97.3（BW：5 MHz） -94.3（BW：10 MHz） -92.5(BW：15MHz） -91.3（BW：20MHz）	- 101.5（BW：1.4MHz） -98.5(BW：3MHz) -96.3（BW：5 MHz） -93.3（BW：10 MHz）
Cell Radius	>5Km	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

Item	Band4 (FDD)	Band12 (FDD)	Band17 (FDD)
Freq. Band[MHz] UL/DL	1710~1755 2110~2155	698~716 728~746	704~716 734~746
ARFCN range UL/DL	19950~20399 1950~2399	23010~23179 5010~5179	23730~23849 5030~5849
Tx/Rx spacing	400MHz	30MHz	30MHz
TCXO Frequency	38.4MHz	38.4MHz	38.4MHz
Modulation	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM	QPSK/16-QAM/64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-101（BW：1.4MHz） -98(BW：3MHz) -96.3（BW：5 MHz） -93.3（BW：10 MHz） -91.5（BW：15MHz） -90.3（BW：20MHz）	-101.5（BW：1.4MHz） -98.5(BW：3MHz) -96.3（BW：5 MHz） -93.3（BW：10 MHz）	-96.3（BW：5 MHz） -93.3（BW：10 MHz）
Cell Radius	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

2-5-2. LTE General Specification (TDD)

Item	Band34 (TDD)	Band38 (TDD)	Band39 (TDD)	Band40 (TDD)	Band41 (TDD)
Freq. Band[MHz] Uplink/Downlink	2010~2025	2570~2620	1880~1920	2300~2400	2555~2655
ARFCN range	36200~36349	37750~38249	38250~38649	38650~39649	40240~41240
Tx/Rx spacing	/	/	/	/	/
TCXO Frequency	38.4MHz	38.4MHz	38.4MHz	38.4MHz	38.4MHz
Modulation	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM	QPSK/16-QAM /64-QAM
MS Power	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm	-40dBm~ 25dBm
Power Class	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)	3 (max: 23 ±2dBm)
Sensitivity	-99.3（BW：5 MHz） -96.3（BW：10MHz） -94.5（BW：15MHz） -93.3（BW：20MHz）	-99.3（BW：5 MHz） -96.3（BW：10MHz） -94.5（BW：15MHz） -93.3（BW：20MHz）	-99.3（BW：5 MHz） -96.3（BW：10MHz） -94.5（BW：15MHz） -93.3（BW：20MHz）	-99.3（BW：5 MHz） -96.3（BW：10MHz） -94.5（BW：15MHz） -93.3（BW：20MHz）	-97.3（BW：5 MHz） -94.3（BW：10MHz） -92.5（BW：15MHz） -91.3（BW：20MHz
Cell Radius	>5Km	>5Km	>5Km	>5Km	>5Km
In/Output Impedance	50Ω	50Ω	50Ω	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃	-30℃ ~ +60℃

2. Specification

2-6. TDSCDMA General Specification

Item	Band34	Band39
Freq. Band[MHz] Uplink/Downlink	2010~2025	1880~1920
ARFCN range	10054~10121	9404~9596
Tx/Rx spacing	/	/
TCXO Frequency	38.4MHz	38.4MHz
Modulation	QPSK/16-QAM	QPSK/16-QAM
MS Power	-49dBm~ 25dBm	-49dBm~ 25dBm
Power Class	2 (max: 24 +1/-3dBm)	2 (max: 24 +1/-3dBm)
Sensitivity	-108dBm/1.28MHz	-108dBm/1.28MHz
Cell Radius	>10Km	>10Km
In/Output Impedance	50Ω	50Ω
Operating Temperature	-30℃ ~ +60℃	-30℃ ~ +60℃

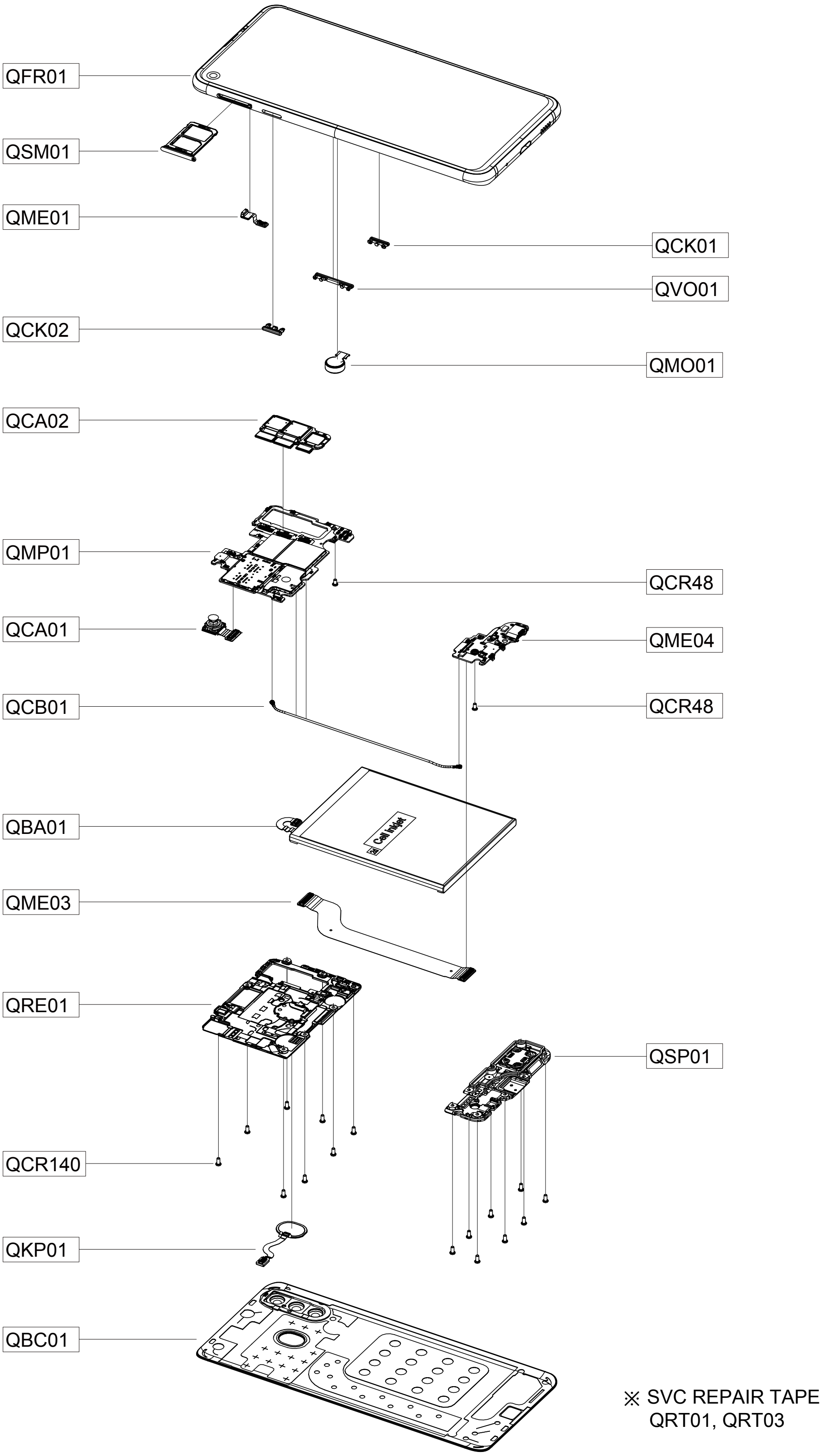
3. Product Function

Main Function

Item	Description
OS	Android V8.1
RF	[2G] CDMA : CDMA800 GSM : GSM850 / GSM900 / GSM1800 / GSM1900 [3G] CDMA : BC0 WCDMA : B1 / B2 /B5 / B8 TD-SCDMA : B34 / B39 [4G] LTE FDD : B1 / B3 / B4 / B5 / B7 / B8 / B12 / B17 LTE TDD : B34 / B38 / B39 / B40 / B41
Battery	3,300mAh
Base Band	SDM710 / 2.2GHz, 1.8GHz
Other RF	GPS, Glonass, Beidou, BT 5.0, USB 2.0, Wi-Fi 802.11 a/b/g/n/ac (2.4G+5GHz), NFC
Camera	3EA Camera (24M+5M+10M) with LED Flash, 24MP FF (Front)
LCD	6.4” FHD+, 2340 x 1080
RAM + ROM	6GB LPDDR4 + 128GB UFS
Sensor	Accelerometer, Fingerprint Sensor, Gyro Sensor, Geomagnetic Sensor, Proximity Sensor, RGB Light Sensor
Accessory	Charger : 5V/2A or 9 V/1.67 A Data cable : USB Type-C Earjack : Type-C

4. Exploded View and Parts List

4-1. Cellular phone Exploded View



※ SVC REPAIR TAPE
QRT01, QRT03

5. MAIN Electrical Parts List

Parts Code	Design LOC	Description
0202-002125	R5037,R6025	PB-SHORT-0603
0401-001110	D5002	DIODE
0404-001250	D5000,D5001	DIODE
0406-001506	ZD1001	DIODE-TVS
0406-001561	ZD5015,ZD6000,ZD6004	DIODE-TVS
0406-001561	ZD6005,ZD6006	DIODE-TVS
0406-001592	ZD3007	DIODE-TVS
0406-001728	ZD5002	DIODE-TVS
0406-001763	ZD3000,ZD3001,ZD3002	DIODE-TVS
0406-001763	ZD3003,ZD3004,ZD3005	DIODE-TVS
0406-001763	ZD3006,ZD3010,ZD6002	DIODE-TVS
0406-001780	ZD5009,ZD5010	DIODE-TVS
0406-001781	ZD5013,ZD5014	DIODE-TVS
0406-001787	ZD5011,ZD5012	DIODE-TVS
0406-001789	ZD6003	DIODE-TVS
0406-001797	ZD5004,ZD5005	DIODE-TVS
0406-001808	ZD5003	DIODE-TVS
0505-002088	TR5000,TR5002,TR7000	FET-SILICON
0505-003234	TR5001	FET-SILICON
0601-003702	LED5000	LED
0801-003639	U6005	IC
1001-001835	U3009,U3010	IC
1001-001970	U1001	IC
1001-002042	U2089	IC
1001-002063	U1000	IC
1003-002951	U5004	IC
1108-000660	UME4000	IC
1201-003869	U3007	IC
1201-003976	U1003,U1004	IC
1201-003984	U2000	IC
1201-004109	PAM2000	IC
1201-004135	U1005	IC
1201-004156	U6000	IC
1203-008249	U3002,U5011	IC
1203-008251	U5003	IC
1203-008269	U7000	IC
1203-008379	U2004	IC
1203-008719	U3000	IC

1203-008797	U5005	IC
1203-008819	U5002	IC
1203-008865	U7001	IC
1203-008867	U3001,U3005,U6006	IC
1203-009035	U7002	IC
1203-009158	U7004	IC
1203-009169	U6003	IC
1203-009201	U5000	IC
1203-009202	U5001	IC
1203-009203	U4001,U5006	IC
1204-003685	U6002	IC
1205-005217	U7006	IC
1205-005731	U3006	IC
1205-005738	U4000	IC
1205-005821	U3008	IC
1205-005833	U2003	IC
1205-006062	UCP4000	IC
1209-002450	U3003	IC
1209-002513	U3004	IC
1404-001724	TH2000,TH3000,TH5000	THERMISTOR
1404-001724	V4000	THERMISTOR
1405-001395	VR1001,VR1002,VR1003	VARISTOR
1405-001395	VR1004,VR1005,VR1006	VARISTOR
1405-001395	VR6013,VR6014,VR6015	VARISTOR
1405-001403	VR6005	VARISTOR
1405-001415	V2000,V2001,V2008	VARISTOR
1405-001415	V3000,V3001	VARISTOR
1405-001458	V7000,V7001	VARISTOR
2007-000171	R3005,R3006	R-CHIP
2007-003015	R5027,R5028	R-CHIP
2007-007142	R3001,R5026,R5035	R-CHIP
2007-007142	R5041,R5042,R5043	R-CHIP
2007-007142	R6003,R7006,R7011	R-CHIP
2007-007308	R5003	R-CHIP
2007-007741	R4000,R4003,R4030	R-CHIP
2007-007741	R4043,R5019,R5039	R-CHIP
2007-007741	R6015	R-CHIP
2007-008056	R1000,R1002	R-CHIP
2007-008263	R4038	R-CHIP

2007-008420	R3017	R-CHIP
2007-008516	R6002,R6021	R-CHIP
2007-008531	R5013,R7005,R7007	R-CHIP
2007-008531	R7008,R7009,R7010	R-CHIP
2007-008587	R6008	R-CHIP
2007-008588	R3015,R3016,R4009	R-CHIP
2007-008588	R4010,R4011,R4012	R-CHIP
2007-008588	R4013,R4014,R4015	R-CHIP
2007-008588	R4016,R4017,R4018	R-CHIP
2007-008588	R4019,R4020,R4021	R-CHIP
2007-008588	R4022,R4023,R4024	R-CHIP
2007-008588	R4025,R4026,R4046	R-CHIP
2007-008588	R4047,R4053,R4054	R-CHIP
2007-008686	R3021,R3024	R-CHIP
2007-008766	R4034,R4035	R-CHIP
2007-008774	R3013,R3014	R-CHIP
2007-008785	R1001	R-CHIP
2007-008808	R6016,R7004	R-CHIP
2007-009111	R3020,R3023,R6014	R-CHIP
2007-009155	R5007,R5010	R-CHIP
2007-009157	R5002,R5008,R5014	R-CHIP
2007-009157	R5020	R-CHIP
2007-009158	R5016,R6020	R-CHIP
2007-009171	R3018	R-CHIP
2007-009201	R4036,R4037,R6005	R-CHIP
2007-009201	R6009	R-CHIP
2007-009212	R4007,R4008,R5012	R-CHIP
2007-009212	R6007	R-CHIP
2007-009223	R6019	R-CHIP
2007-009315	R5009,R6004,R6011	R-CHIP
2007-009315	R6012,R6013,R6027	R-CHIP
2007-009315	R6028	R-CHIP
2007-009352	R4004,R7013	R-CHIP
2007-009408	R4005,R4006,R5021	R-CHIP
2007-009408	R5022,R5030	R-CHIP
2007-009766	R6000	R-CHIP
2007-009801	R4039,R4040,R7012	R-CHIP
2007-009805	R1003,R1004,R1006	R-CHIP
2007-009805	R1007,R2000,R3019	R-CHIP

2007-009805	R3022,R3025	R-CHIP
2007-009838	R6017	R-CHIP
2007-009866	R5011	R-CHIP
2007-009920	R4032,R4044,R4045	R-CHIP
2007-009920	R5024,R5025	R-CHIP
2007-009954	R5004,R5005,R5006	R-CHIP
2007-009954	R6018	R-CHIP
2007-009969	R3004,R3007,R6006	R-CHIP
2007-010202	R3000,R5023	R-CHIP
2007-010685	R1009,R6001	R-CHIP
2007-012068	R5036	R-CHIP
2203-000254	C5089,C5094	C-CERAMIC,CHIP
2203-000386	C3040	C-CERAMIC,CHIP
2203-000425	C7005	C-CERAMIC,CHIP
2203-000812	C7003	C-CERAMIC,CHIP
2203-000995	C5105	C-CERAMIC,CHIP
2203-001153	C3034,C3035,C7057	C-CERAMIC,CHIP
2203-001239	C3036	C-CERAMIC,CHIP
2203-005344	C5079	C-CERAMIC,CHIP
2203-005682	C1005,C1007,C1009	C-CERAMIC,CHIP
2203-005682	C1015,C1067,C2034	C-CERAMIC,CHIP
2203-005682	C2077,C2103,C2121	C-CERAMIC,CHIP
2203-005682	C5120,C5123,C5124	C-CERAMIC,CHIP
2203-005682	C5125,C7012,C7013	C-CERAMIC,CHIP
2203-005682	C7042,C7048,C7055	C-CERAMIC,CHIP
2203-005717	C1049,C2051,C2076	C-CERAMIC,CHIP
2203-005719	C3081	C-CERAMIC,CHIP
2203-005725	C1090,C3079,C7017	C-CERAMIC,CHIP
2203-005725	C7041,C7050	C-CERAMIC,CHIP
2203-005726	C5121,C5122,C5143	C-CERAMIC,CHIP
2203-005726	C5144,C5146,C5147	C-CERAMIC,CHIP
2203-005726	C6040,C7073,C7074	C-CERAMIC,CHIP
2203-005726	C7082	C-CERAMIC,CHIP
2203-005729	C7045	C-CERAMIC,CHIP
2203-005731	C3090,C3092,C3098	C-CERAMIC,CHIP
2203-005731	C3099,C6023,C7010	C-CERAMIC,CHIP
2203-005731	C7011,C7078	C-CERAMIC,CHIP
2203-005732	C5106,C6038,C6043	C-CERAMIC,CHIP
2203-005736	C1012,C1013,C1041	C-CERAMIC,CHIP

2203-005736	C1044,C1045,C1052	C-CERAMIC,CHIP
2203-005736	C1055,C1075,C1091	C-CERAMIC,CHIP
2203-005736	C1092,C1096,C1100	C-CERAMIC,CHIP
2203-005736	C1102,C2002,C2042	C-CERAMIC,CHIP
2203-005736	C2046,C2048,C2054	C-CERAMIC,CHIP
2203-005736	C2055,C2062,C2064	C-CERAMIC,CHIP
2203-005736	C2065,C2068,C2069	C-CERAMIC,CHIP
2203-005736	C2070,C2071,C2072	C-CERAMIC,CHIP
2203-005736	C2078,C2080,C2107	C-CERAMIC,CHIP
2203-005736	C2113,C2115,C3044	C-CERAMIC,CHIP
2203-005736	C3045,C3049,C3050	C-CERAMIC,CHIP
2203-005736	C3051,C3052,C3053	C-CERAMIC,CHIP
2203-005736	C3064,C3080,C3085	C-CERAMIC,CHIP
2203-005736	C5090,C5091	C-CERAMIC,CHIP
2203-005777	C1002,C1028,C1046	C-CERAMIC,CHIP
2203-005777	C1047,C1080,C1094	C-CERAMIC,CHIP
2203-005777	C1103,C2001,C2044	C-CERAMIC,CHIP
2203-005777	C2053,C2056,C3075	C-CERAMIC,CHIP
2203-005789	C1032,C1035,C1048	C-CERAMIC,CHIP
2203-005789	C1074,C1104,C2032	C-CERAMIC,CHIP
2203-005789	C2039,C2041,C2081	C-CERAMIC,CHIP
2203-005789	C2086	C-CERAMIC,CHIP
2203-005792	C1105,C2043,C3083	C-CERAMIC,CHIP
2203-005806	C2050,C5080	C-CERAMIC,CHIP
2203-006121	C6024	C-CERAMIC,CHIP
2203-006123	C3086	C-CERAMIC,CHIP
2203-006187	C1106	C-CERAMIC,CHIP
2203-006194	C3054,C3060,C3061	C-CERAMIC,CHIP
2203-006194	C3062,C3063,C3068	C-CERAMIC,CHIP
2203-006194	C3069,C3071,C3072	C-CERAMIC,CHIP
2203-006194	C3073,C3074,C3078	C-CERAMIC,CHIP
2203-006194	C6011	C-CERAMIC,CHIP
2203-006305	C3093,C3094,C3101	C-CERAMIC,CHIP
2203-006305	C3102,C3107,C5095	C-CERAMIC,CHIP
2203-006305	C7024	C-CERAMIC,CHIP
2203-006318	C3026	C-CERAMIC,CHIP
2203-006379	C6034	C-CERAMIC,CHIP
2203-006400	C2060,C3024,C3025	C-CERAMIC,CHIP
2203-006400	C5028,C6003	C-CERAMIC,CHIP

2203-006410	C2110	C-CERAMIC,CHIP
2203-006423	C2003,C2004,C2005	C-CERAMIC,CHIP
2203-006423	C2006,C2007,C2015	C-CERAMIC,CHIP
2203-006423	C2016,C2017,C2018	C-CERAMIC,CHIP
2203-006423	C2019,C2074,C2093	C-CERAMIC,CHIP
2203-006423	C2094,C4044,C4090	C-CERAMIC,CHIP
2203-006423	C4094,C5049,C7037	C-CERAMIC,CHIP
2203-006556	C1010,C1011,C1020	C-CERAMIC,CHIP
2203-006556	C1071,C1086,C2036	C-CERAMIC,CHIP
2203-006556	C7058,C7059,C7060	C-CERAMIC,CHIP
2203-006556	C7061,C7063,C7064	C-CERAMIC,CHIP
2203-006556	C7065,C7066,C7067	C-CERAMIC,CHIP
2203-006556	C7068,C7069,C7070	C-CERAMIC,CHIP
2203-006556	C7071	C-CERAMIC,CHIP
2203-006562	C2031,C5010,C5012	C-CERAMIC,CHIP
2203-006562	C5013,C5014,C5015	C-CERAMIC,CHIP
2203-006562	C5016	C-CERAMIC,CHIP
2203-006604	C3084,C3108	C-CERAMIC,CHIP
2203-006611	C2087,C3088	C-CERAMIC,CHIP
2203-006707	C1107	C-CERAMIC,CHIP
2203-006839	C1000,C1058,C1066	C-CERAMIC,CHIP
2203-006839	C2009,C2010,C2011	C-CERAMIC,CHIP
2203-006839	C2012,C2020,C2059	C-CERAMIC,CHIP
2203-006839	C3008,C3011,C3012	C-CERAMIC,CHIP
2203-006839	C3018,C3048,C4018	C-CERAMIC,CHIP
2203-006839	C4070,C4121,C5006	C-CERAMIC,CHIP
2203-006839	C5007,C5033,C5034	C-CERAMIC,CHIP
2203-006839	C5036,C5039,C5066	C-CERAMIC,CHIP
2203-006839	C5073,C5114,C5118	C-CERAMIC,CHIP
2203-006839	C6008,C6026,C6029	C-CERAMIC,CHIP
2203-006839	C6031,C6039	C-CERAMIC,CHIP
2203-006846	C1030,C1088,C3029	C-CERAMIC,CHIP
2203-006872	C7047	C-CERAMIC,CHIP
2203-007143	C5088	C-CERAMIC,CHIP
2203-007210	C3020,C3057,C4005	C-CERAMIC,CHIP
2203-007210	C4009,C4010,C4082	C-CERAMIC,CHIP
2203-007210	C4083,C4095,C4096	C-CERAMIC,CHIP
2203-007270	C5074	C-CERAMIC,CHIP
2203-007271	C1023,C2028,C3056	C-CERAMIC,CHIP

2203-007271	C5000,C5002,C5065	C-CERAMIC,CHIP
2203-007271	C5115	C-CERAMIC,CHIP
2203-007317	C3055,C3065,C3070	C-CERAMIC,CHIP
2203-007317	C3076,C3077,C7025	C-CERAMIC,CHIP
2203-007317	C7026,C7032,C7034	C-CERAMIC,CHIP
2203-007391	C3002,C3003,C3033	C-CERAMIC,CHIP
2203-007391	C3059,C3066,C3067	C-CERAMIC,CHIP
2203-007391	C6037	C-CERAMIC,CHIP
2203-007393	C2000,C2008,C2013	C-CERAMIC,CHIP
2203-007393	C2014,C2021,C2023	C-CERAMIC,CHIP
2203-007393	C2024,C3038,C3039	C-CERAMIC,CHIP
2203-007393	C4114,C4122,C5001	C-CERAMIC,CHIP
2203-007393	C5005,C5011,C5017	C-CERAMIC,CHIP
2203-007393	C5032,C5035,C5050	C-CERAMIC,CHIP
2203-007393	C5053,C5055,C5056	C-CERAMIC,CHIP
2203-007393	C5063,C5077,C5084	C-CERAMIC,CHIP
2203-007393	C5099,C5101,C5119	C-CERAMIC,CHIP
2203-007393	C6030	C-CERAMIC,CHIP
2203-007403	C2111	C-CERAMIC,CHIP
2203-007425	C5070	C-CERAMIC,CHIP
2203-007486	C5096	C-CERAMIC,CHIP
2203-007544	C6010	C-CERAMIC,CHIP
2203-007775	C6025,C7027	C-CERAMIC,CHIP
2203-007796	C3005,C3015,C3016	C-CERAMIC,CHIP
2203-007796	C3032,C3041,C3042	C-CERAMIC,CHIP
2203-007796	C4003,C4004,C4007	C-CERAMIC,CHIP
2203-007796	C4008,C4014,C4017	C-CERAMIC,CHIP
2203-007796	C4022,C4028,C4029	C-CERAMIC,CHIP
2203-007796	C4030,C4031,C4032	C-CERAMIC,CHIP
2203-007796	C4033,C4036,C4037	C-CERAMIC,CHIP
2203-007796	C4038,C4039,C4042	C-CERAMIC,CHIP
2203-007796	C4046,C4047,C4048	C-CERAMIC,CHIP
2203-007796	C4049,C4052,C4058	C-CERAMIC,CHIP
2203-007796	C4059,C4062,C4063	C-CERAMIC,CHIP
2203-007796	C4068,C4072,C4073	C-CERAMIC,CHIP
2203-007796	C4074,C4077,C4079	C-CERAMIC,CHIP
2203-007796	C4080,C4081,C4087	C-CERAMIC,CHIP
2203-007796	C4089,C4091,C4092	C-CERAMIC,CHIP
2203-007796	C4093,C4098,C4099	C-CERAMIC,CHIP

2203-007796	C4100,C4101,C4102	C-CERAMIC,CHIP
2203-007796	C4104,C4105,C4106	C-CERAMIC,CHIP
2203-007796	C4108,C4111,C4115	C-CERAMIC,CHIP
2203-007796	C4116,C4117,C4126	C-CERAMIC,CHIP
2203-007796	C4130,C4132,C4133	C-CERAMIC,CHIP
2203-007796	C4134,C5008,C5009	C-CERAMIC,CHIP
2203-007796	C5018,C5019,C5029	C-CERAMIC,CHIP
2203-007796	C5052,C5054,C5057	C-CERAMIC,CHIP
2203-007796	C5067,C5075,C5103	C-CERAMIC,CHIP
2203-007796	C5104,C5142,C6002	C-CERAMIC,CHIP
2203-007796	C6033,C6035,C6041	C-CERAMIC,CHIP
2203-007796	C6042,C7015,C7016	C-CERAMIC,CHIP
2203-007796	C7019,C7022,C7028	C-CERAMIC,CHIP
2203-007796	C7031,C7044,C7051	C-CERAMIC,CHIP
2203-007796	C7052,C7079	C-CERAMIC,CHIP
2203-008095	C4013	C-CERAMIC,CHIP
2203-008097	C1019,C1022,C1087	C-CERAMIC,CHIP
2203-008097	C3021,C6001	C-CERAMIC,CHIP
2203-008242	C7038,C7039	C-CERAMIC,CHIP
2203-008529	C7035	C-CERAMIC,CHIP
2203-008572	C5086,C5087	C-CERAMIC,CHIP
2203-008749	C5098,C7004	C-CERAMIC,CHIP
2203-008844	C7036,C7080,C7081	C-CERAMIC,CHIP
2203-008860	C3001,C3014,C3017	C-CERAMIC,CHIP
2203-008860	C4024,C4025,C5020	C-CERAMIC,CHIP
2203-008860	C5021,C5022,C5023	C-CERAMIC,CHIP
2203-008860	C5024,C5025,C5027	C-CERAMIC,CHIP
2203-008860	C5031,C5037,C5038	C-CERAMIC,CHIP
2203-008860	C5043,C5044,C5045	C-CERAMIC,CHIP
2203-008860	C5046,C5047,C5064	C-CERAMIC,CHIP
2203-008860	C5076,C5078,C5092	C-CERAMIC,CHIP
2203-008860	C5093,C5137,C5139	C-CERAMIC,CHIP
2203-008860	C5140,C5145,C6005	C-CERAMIC,CHIP
2203-008860	C6027,C6028,C6032	C-CERAMIC,CHIP
2203-008860	C7000,C7001,C7002	C-CERAMIC,CHIP
2203-008860	C7023,C7049,C7056	C-CERAMIC,CHIP
2203-008860	C7072,C7077	C-CERAMIC,CHIP
2203-008876	C3000,C3009,C3019	C-CERAMIC,CHIP
2203-008876	C3022,C3030,C3031	C-CERAMIC,CHIP

2203-008876	C4109,C4110,C4112	C-CERAMIC,CHIP
2203-008876	C4113,C5003,C5004	C-CERAMIC,CHIP
2203-008876	C5042,C5048,C5068	C-CERAMIC,CHIP
2203-008876	C5069,C7040	C-CERAMIC,CHIP
2203-009064	C4006,C4011,C4012	C-CERAMIC,CHIP
2203-009064	C4015,C4016,C4078	C-CERAMIC,CHIP
2203-009167	C3027,C3028	C-CERAMIC,CHIP
2203-009328	C4135,C4136,C5116	C-CERAMIC,CHIP
2203-009328	C5117	C-CERAMIC,CHIP
2203-009618	C5058,C5059,C5060	C-CERAMIC,CHIP
2203-009618	C5061,C5138	C-CERAMIC,CHIP
2203-009733	C1018,C2025,C3004	C-CERAMIC,CHIP
2203-009733	C5071,C5072,C5107	C-CERAMIC,CHIP
2203-009733	C5108,C5109,C5110	C-CERAMIC,CHIP
2203-009733	C5111,C5112,C6004	C-CERAMIC,CHIP
2203-009733	C6006,C7006,C7007	C-CERAMIC,CHIP
2203-009733	C7008,C7009,C7014	C-CERAMIC,CHIP
2203-009733	C7020,C7021,C7030	C-CERAMIC,CHIP
2203-009733	C7033,C7043,C7046	C-CERAMIC,CHIP
2203-009733	C7053,C7054,C7062	C-CERAMIC,CHIP
2203-009734	C5026	C-CERAMIC,CHIP
2203-009735	C4019,C4020	C-CERAMIC,CHIP
2203-009736	C4040,C4050,C4054	C-CERAMIC,CHIP
2203-009736	C4131	C-CERAMIC,CHIP
2203-009738	C4055,C4097,C5113	C-CERAMIC,CHIP
2203-009812	C6007	C-CERAMIC,CHIP
2203-009917	C5040,C5041	C-CERAMIC,CHIP
2203-009969	C6000	C-CERAMIC,CHIP
2203-010085	C4118,C4120	C-CERAMIC,CHIP
2703-002649	L2029	INDUCTOR-SMD
2703-002858	L1005	INDUCTOR-SMD
2703-002900	L2038	INDUCTOR-SMD
2703-002901	L1002,L1009,L1012	INDUCTOR-SMD
2703-002903	L2057	INDUCTOR-SMD
2703-002907	L2052,L2056	INDUCTOR-SMD
2703-002951	L1017	INDUCTOR-SMD
2703-002953	L1014	INDUCTOR-SMD
2703-002958	L1056,L3026	INDUCTOR-SMD
2703-002959	L3029	INDUCTOR-SMD

2703-002999	L2007	INDUCTOR-SMD
2703-003004	L3022	INDUCTOR-SMD
2703-003921	L2048,L3010	INDUCTOR-SMD
2703-003970	L1038	INDUCTOR-SMD
2703-004001	L1015	INDUCTOR-SMD
2703-004012	L2008,L2064	INDUCTOR-SMD
2703-004013	L1024,L1048,L2040	INDUCTOR-SMD
2703-004013	L3011	INDUCTOR-SMD
2703-004014	L1011,L1023,L2034	INDUCTOR-SMD
2703-004014	L3013,L3023	INDUCTOR-SMD
2703-004018	L1050,L2053	INDUCTOR-SMD
2703-004024	L1028	INDUCTOR-SMD
2703-004030	L1043	INDUCTOR-SMD
2703-004032	L1032,L2015	INDUCTOR-SMD
2703-004033	L1006,L1025	INDUCTOR-SMD
2703-004034	L2017,L2047,L2051	INDUCTOR-SMD
2703-004034	L2055,L3015,L3019	INDUCTOR-SMD
2703-004035	L1026,L1029,L1037	INDUCTOR-SMD
2703-004035	L1040,L1051,L2009	INDUCTOR-SMD
2703-004035	L2014,L2016,L2019	INDUCTOR-SMD
2703-004035	L2021,L2025,L2027	INDUCTOR-SMD
2703-004035	L3014,L3031	INDUCTOR-SMD
2703-004038	L1047,L2020,L2022	INDUCTOR-SMD
2703-004286	L2003,L2030	INDUCTOR-SMD
2703-004287	L1004,L1020,L1022	INDUCTOR-SMD
2703-004288	L2028	INDUCTOR-SMD
2703-004289	L1021,L2006,L3033	INDUCTOR-SMD
2703-004302	L1031,L1036	INDUCTOR-SMD
2703-004317	L3009,L3012,L3020	INDUCTOR-SMD
2703-004317	L3021,L3035	INDUCTOR-SMD
2703-004328	L1018,L2012,L2013	INDUCTOR-SMD
2703-004328	L3002,L3016,L3017	INDUCTOR-SMD
2703-004328	L3036	INDUCTOR-SMD
2703-004366	L1044	INDUCTOR-SMD
2703-004703	L2039,L2045	INDUCTOR-SMD
2703-004764	L1046,L2041,L2061	INDUCTOR-SMD
2703-004853	L3028	INDUCTOR-SMD
2703-004911	L1013,L1039,L1042	INDUCTOR-SMD
2703-004911	L1059	INDUCTOR-SMD

2703-005066	L5001	INDUCTOR-SMD
2703-005085	L1052,L2002,L2018	INDUCTOR-SMD
2703-005085	L2063,L3008	INDUCTOR-SMD
2703-005087	L1049	INDUCTOR-SMD
2703-005089	L1060	INDUCTOR-SMD
2703-005101	L2001	INDUCTOR-SMD
2703-005118	L7001	INDUCTOR-SMD
2703-005200	L7002	INDUCTOR-SMD
2703-005295	L1019,L2011,L3005	INDUCTOR-SMD
2703-005296	L1008,L1034,L1057	INDUCTOR-SMD
2703-005296	L3003	INDUCTOR-SMD
2703-005366	L2010	INDUCTOR-SMD
2703-005505	L5002,L5003,L6000	INDUCTOR-SMD
2703-005555	L3000,L3001	INDUCTOR-SMD
2703-005581	L4000,L5018	INDUCTOR-SMD
2703-005660	L5000,L5004,L5005	INDUCTOR-SMD
2703-005660	L5006,L5007,L5008	INDUCTOR-SMD
2703-005660	L5009,L5010,L5011	INDUCTOR-SMD
2703-005660	L5013	INDUCTOR-SMD
2801-005481	OSC5000	CRYSTAL-UNIT
2901-001690	C4021	C-CERAMIC,CHIP
2901-001798	F7010,F7011,F7012	FILTER-EMI SMD
2901-001798	F7013,F7014	FILTER-EMI SMD
2903-001576	F3005,F3006	FILTER-SAW
2904-002134	F1007	FILTER-SAW
2904-002190	F1006	FILTER-SAW
2904-002253	F1001,F2004	FILTER-SAW
2904-002257	F2002	FILTER-SAW
2904-002350	F2000	FILTER-SAW
2904-002356	F3000	FILTER-SAW
2904-002360	F3004	FILTER-SAW
2904-002397	F2003,F2007	FILTER-SAW
2910-000390	F2001	FILTER
2911-000430	U2006	FILTER
2911-000452	F1005	FILTER
3003-001232	MIC6000	MIC-CONDENSOR
3301-001885	L6002	CORE-FERRITE BEAD
3301-001901	L7000	CORE-FERRITE BEAD
3301-002223	L7006,L7007,L7008	CORE-FERRITE BEAD

5. MAIN Electrical Parts List

3301-002223	L7009	CORE-FERRITE BEAD
3301-002235	L2037	CORE-FERRITE BEAD
3301-002237	L1001,L6001	CORE-FERRITE BEAD
3301-002238	L3004,L3007	CORE-FERRITE BEAD
3301-002243	L7003,L7004,L7005	CORE-FERRITE BEAD
3301-002243	L7010	CORE-FERRITE BEAD
3301-002312	L3006,L5012	CORE-FERRITE BEAD
3301-002331	L5014,L5015	CORE-FERRITE BEAD
3301-002372	L6003	CORE-FERRITE BEAD
3404-001567	TAC5000	SWITCH-TACT
3705-001937	RFS1000	CONNECTOR-COAXIAL
3709-001884	SIM6000	CONNECTOR-CARD EDGE
3711-008508	HDC7000,HDC7001	CONNECTOR-HEADER
3711-008511	HDC7002	CONNECTOR-HEADER
3711-008847	HDC5000	CONNECTOR-HEADER
3711-008931	HDC7003	CONNECTOR-HEADER
3711-008997	HDC3000,HDC3001	CONNECTOR-HEADER
3711-009073	HDC5001,HDC7004	CONNECTOR-HEADER
3712-001633	ANT5000,ANT5001	CONNECTOR
3712-001694	ANT1001,ANT1002	CONNECTOR
3712-001694	ANT1003,ANT1004	CONNECTOR
3712-001694	ANT1005,ANT1006	CONNECTOR
3712-001694	ANT2002,ANT3000	CONNECTOR
3712-001694	ANT3001,ANT3003	CONNECTOR
3712-001694	ANT3004,ANT5004	CONNECTOR
3712-001694	ANT5005,ANT5006	CONNECTOR
3712-001694	ANT5007,ANT6000	CONNECTOR
3712-001694	ANT6001	CONNECTOR
3712-001731	ANT2001,ANT3002	CONNECTOR
4709-002150	F3002,F3003	RF-MODULE
4709-002226	F1003	RF-MODULE
4709-002285	DIF3000	RF-MODULE
4709-002464	F1004	RF-MODULE
4709-002484	U2005	RF-MODULE
4709-002566	F3001	RF-MODULE
4709-002589	F1002	RF-MODULE
GH61-14072A	SUS6000	SUS
GH61-14073A	SUS6001	SUS
GH61-14174A	SUS6002	SUS

5. MAIN Electrical Parts List

GH62-00042A	GA7000,GA7001	GASKET
GH63-16694A	SC6000	SHIELD-CAN
GH63-16695A	SC6004	SHIELD-CAN
GH63-16696A	SC6002	SHIELD-CAN
GH63-16699A	SC6001	SHIELD-CAN
GH98-43873A	SC6003	SHIELD-CAN
GH98-43942A	SC6005	SHIELD-CAN
rf-2D-BARCO	CON6000,CON6001	2D-BARCODE

Please consult the GSPN website (Samsung Portal) for the most recent version of the product's part list.

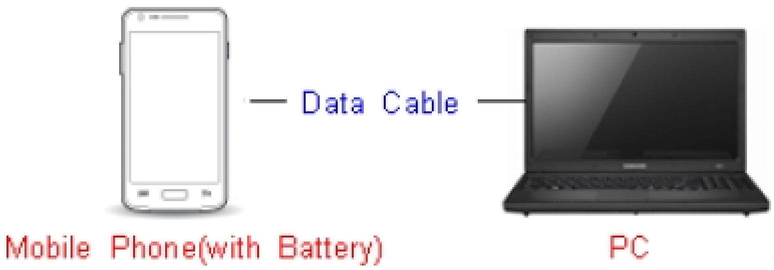
6. Level 1 Repair

6-1. S/W Update

6-1-1. Preparation

- S/W Update program : **Fenrir 5.17.xxxx**
- Mobile Phone
- Data Cable

※ Settings



Data Cable : GH39-01971A

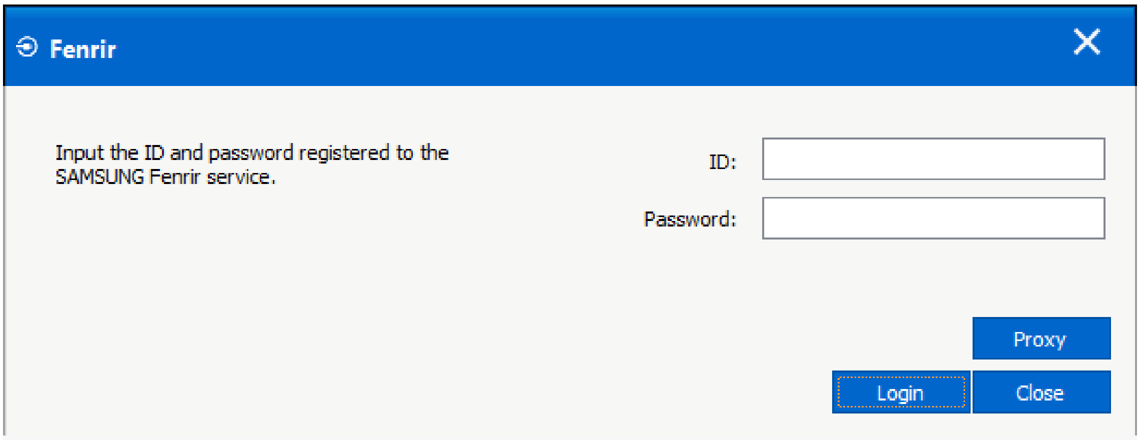
6-1-2. How to use 'Fenrir' S/W update program.

1) Launch Fenrir by clicking on the icon on the desktop

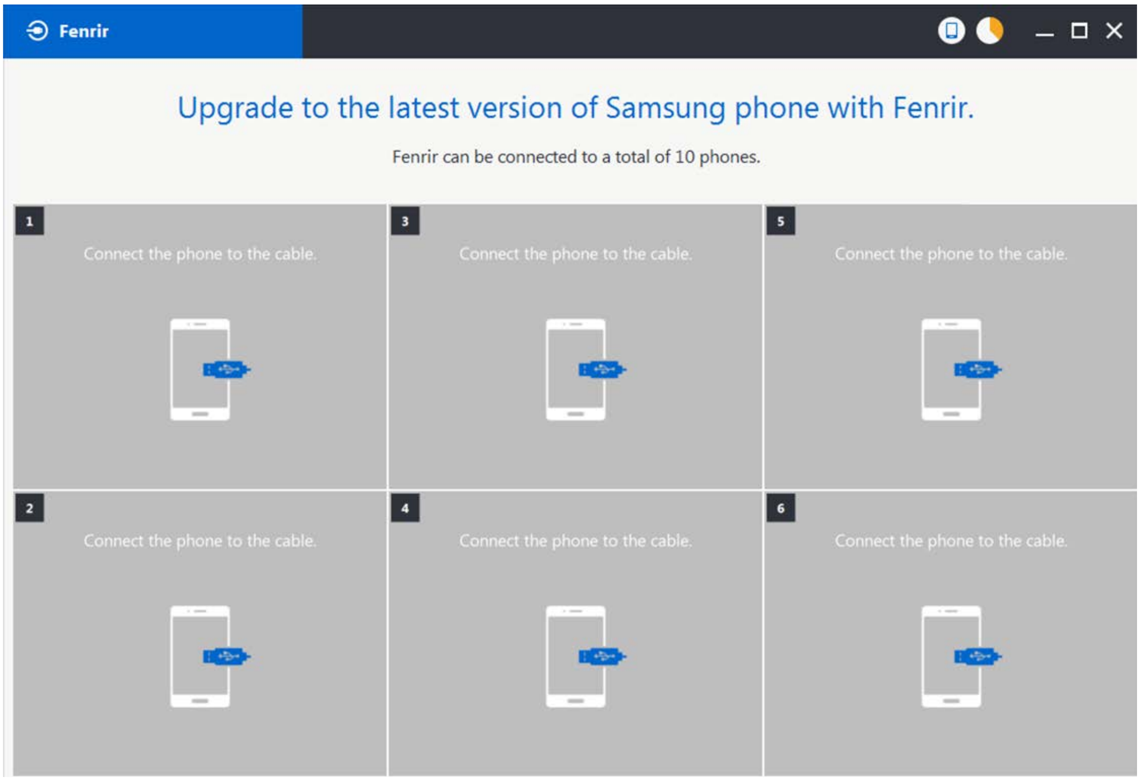
- SVH (Fenrir_Home) : It uses Home binary which does not have user data area in the memory when flashed to a device. (Keep user data)

- SVC (Fenrir_Factory) : It uses Factory binary which erases all user data in the memory when flashed to a device. (Clear user data)
- SVA (Fenrir_All) : It uses Factory and Home binaries. you can download Home and Factory binary in a PC(but requires double HDD storage and NW traffic)

2) Input ID & password
※You need to reset the ID information in case of PC change and format and repair, hard disk change

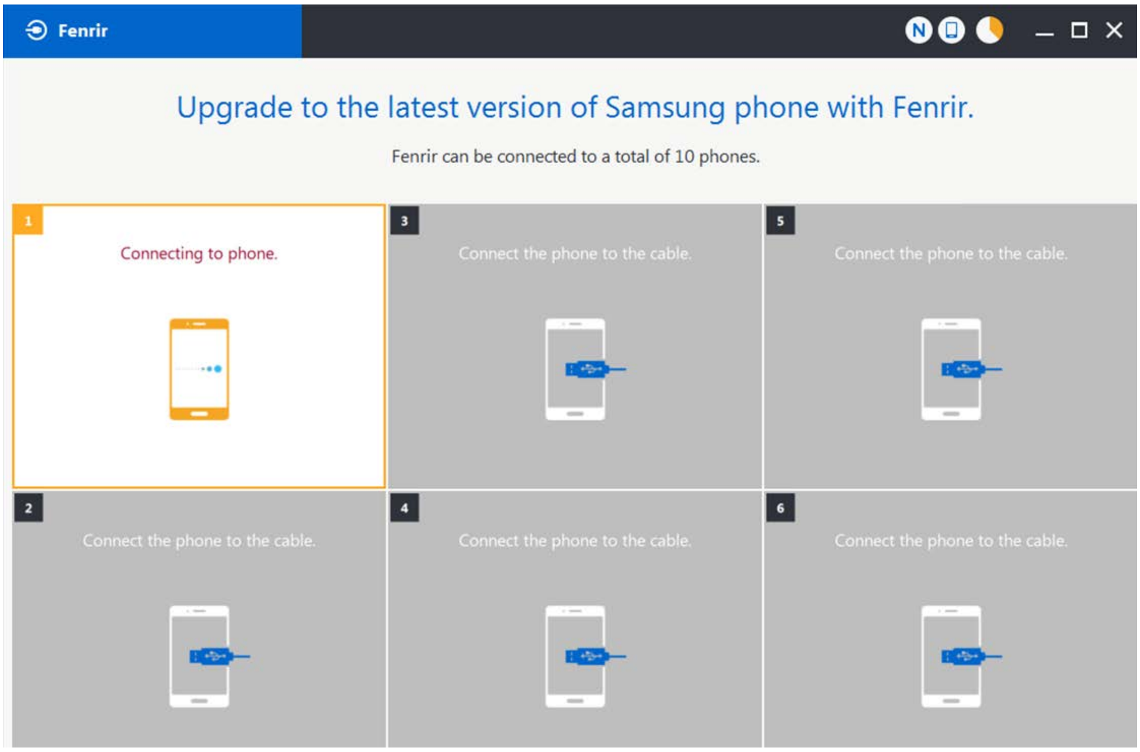


3) Ensure device has sufficient charge (at least 20%) to start firmware update.

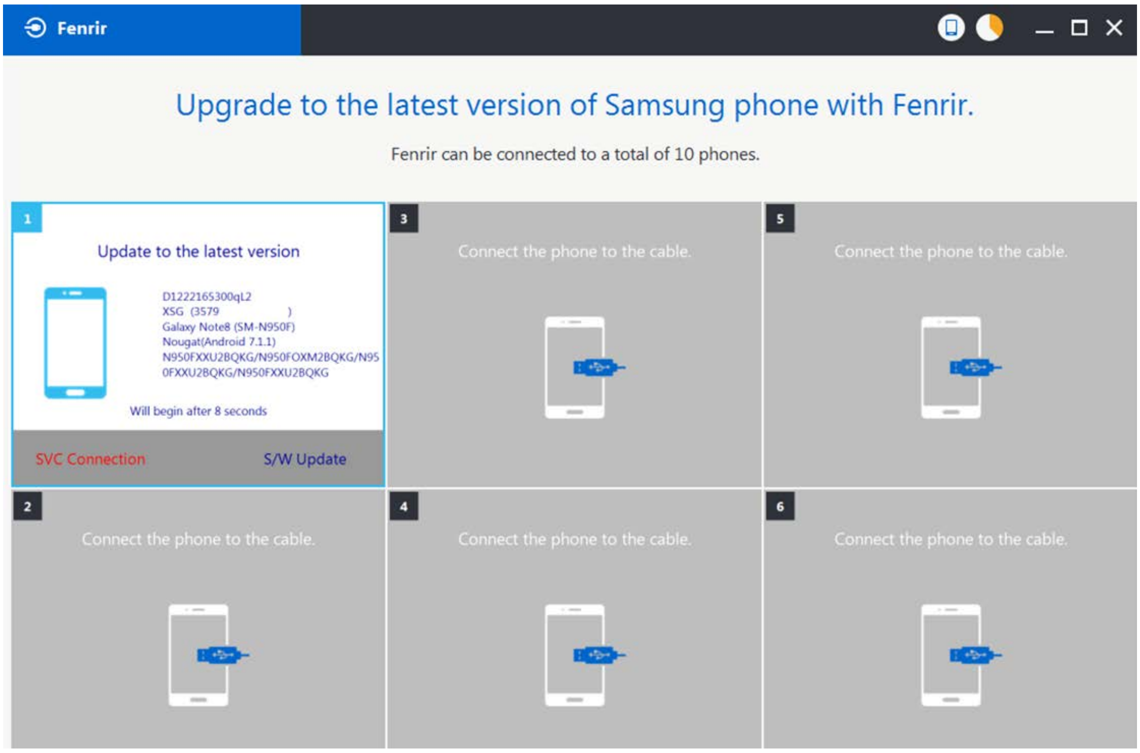


4) Connect the device to PC via data cable.

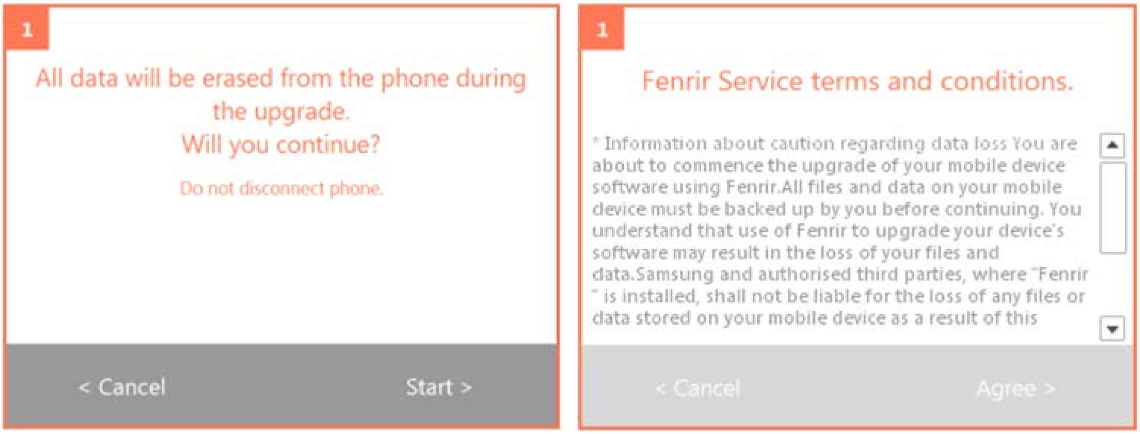
5) Upon USB connection, you will be presented with below screen.



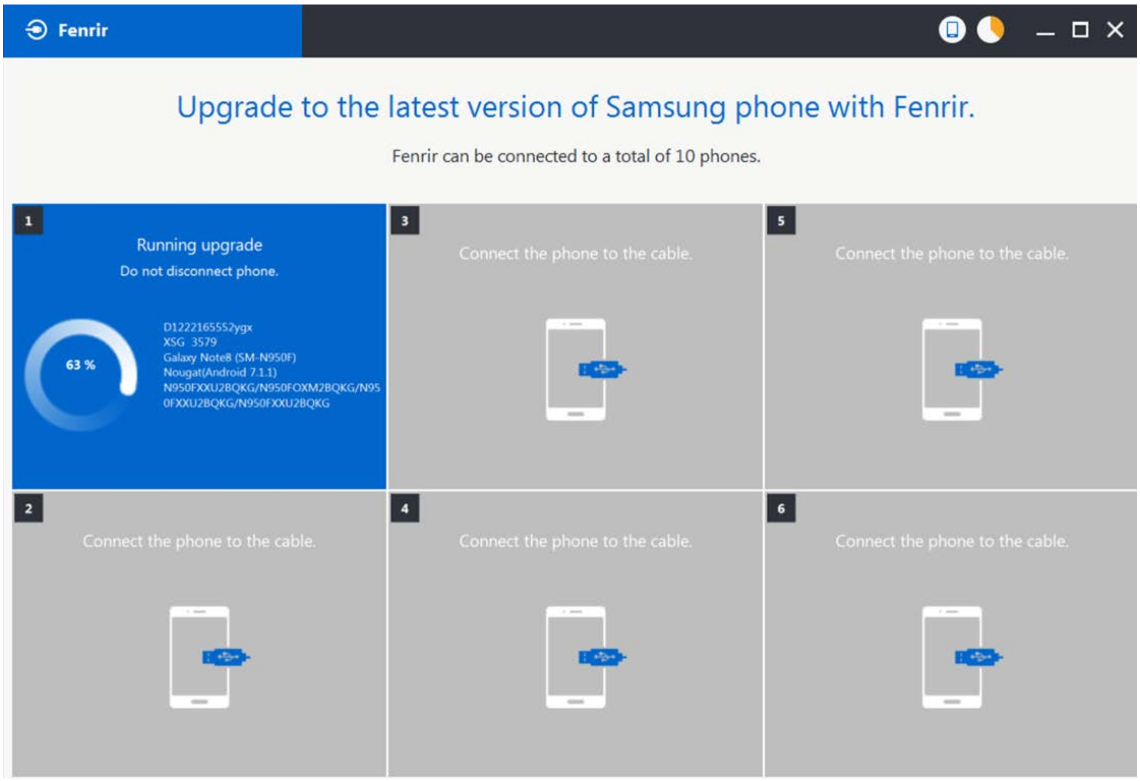
6) Once device is detected, you will be presented with below screen. To update S/W, select "S/W Update" or to exit select "SVC Connection". If you select "SVC Connection", only Fenrir connection history (record) will be stored in the FUS server to support warranty validation. (This is known as "Service Connection" history)



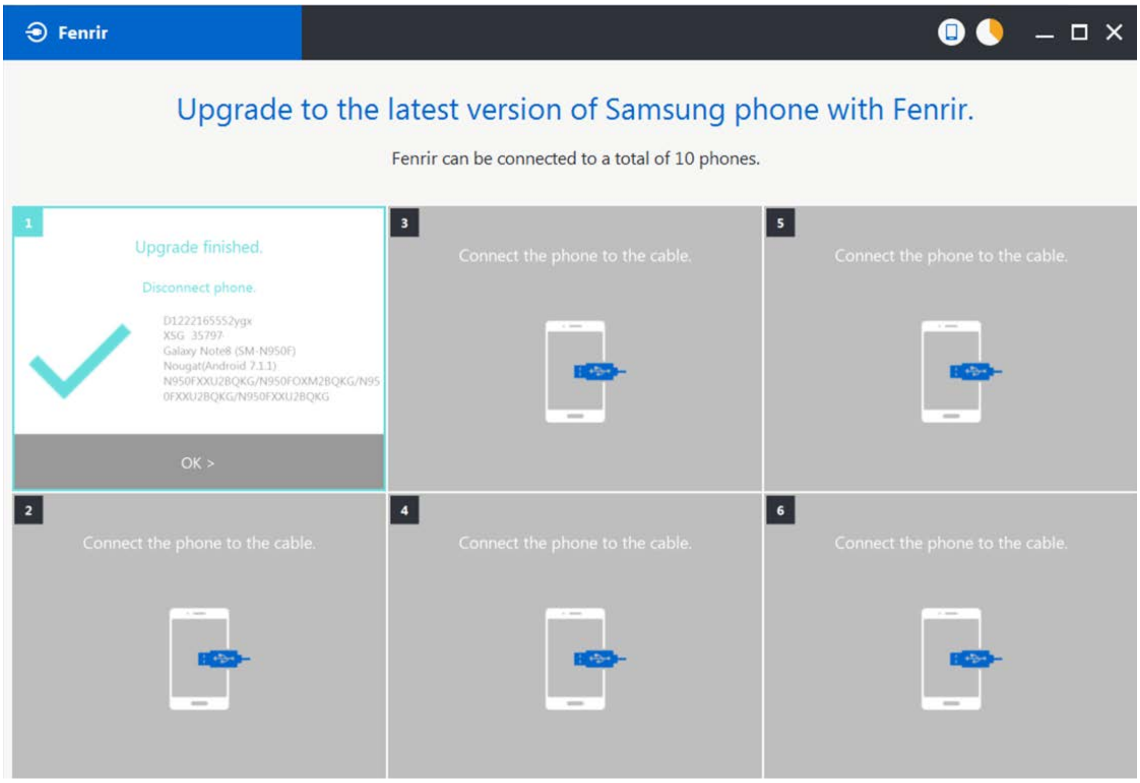
7) Once Fenrir starts, application will display the below screen. And select the Start button & Agree button.



8) The status circle increases as the update installs. The update process takes approximately 5-10 minutes to complete. Do not disconnect the device from USB during processing.



9) Once complete, application will present the below screen indicating update complete.
Click Ok and detach device from USB.



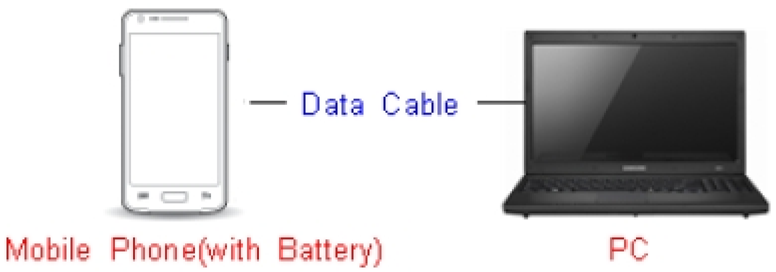
6-2. How to use 'Odin' program

※ S/W Update via Fenrir is mandatory.
Below is the method to use 'Odin' program in any specific case.

6-2-1. Preparation

- Installation program : **Odin3 v3.13.2.exe or above**
- Mobile Phone
- Data Cable
- S/W Binary files (downloaded from GSPN)

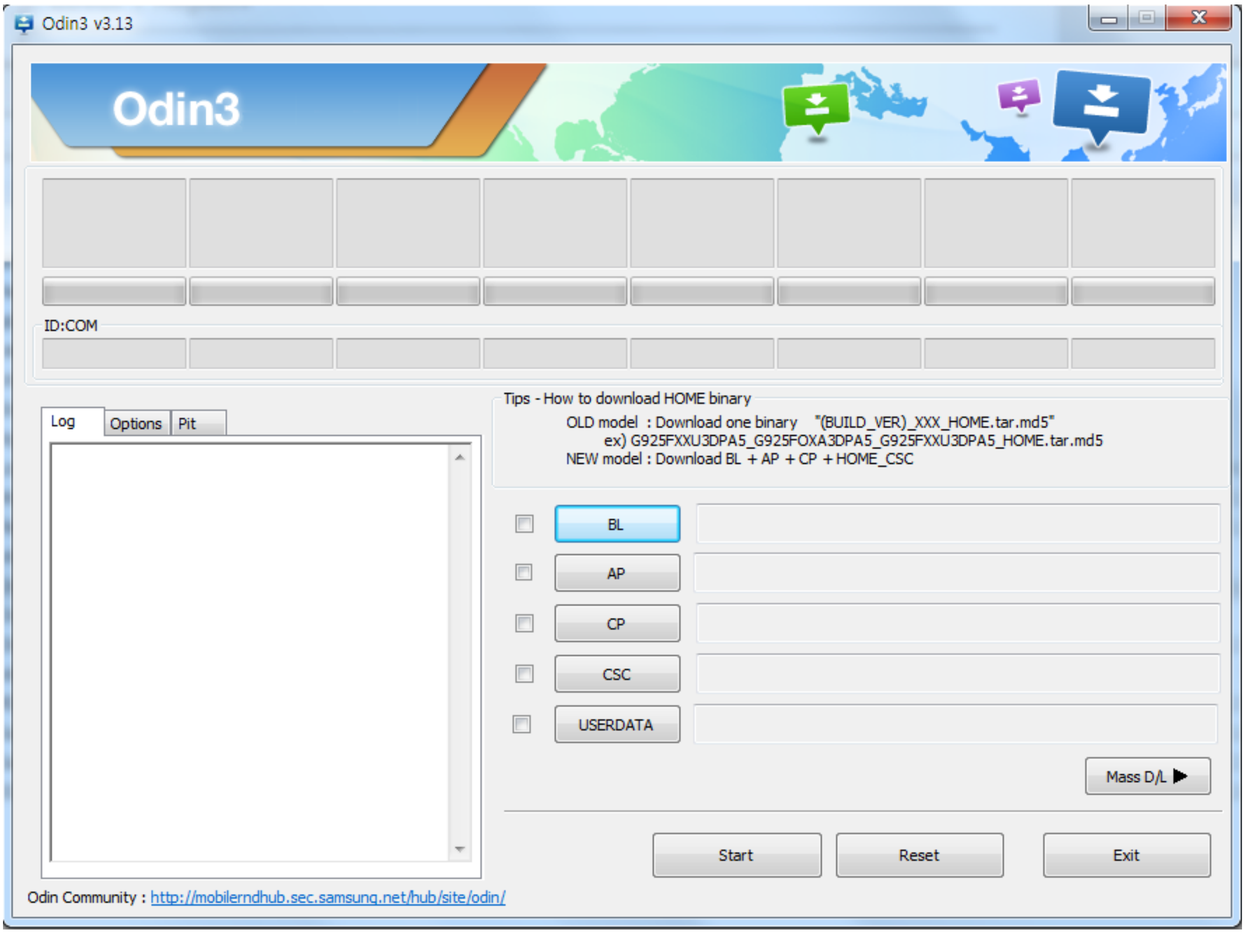
※ Settings



Data Cable : GH39-01971A

6-2-2. S/W Installation Program (Downloader program)

Open up the S/W Installation Program by executing the **"Odin3 v3.13.2.exe"**

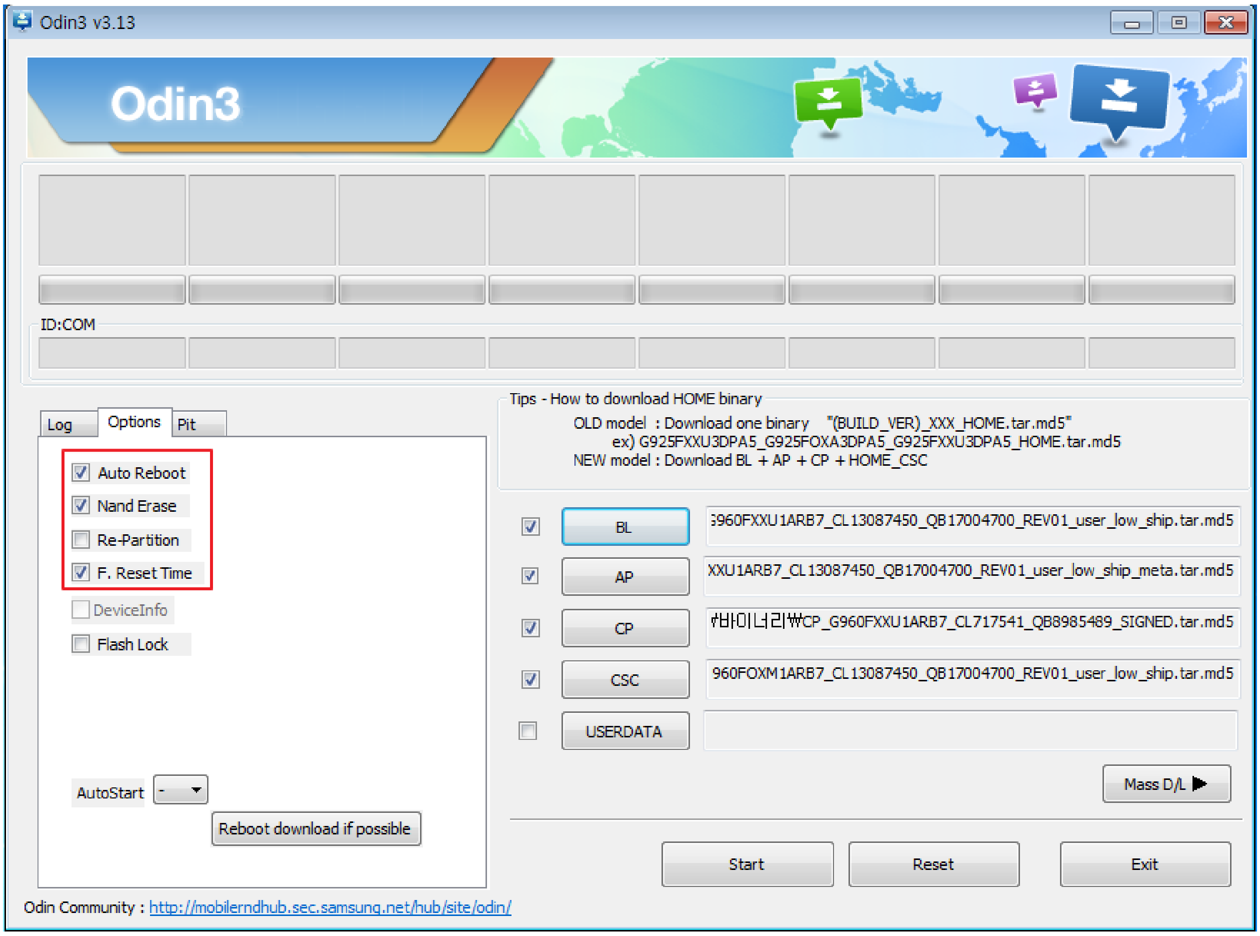


6. Level 1 Repair

1. Enable the check mark by click on the following options

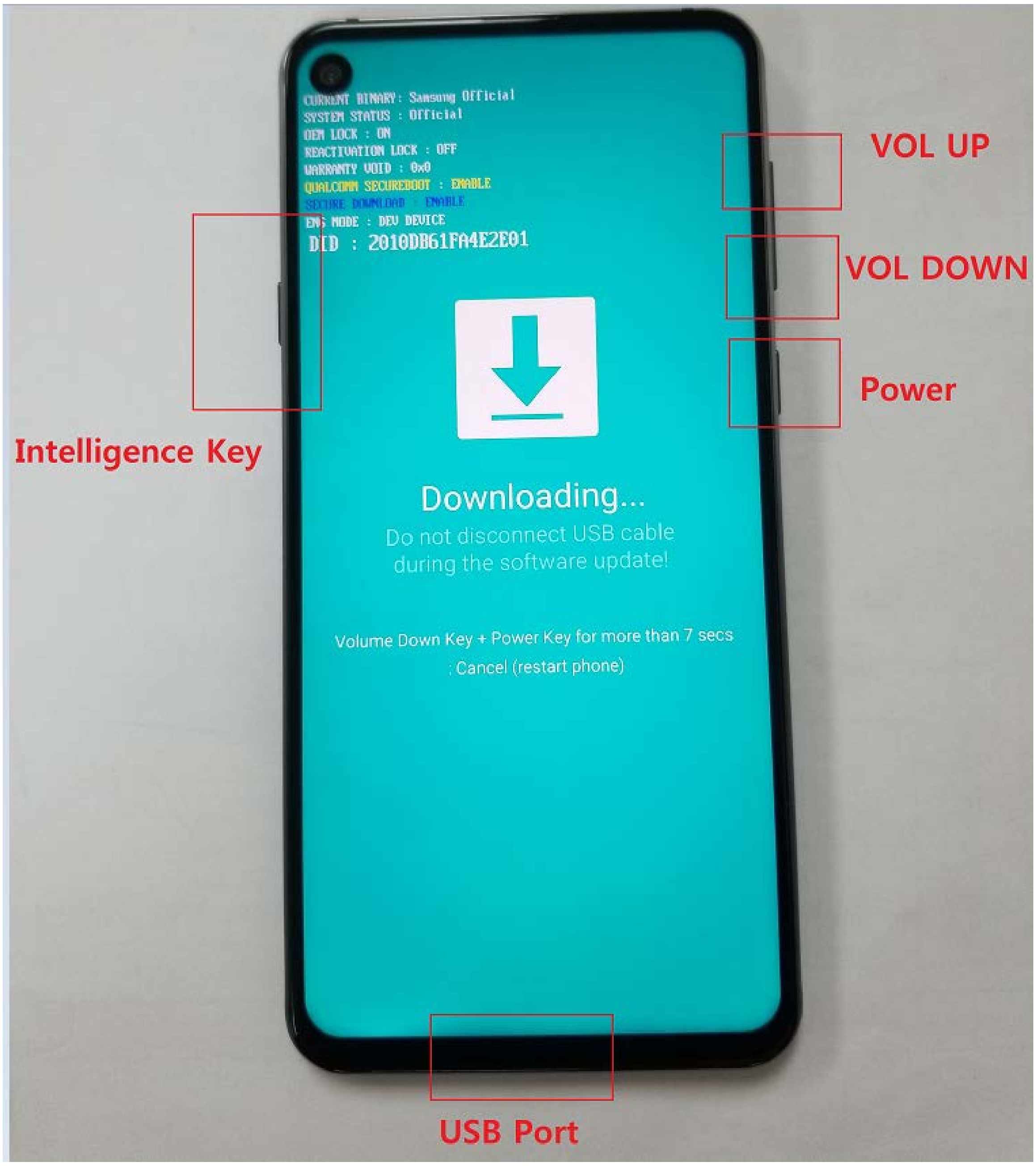
- Check   Auto Reboot, F. Reset Time, Nand Erase
- Check BL, AP, CP, CSC Files

* Note : "Odin v3.13.2 or above" checks MD5 checksum just after file selection.

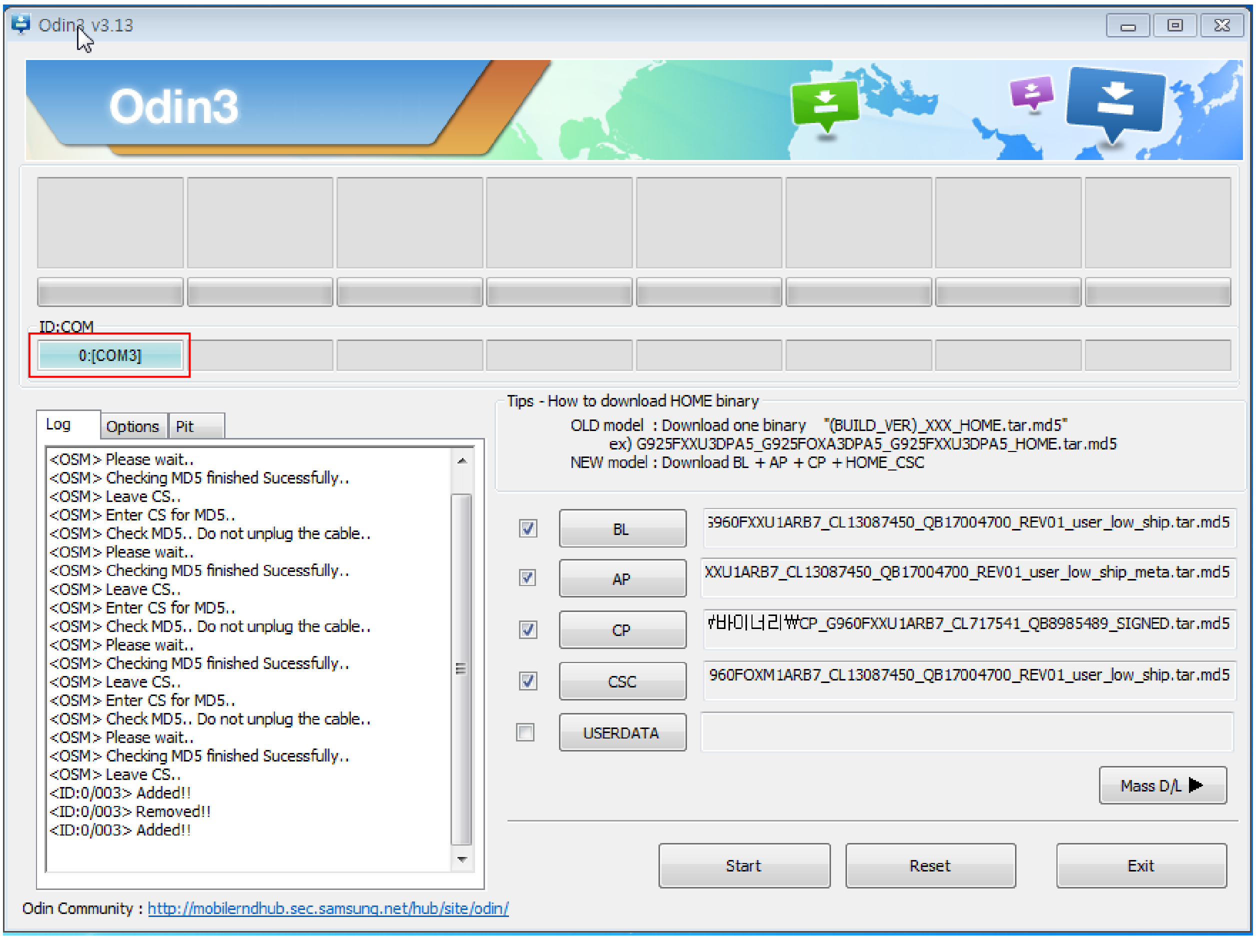


6. Level 1 Repair

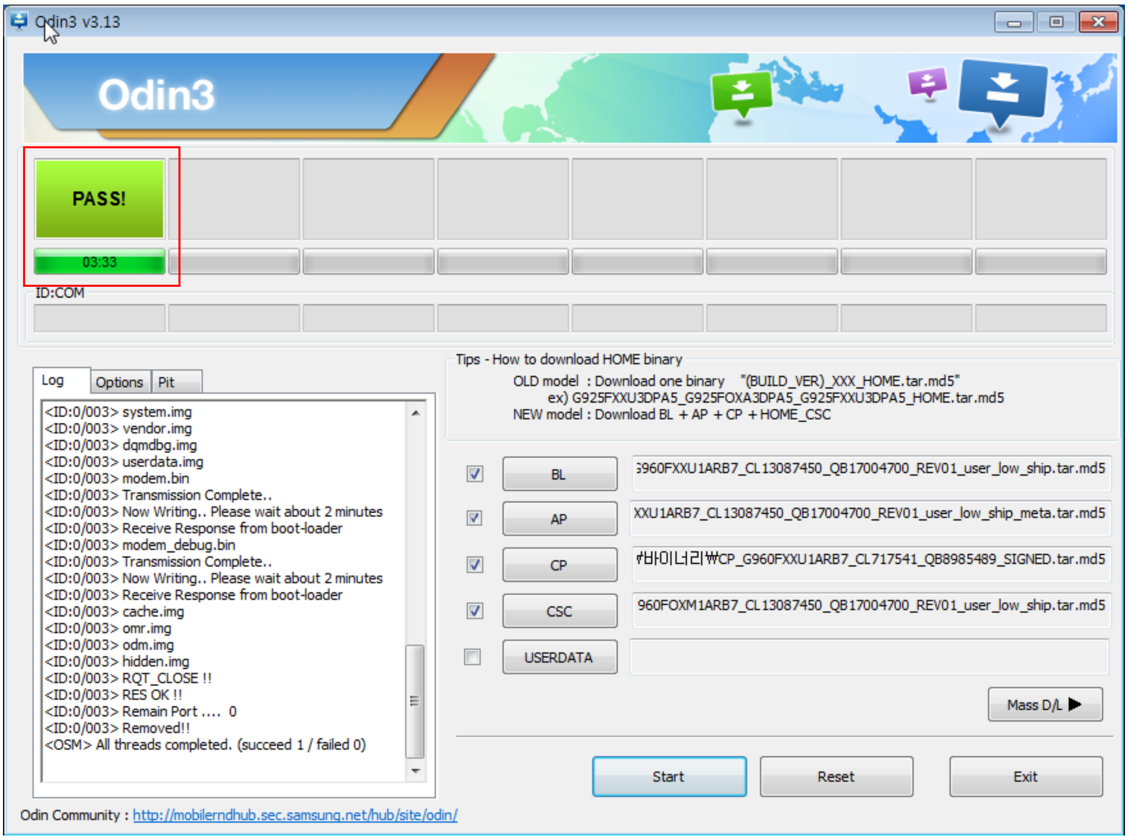
2. Enter into Download Mode
- Enter into Download Mode by pressing Volume Down button, Intelligence button and connecting USB cable
-Press Volume Up button to download mode.



3. Connect the device to PC via Data Cable.
Make sure that the one of communication ports [ID:COM] box is highlighted in sky blue.
The device is now connected with the PC and ready to download the binary files in it.



4. Start downloading the binary files into the device by clicking Start button on the screen.
The green colored "PASS!" sign will appear on the upper-left box if the binary files have been successfully downloaded into the device.



5. Disconnect the device from the Data cable.

6. Once the device boots up, you can check the version of the binary file or name by pressing the following code in sequence; ***#1234#**

You can perform Factory data Reset by Settings → General Management → Reset

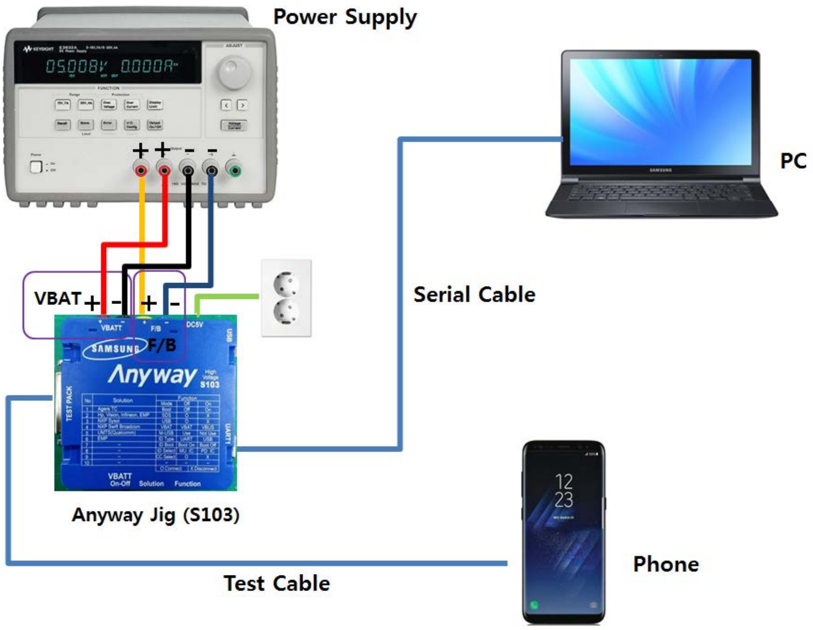
※ Caution. Never disconnect during the S/W downloading.

6-3. IMEI writing

6-3-1. Preparation

- New IMEI writing Program has been released.
- Supported Model : Models which CAB files are uploaded on HHPsvc INI File category, instead of ini file.
- Refer to below IMEI writing procedure.

- H/W



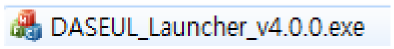
- S/W

① Library Install	To use Daseul, library files should be installed. Refer to SVC Bulletin “(11-82) Daseul (New IMEI writing Program) Library Install guide_rev1.0”
②Launcher	**DASEUL_Launcher_v4.0.0** or higher -Uploaded on HHPsvc Notice
③ Runtime File	1. **DASEUL_IMEI_ALL_Runtime_3.1.386.0_r00573.CAB** or higher -Uploaded on HHPsvc Notice 2. Make ‘ModelName’ folder at the same position with launcher & Runtime file. DASEUL_IMEI_ALL_Runtime_3.1.386.0_r00573.CAB DASEUL_Launcher_v4.0.0.exe SM-A920F_128DS(CSC)_IMEI_Ver_3.1.385.1.CAB
④Model File	Copy Model File under the ‘SM-G8870’ folder

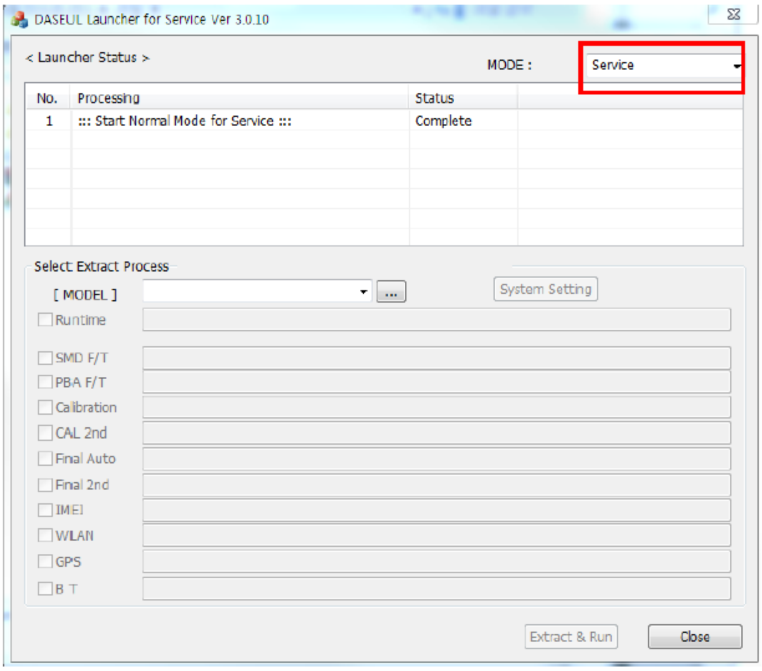
6. Level 1 Repair

6-3-2. IMEI writing Process

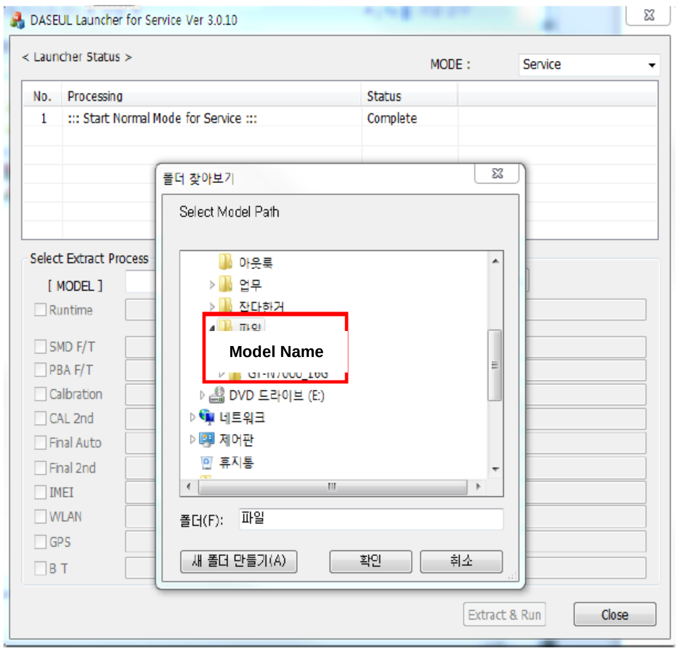
1. Run **DASEUL_Launcher_v4.0.0**



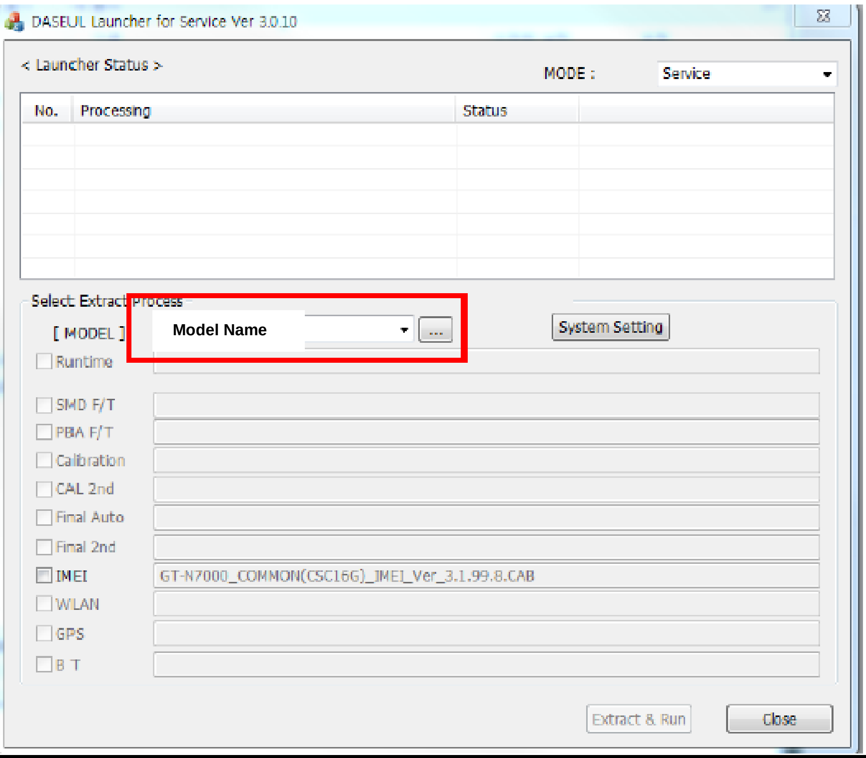
2. Select Service Mode



3. Click  and Select folder where the Launcher exists

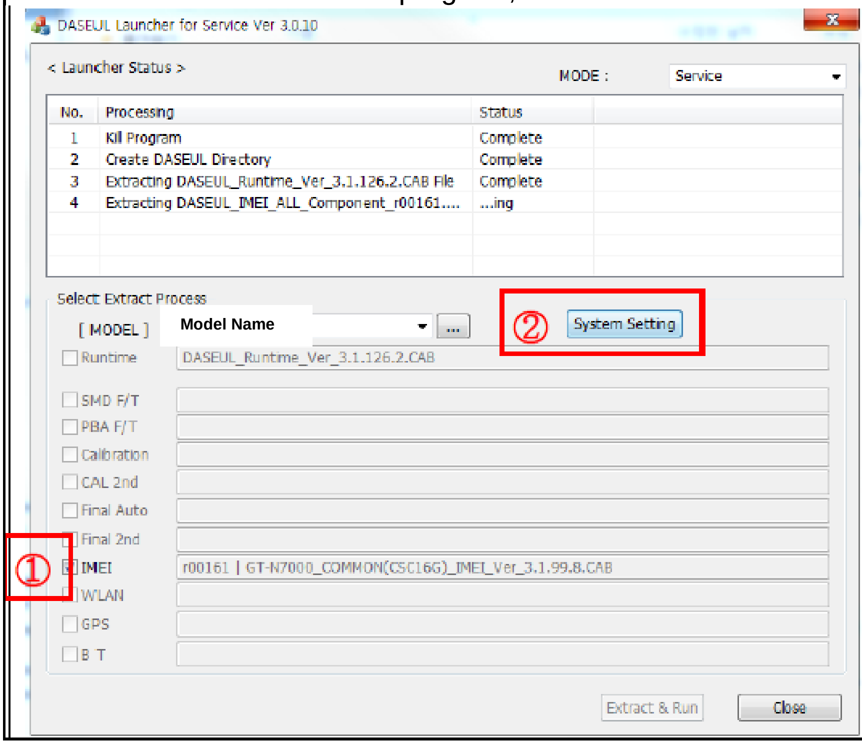


4. Select Model

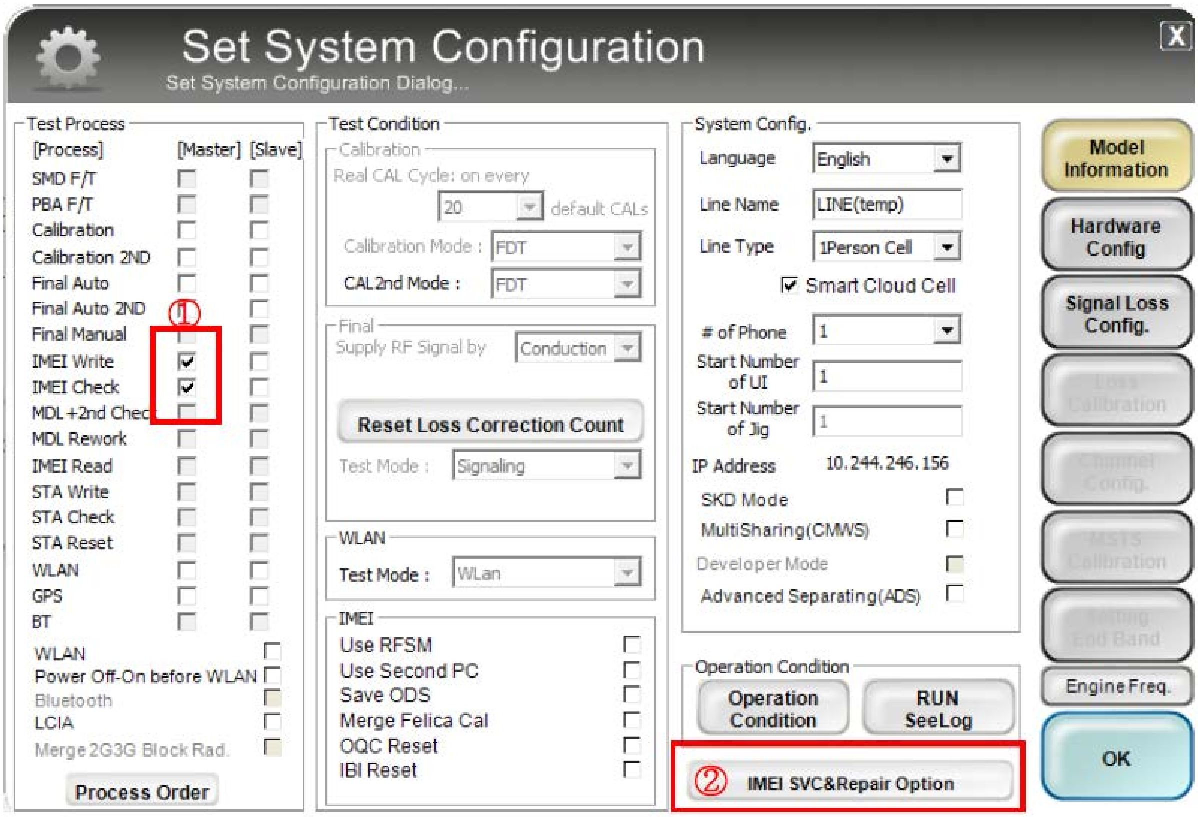


5. Check IMEI and click System Setting

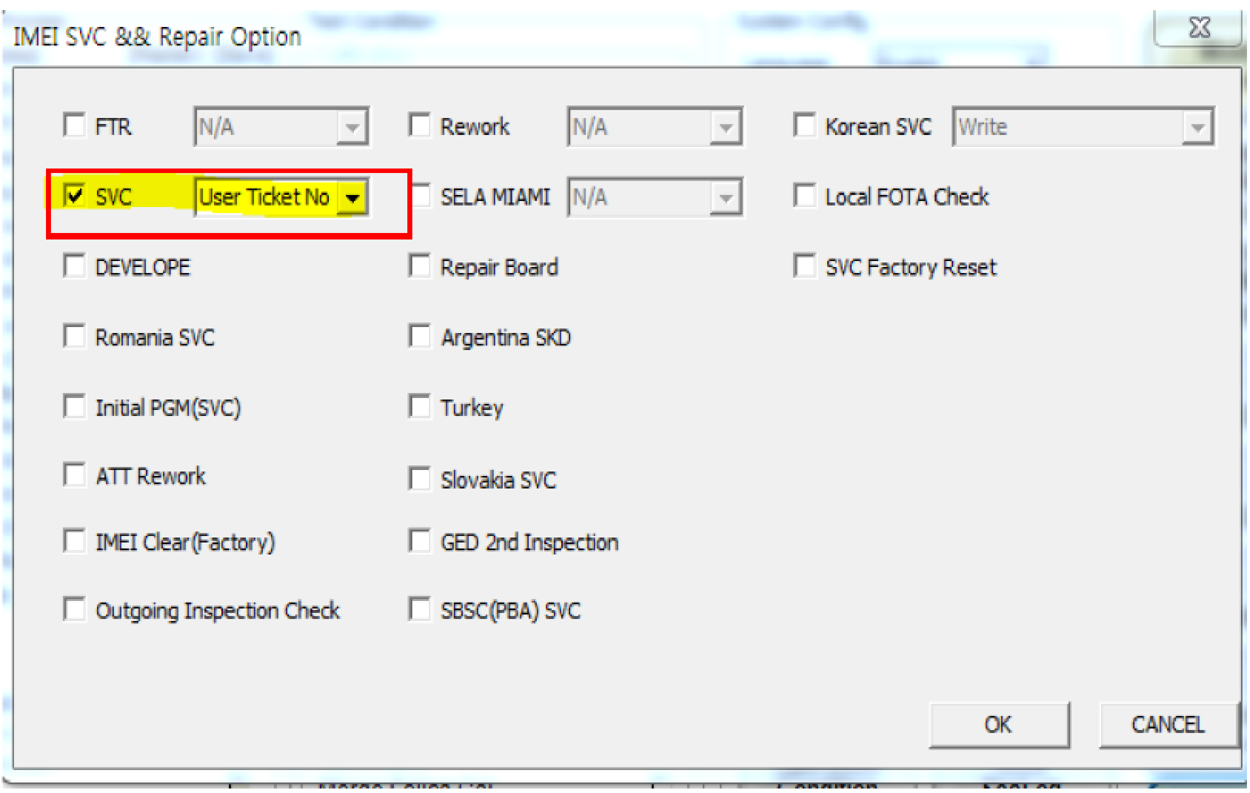
※Once you setup the setting, you don t have to do it again, unless there is change.
From second run of the IMEI program, check IMEI and click Extract & Run.



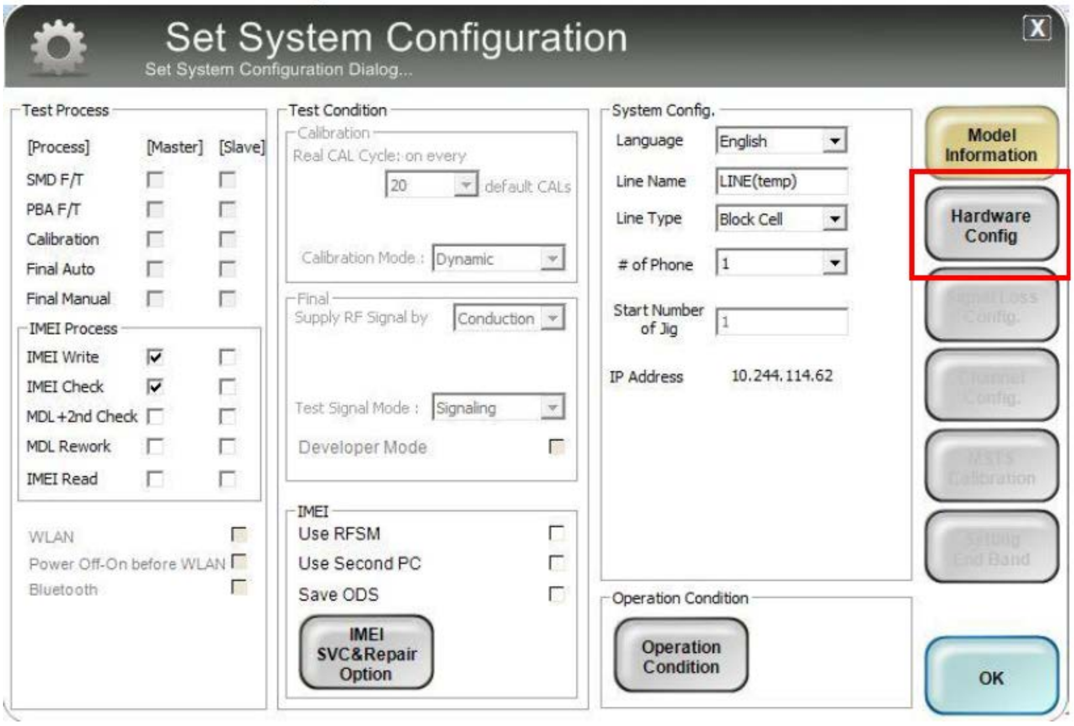
6. Check IMEI Write / IMEI Check and click IMEI SVC & Repair Option.



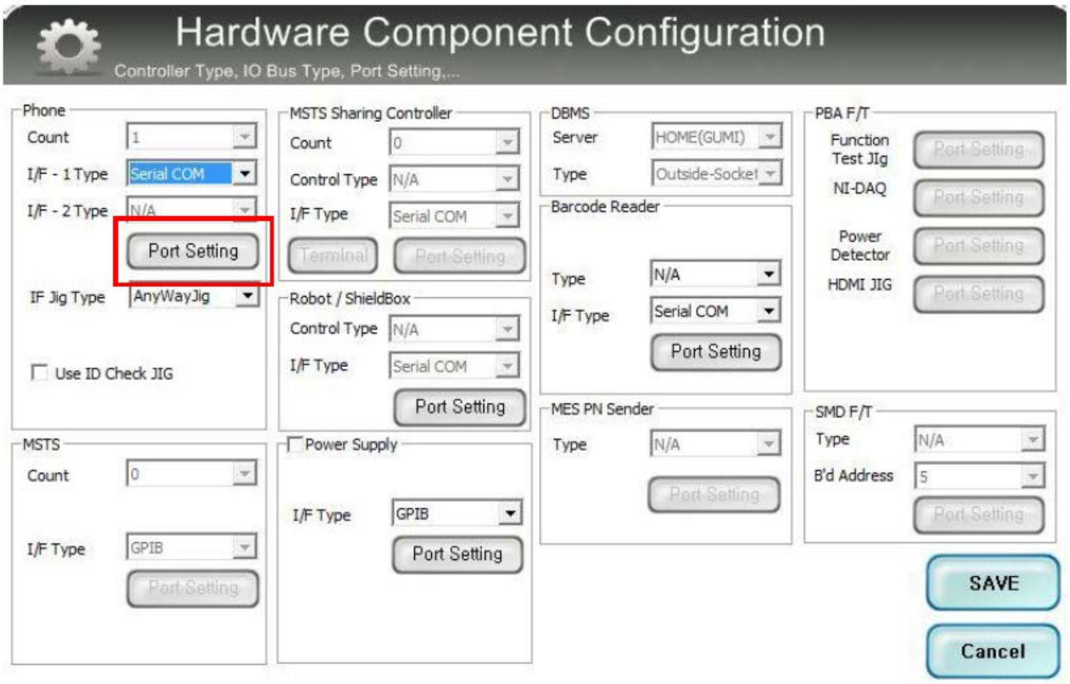
7. Check 'SVC , User Ticket No' and click OK



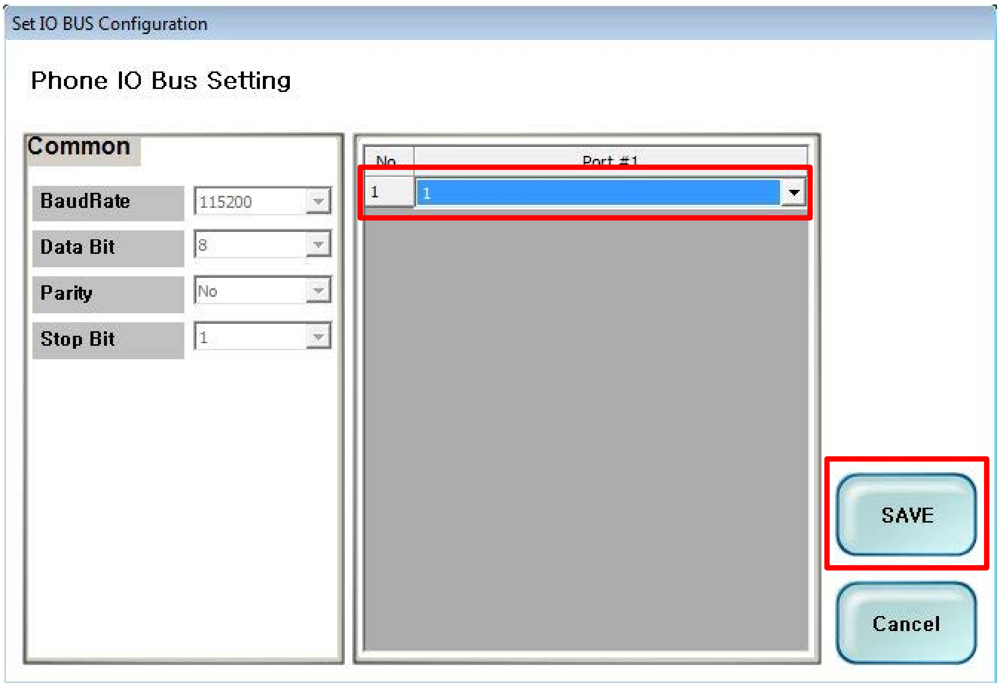
8. Click 'Hardware Config'



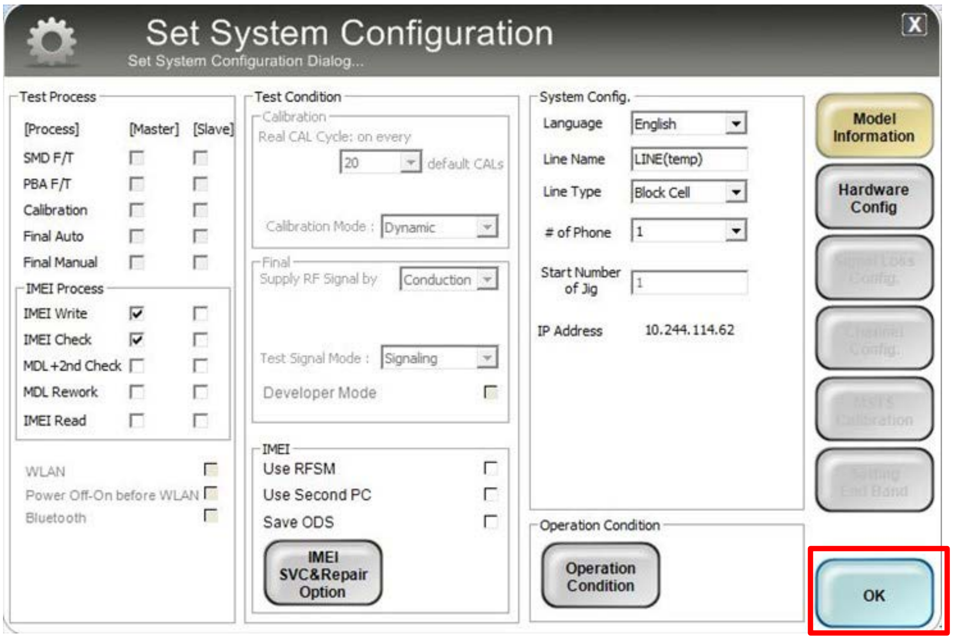
9. Click 'Port Setting'



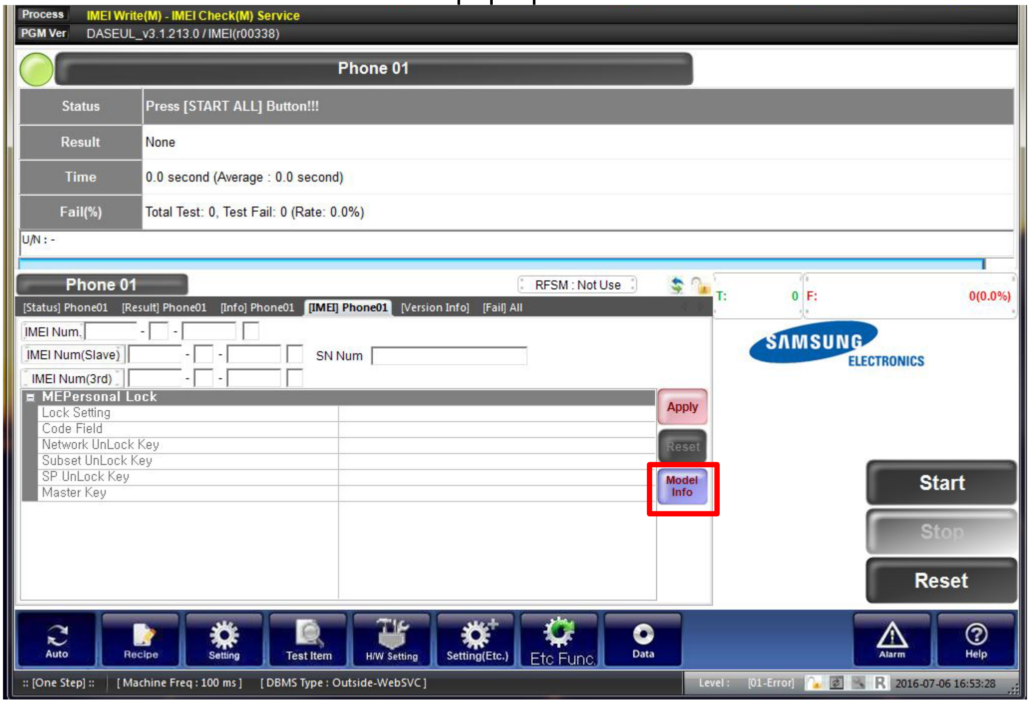
10. Select Port Number and SAVE



11.Click OK to proceed



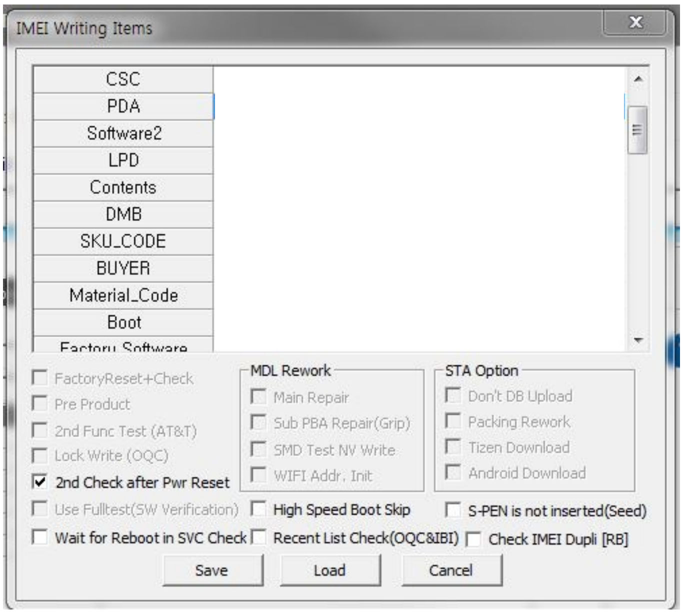
12. Click Model Info and OK when pop-up shows



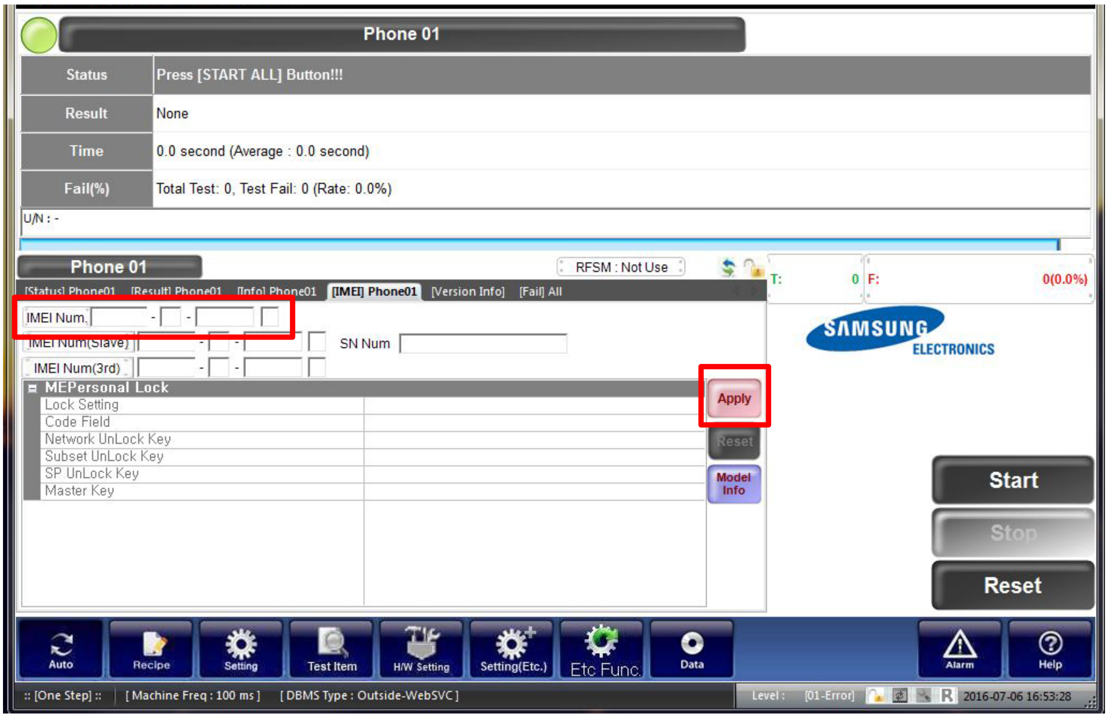
13. Click OK

14. Input SKU_CODE and BUYER, then click Save button.

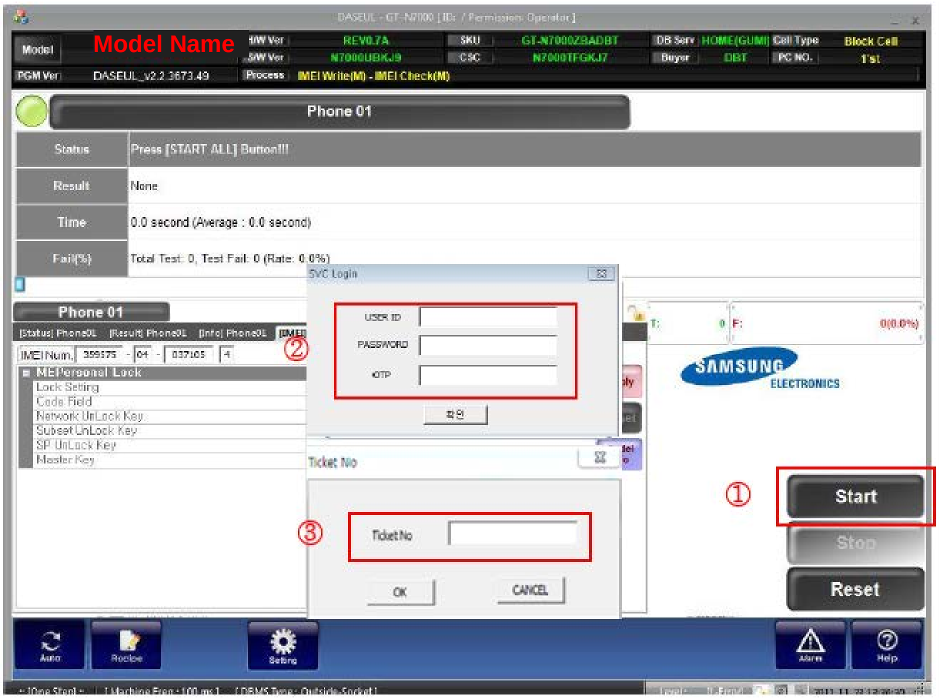
※ Refer to HHPsvc→IMEI Review to check SKU Code and buyer



15. Input IMEI Number and click Apply



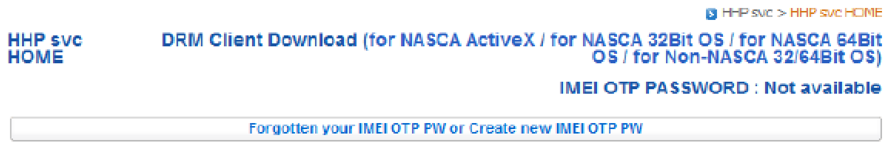
16. ① Click Start → ②Input IMEI writing ID and Password & OTP → ③Input Ticket No

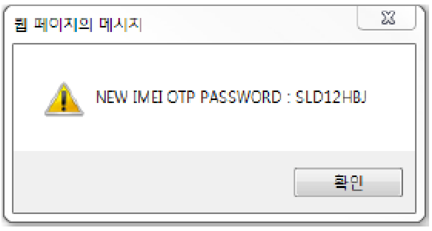


※ OTP(One time Password) : OTP is valid for 6 hours.

After that, you can get new OTP by click the "Forgotten your IMEI OTP PW or Crete new IMEI OTP PW" button.

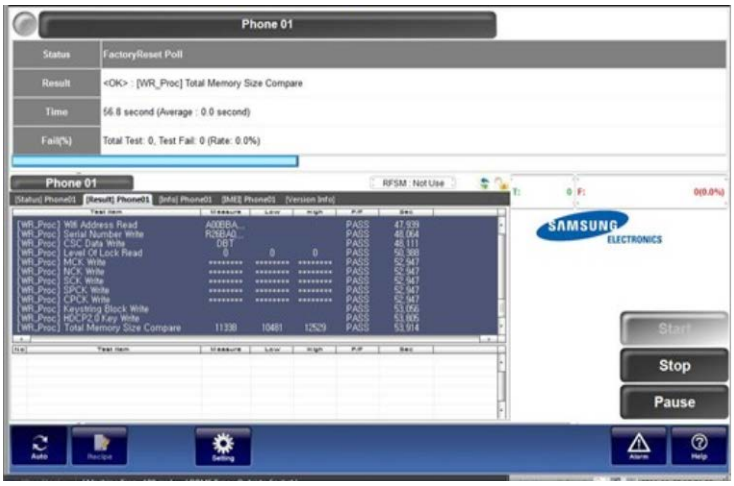
☞ OTP Location : GSPN → Knowledge → HHP svc → Home



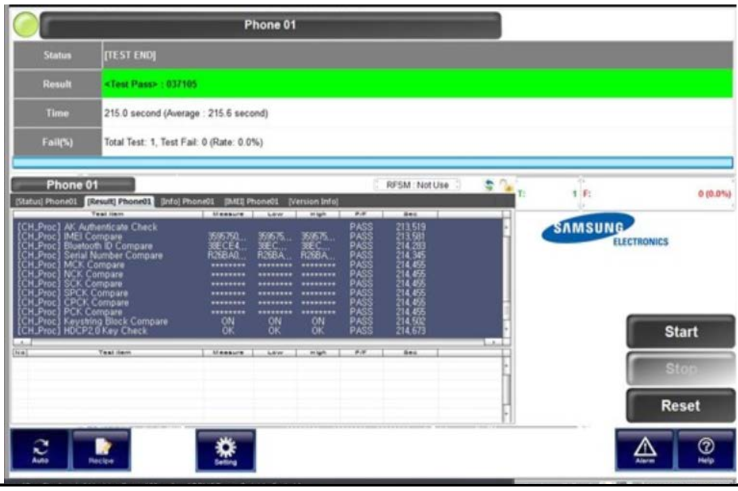


17. Connect the phone to Anyway JIG
※ When you connect the phone, the phone should be turned off.
After connecting the phone, the phone will be booted automatically.

18. IMEI Writing Proceeding



19. IMEI Writing Success



6-4. RF Calibration

6-4-1. Required items in order to calibrate RF

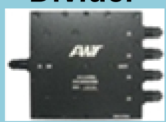
• Installation program: RF Calibration Program
- Daseul_Launcher_vx.x.xx.exe
- Daseul_CAL_ALL_Runtime_x.x.xxx.x.CAB
- Model File

: SM-xxxx_OPEN_CALIBRATION_Ver_x.x.xxx.x.CAB

※ It is required to use the latest program.

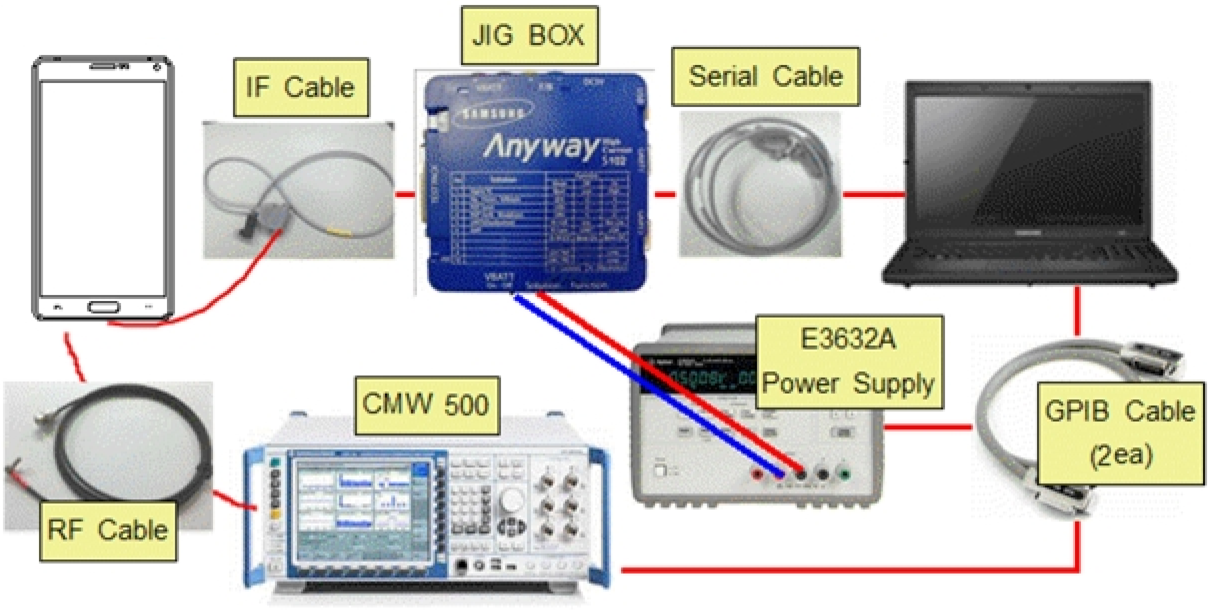
- Mobile Phone
- R&S CMW500
- E3632A Power Supply
- GPIB Cable (2ea)
- JIG BOX (S103)
- Adapter (GH81-11888K)
- UART Serial Cable
- IF Cable (GH81-11962W)

❖ Table of test cables

RF Cable (Manual)	**GH81-11962M**	**GH81-11962U**	
	1.2T, male, 102mm 	**1.4M, Long** 	
4 Port Divider	**GH81-11962A**	**GH81-11962B**	**GH81-11962E**
	Divider 	**Divider Cable** 	**50Ω terminator** 

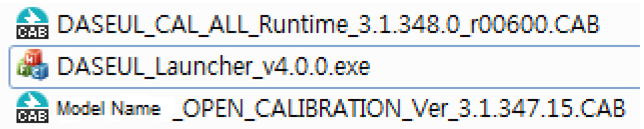
6. Level 1 Repair

❖ Setting

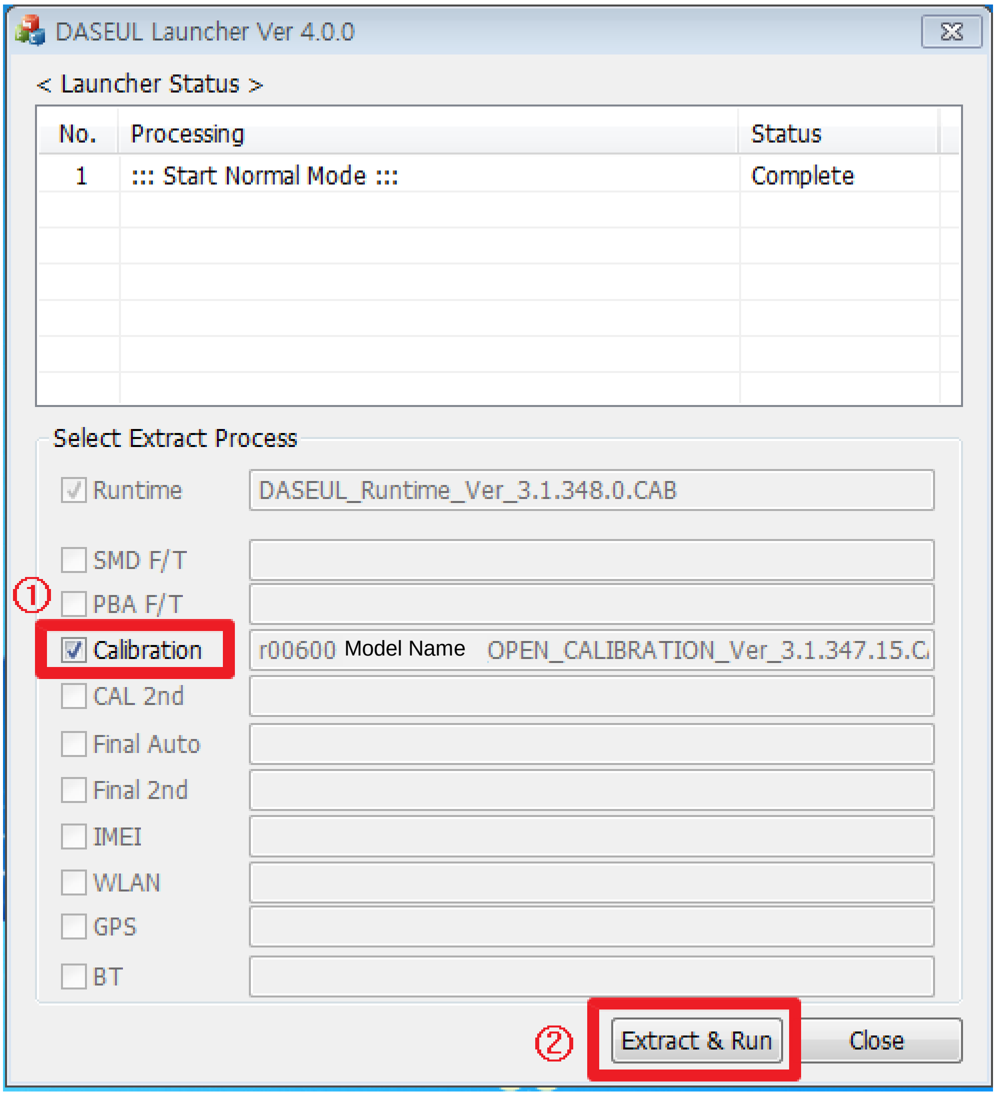


6-4-2. RF Calibration Program

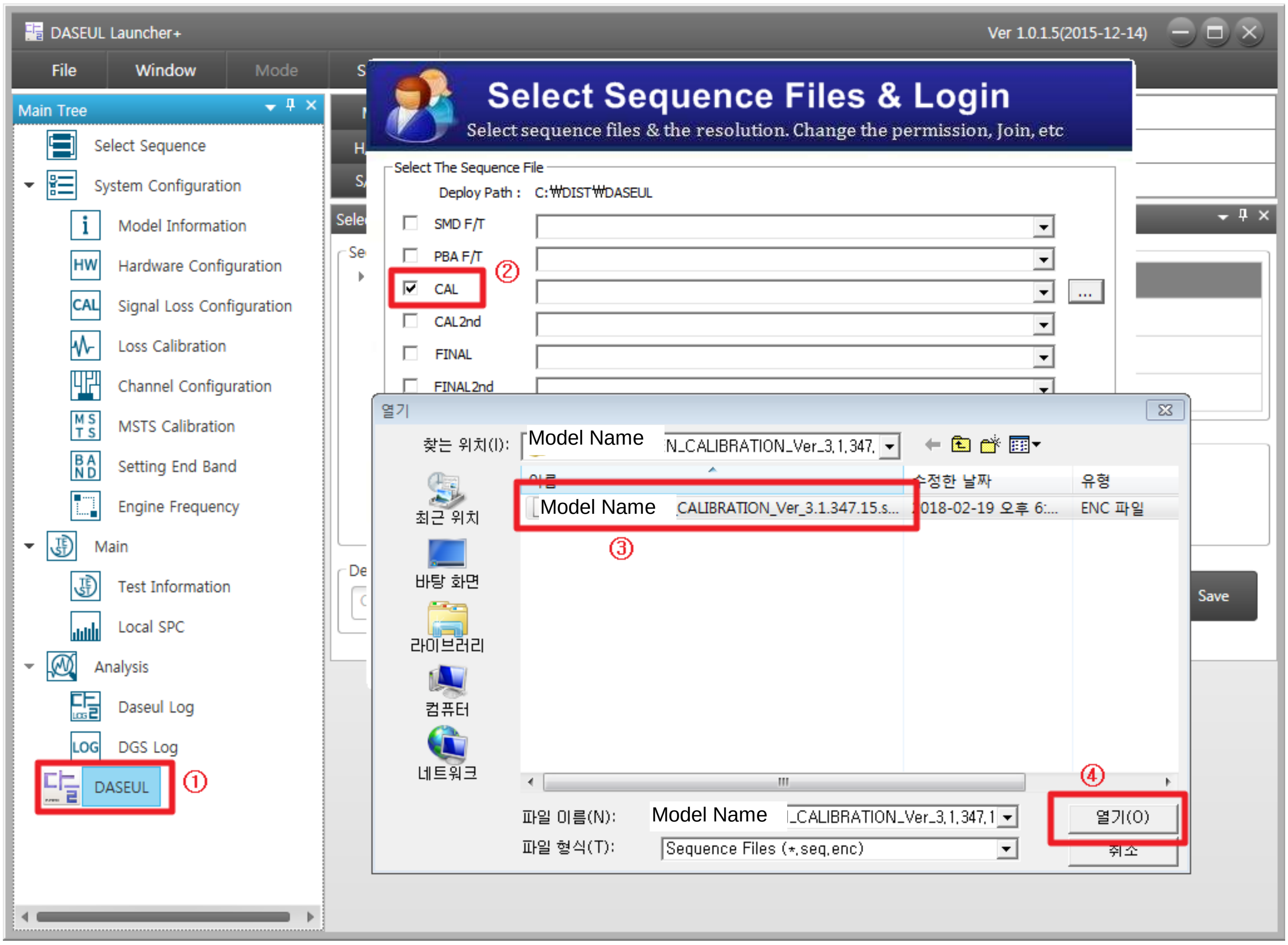
1. Run the RF Calibration Program Launcher, 'DASEUL_Launcher_vx.x.xx.exe'.



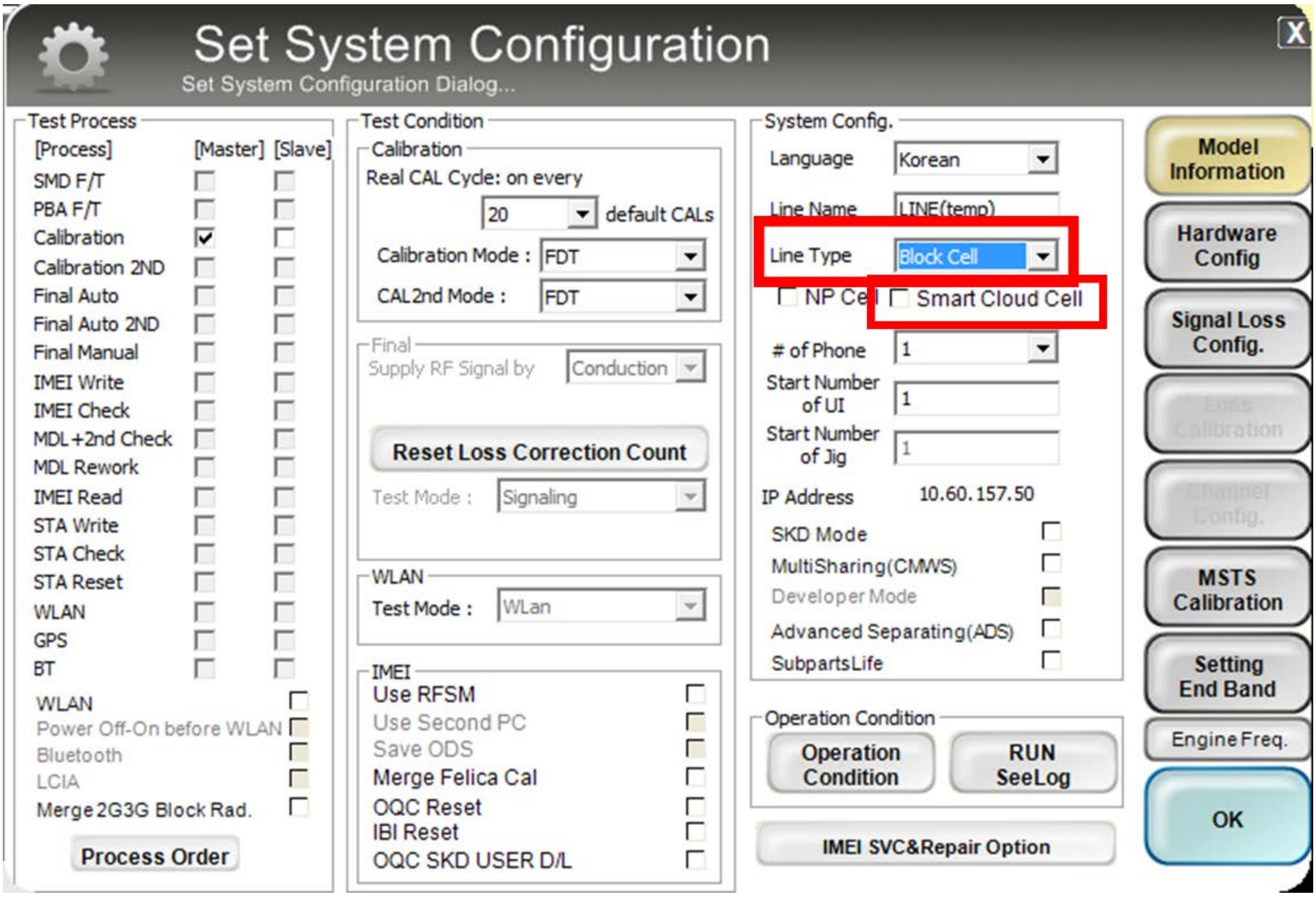
2. Check the 'Calibration' option and Click 'Extract & Run'.



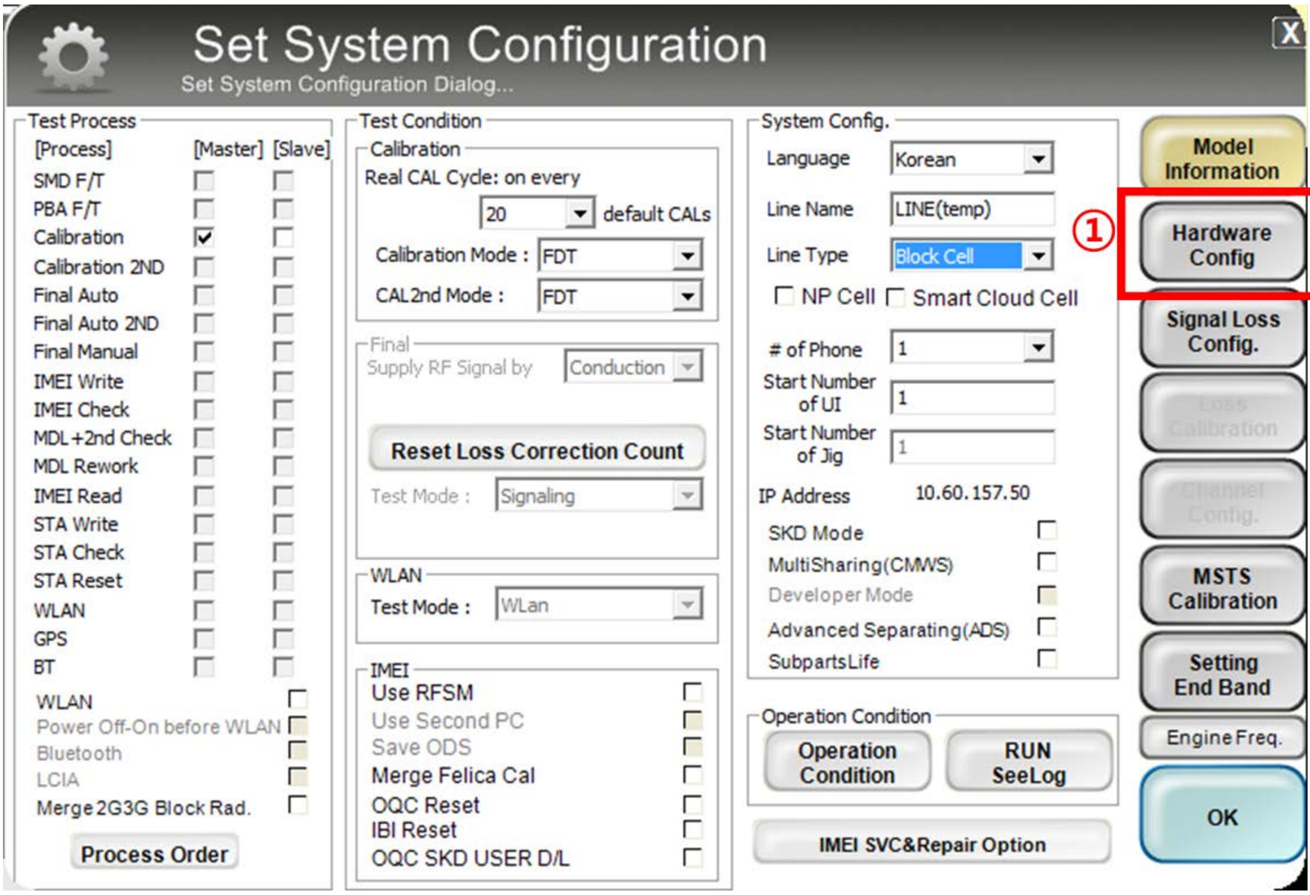
3. Check the 'CAL' and open the model file, then select 'Start' button.

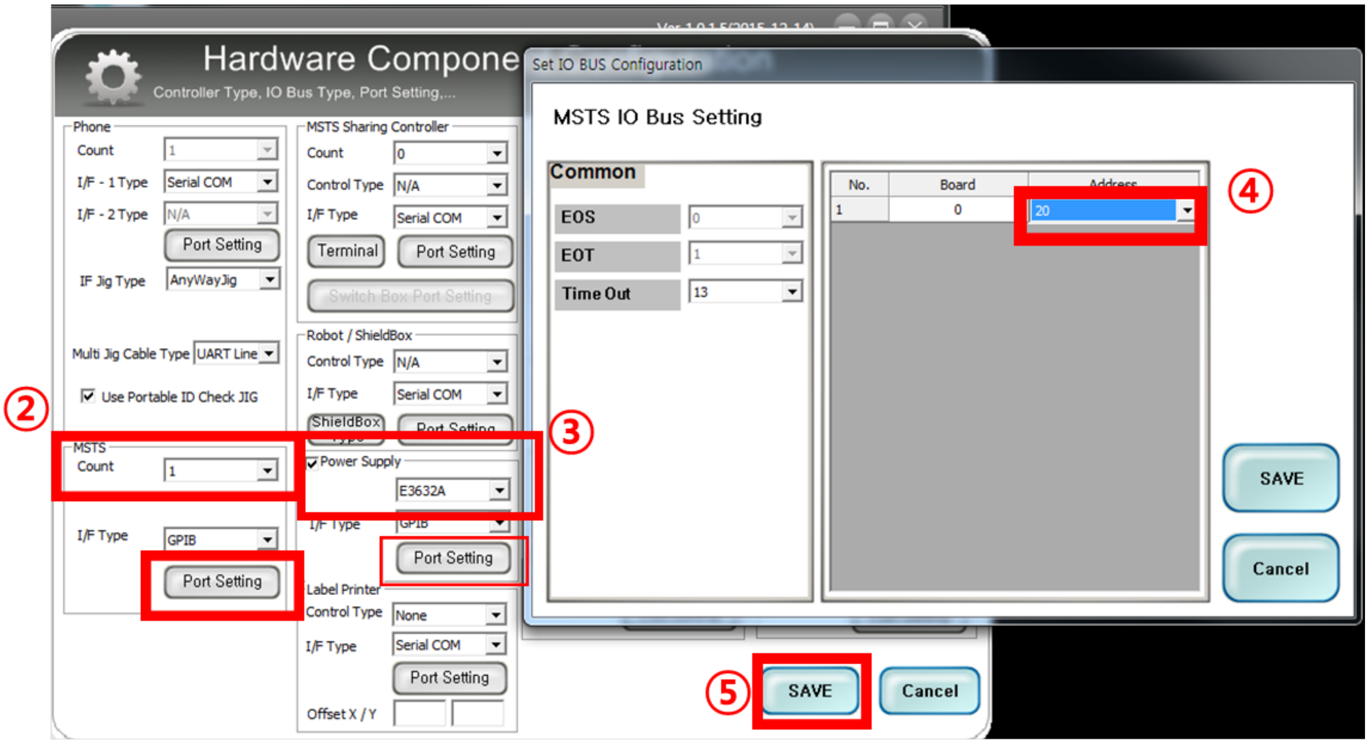


4. Change the Line Type to 'Block Cell' and disable 'Smart Cloud Cell'.

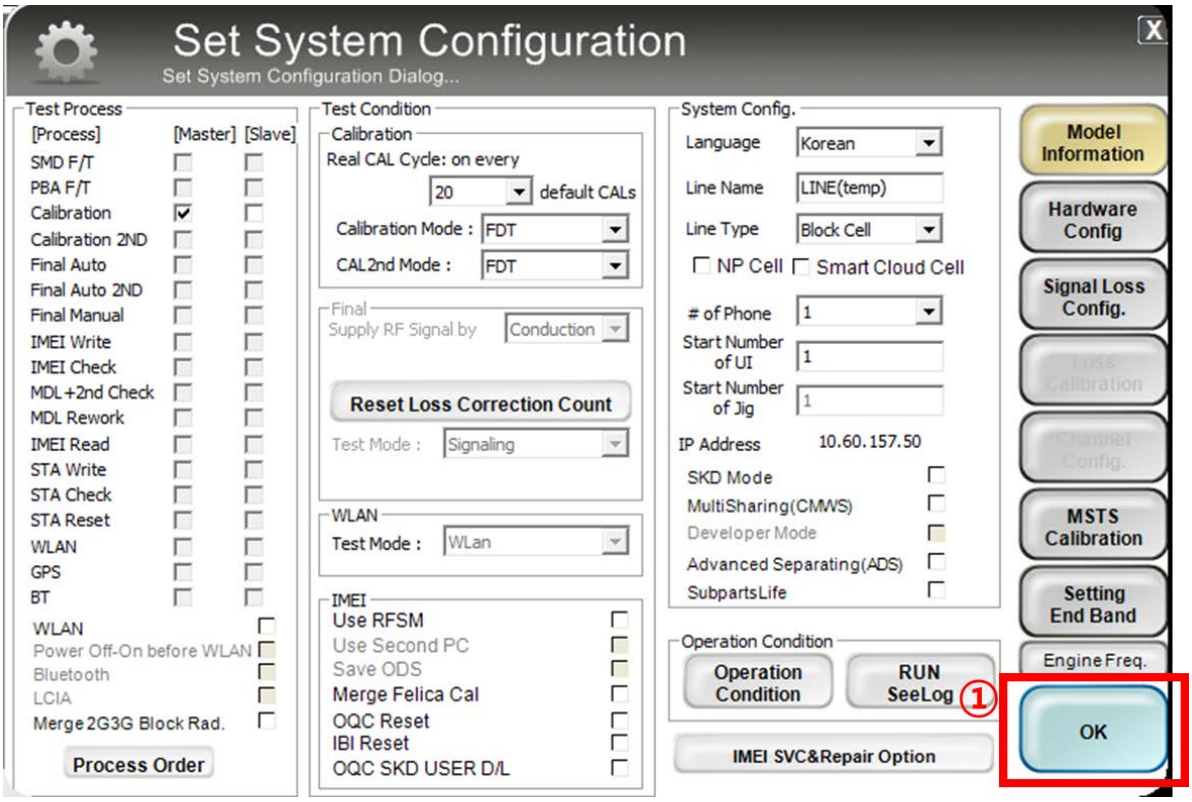


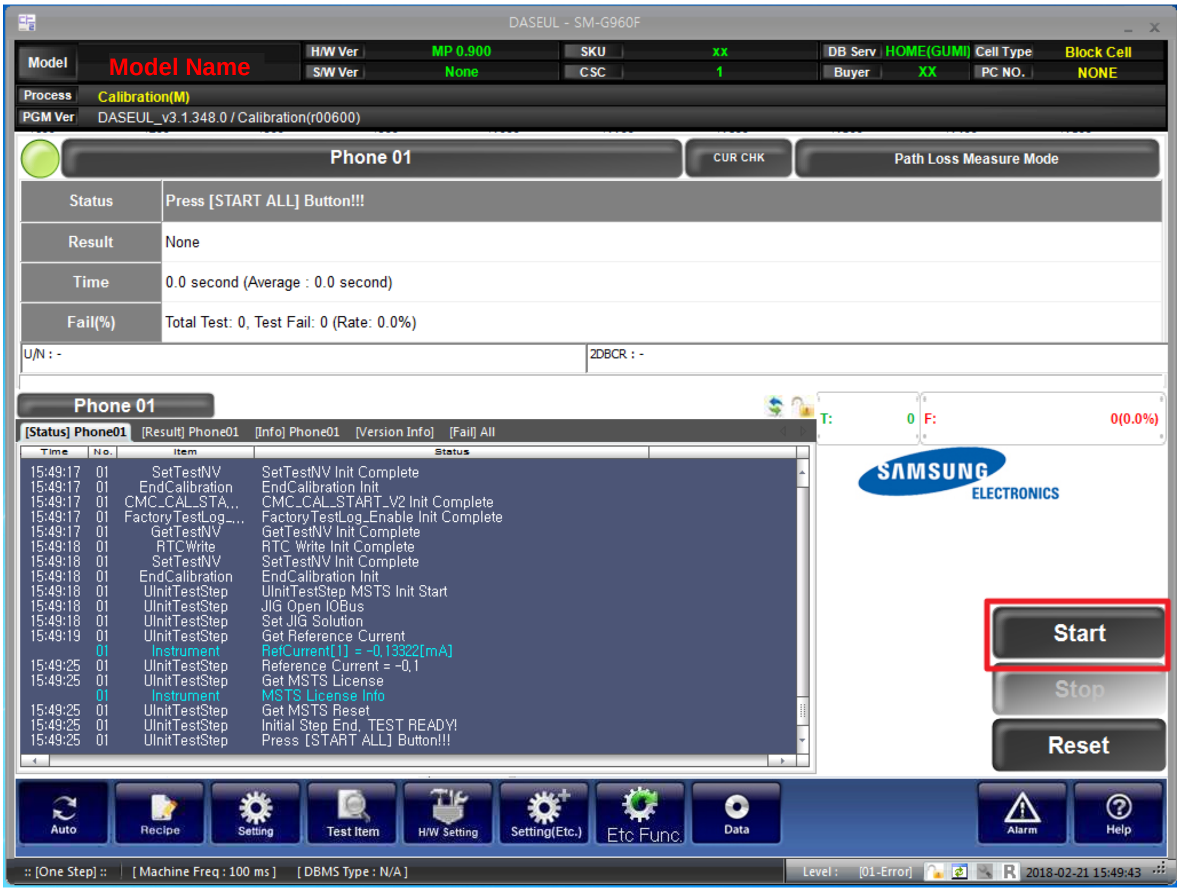
5. Set the GPIB address of MSTS(CMW500) and Power Supply(E3632A) to enter 'Hardware Config' and 'Save'. (Check the GPIB address of equipments in advance)





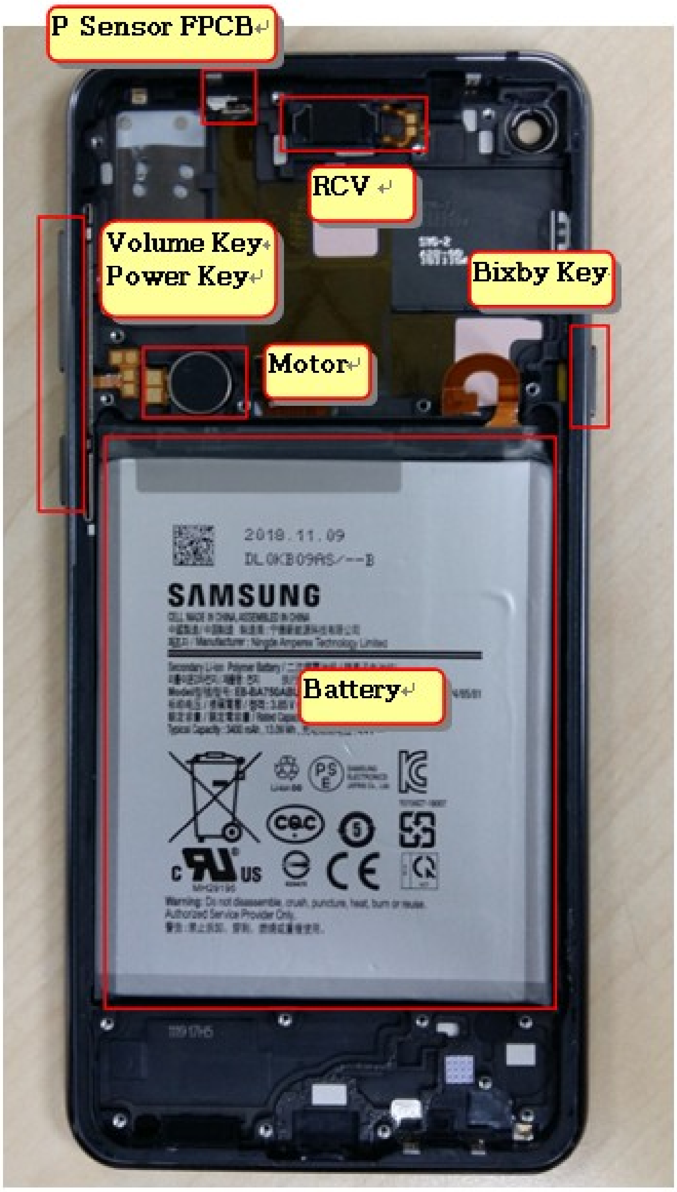
6. Press 'OK' to start RF Calibration after completing all settings.





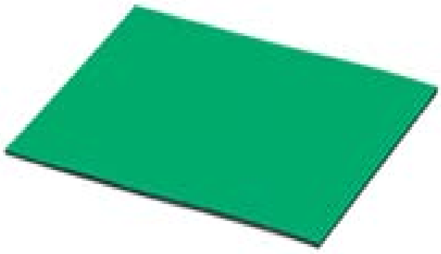
7. Level 2 Repair

7-1. Components on the Rear Case



7. Level 2 Repair

7-2. Pre-requisite

	
Tweezers / Disass'y Stick / Screw Driver	**Anti-static Gloves**
	
Anti-static Mat	**Hot Plate**
	
A OCTA Disassembly Holder	**OCTA Disassembly Upper**
	
Ethyl Alcohol	**Cotton Swab**

7. Level 2 Repair

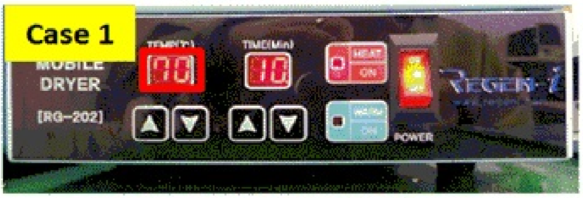
7-3. Parts which must be changed after repair

BOM description & part code	Image	Remarks
TAPE DOUBLE FACE BACK GLASS OUTER [GH02-16583A]		Replace for Back Glass Or Finger print sensor repair

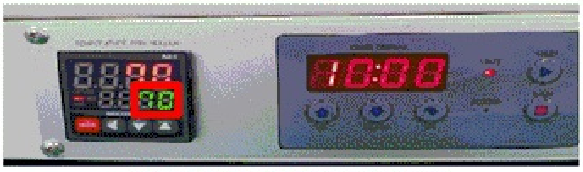
7-4. Disassembly

1

1) Put the device in the chamber as following below heating condition
 - SOC 68%↓: 70℃/10~20 minute
 - SOC 68%↑: 60℃/10~20 minute
※ Please confirm the heating condition released lastly, and follow it.



Put the device into Mobile Dryer.(70℃/10mins)

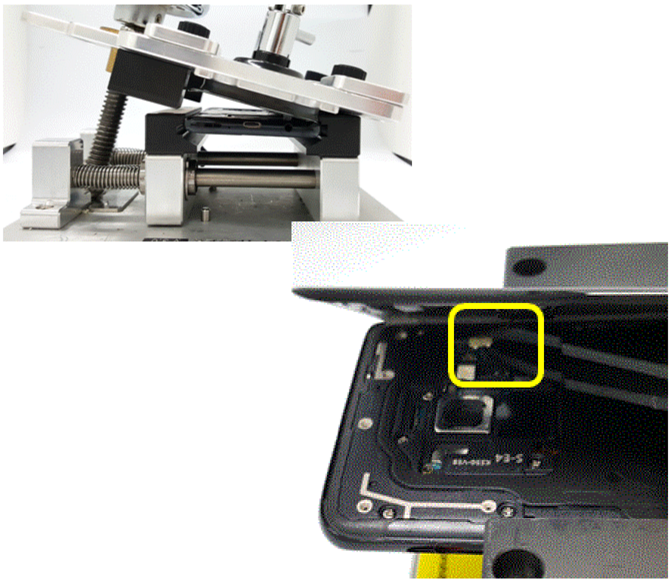


Put the device in to HOT PLATE.(70℃/10mins)

※ Caution
Be care of scratch

2

Detach the Back Glass. And Finger sensor
1) Detach the left side of Back glass
2) Detach the Finger sensor connector



※ Caution
Be care of scratch

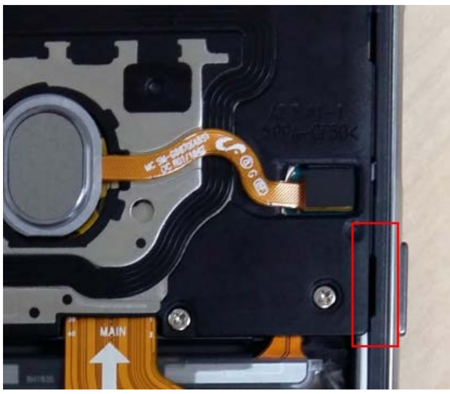
3

Unscrew 16 Point and disassemble SIM Tray from device

※ Caution
Be care of Rear damage

4

Disassemble Upper Rear.
(Use disassemble hole left/right side of Rear to detach it)



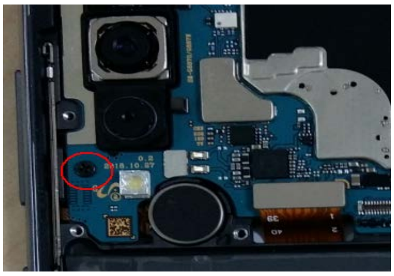
Rear Top disassemble hole.

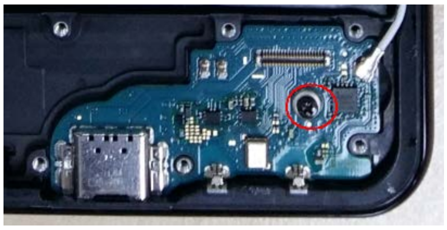
※ Caution
1) Be care of scratch
2) Be care of Rear and connector damage

5 Finally Disassemble Rear of bottom side.
(Use disassemble hole right side of Rear to detach it)

※ Caution
1) Be care of scratch
2) Be care of Rear and connector damage
3) Be careful not to damage the PBA

6 Unscrew PBA 3points.





※ Caution
Be careful not to damage the PBA

7 Disassemble 7 Point connector of PBA.

※ Caution
1) Be care of scratch
2) Be care of connector/cable damage

8 Disassemble Main PBA and Main FPCB.

※ Caution
Be care of Main PBA damage

9

Detach Ear Jack
1) Unscrew PBA 1points.
2) Separation of Ear jack Module using pinset.

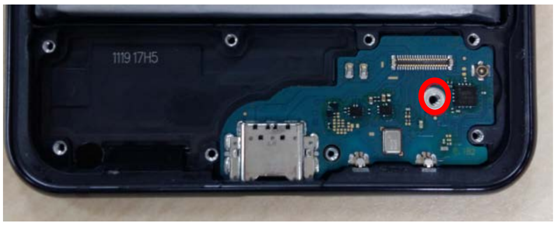
※ Caution
Be care of SUB PBA damage

7-5. Assembly

1 Attach SUB PBA and Ear jack on Front.

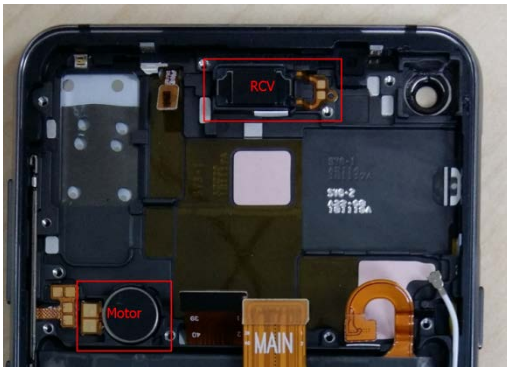
※ Caution
Be care of coaxial cable damage

2 Screw 1 point in SUB PBA.



※ Caution
Be care of chip damage nearby screw point.

3 Attach components on Front.
(RCV module, Motor)



※ Caution
Be care of FPCB damage and tilt

4 Attach coaxial cable (SUB PBA Sida).

※ Caution
Be care of Push SUB PBA connector.

7. Level 2 Repair

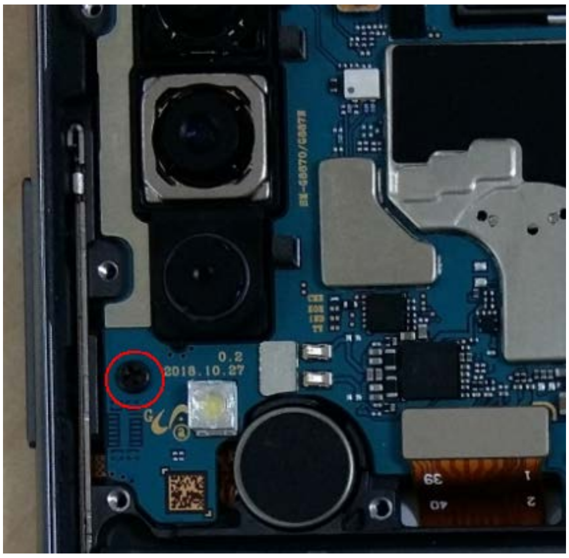
5 Attach Main PBA On Front.

※ Caution
Be care of PBA and Connector damage

6 Attach coaxial cable, Battery connect and the other connectors.
Assemble Main FPCB and Front Camera.

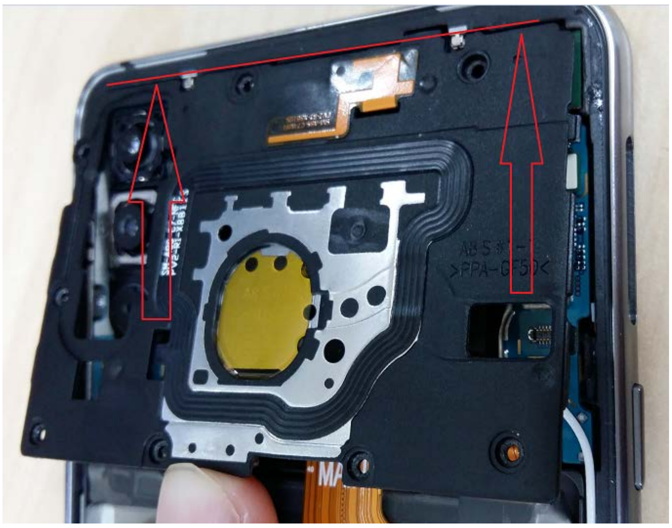
※ Caution
Be care of FPCB damage

7 Screw 1 point in Main PBA.
* Torque Value : 1.2±0.1kgfcm



※ Caution
Be care of PBA damage

8 Assemble the TOP REAR Assy in sequence.



※ Caution
Be care of scratch and REAR damage

9

Assemble the Bottom REAR Assy in sequence.

※ Caution
Be care of scratch and Bottom REAR damage

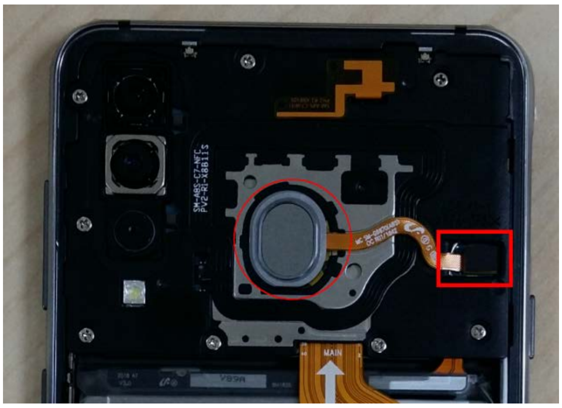
10

Attach 16EA Screws on the Rear.
* Torque Value : 1.2±0.1kgfcm

※ Caution
Be care of scratch and tilt.

11

Attach finger sensor on the TOP Rear.

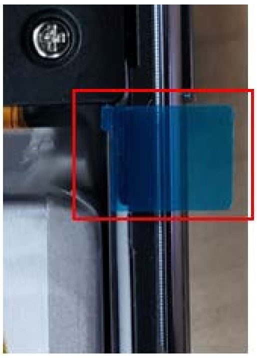
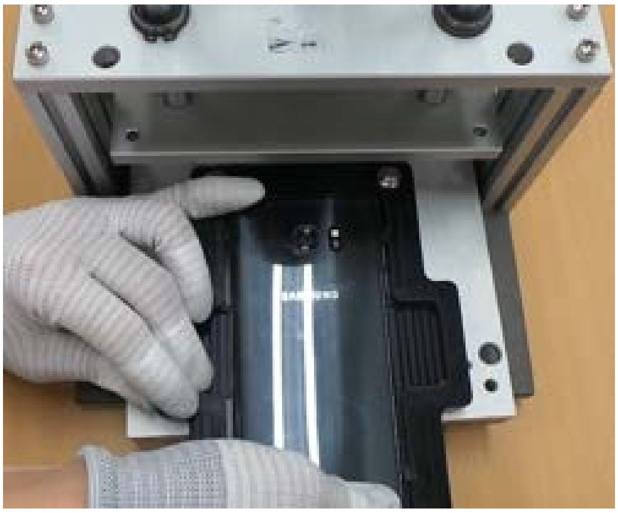


※ Caution
Be care of scratch and tilt.

12

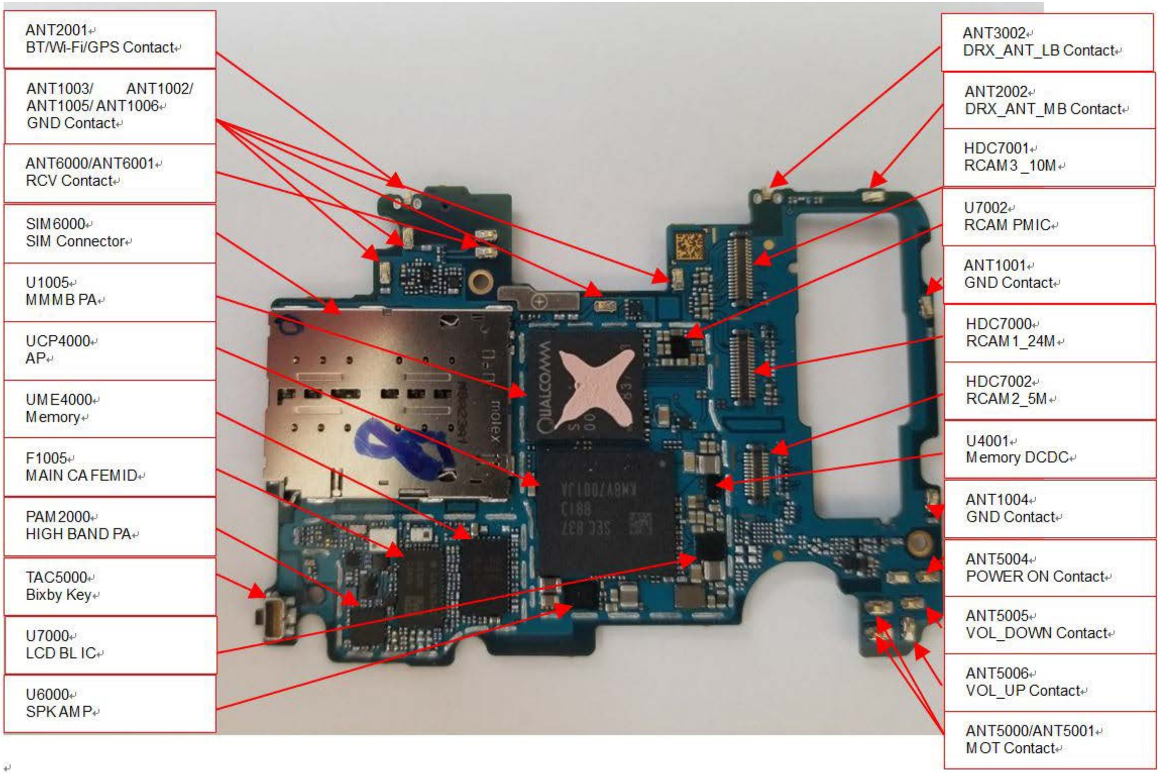
Attach the back glass outside tape on the rear case.

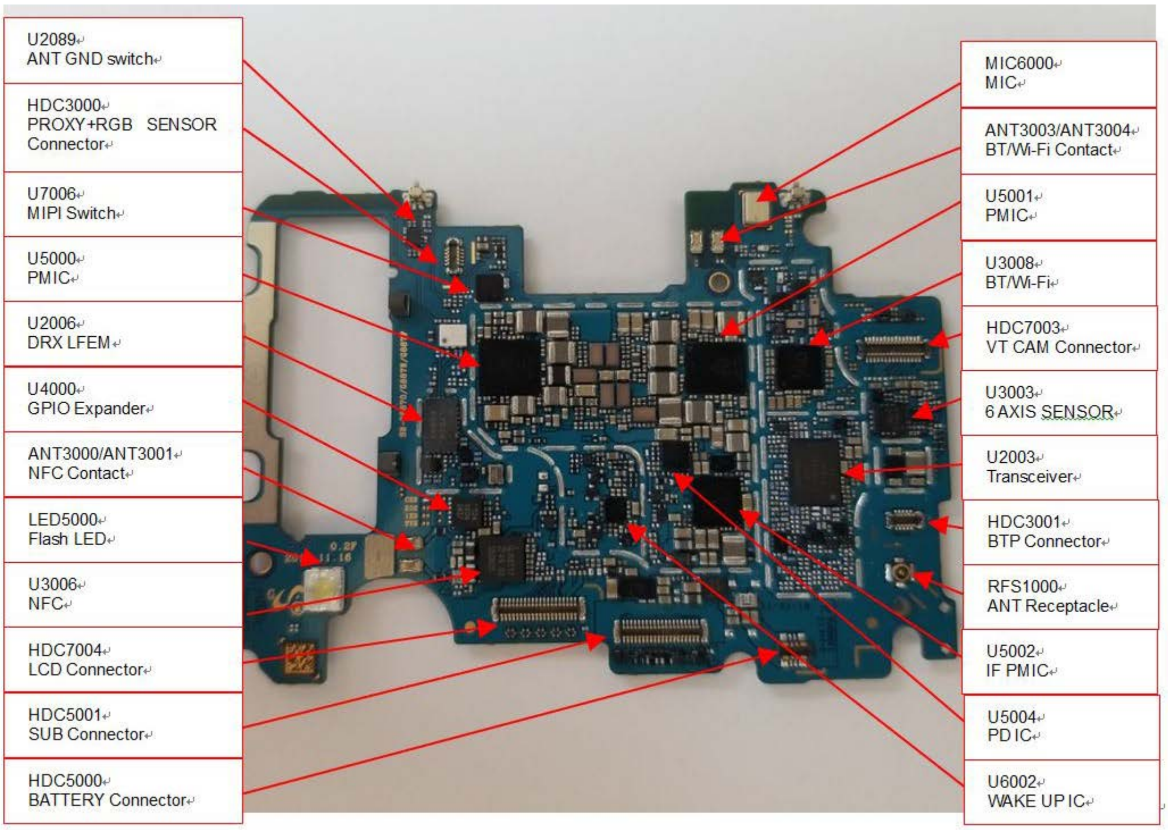
※ Caution
Be care of scratch and tilt.

13 Attach the finger tape on the back glass.	14 Attach tape sponge-cable line.
	 
※ Caution Be care of scratch and tilt.	※ Caution Be care of scratch and tilt.
14 Attach the back glass on the Top REAR side.	15 Press Back Glass. - Pressing force : 1 N - Pressing time : 1 minute SIM Tray must be seperated before pressing the backglass or the OCTA
	
※ Caution Be care of scratch.	※ Caution Be care of scratch.

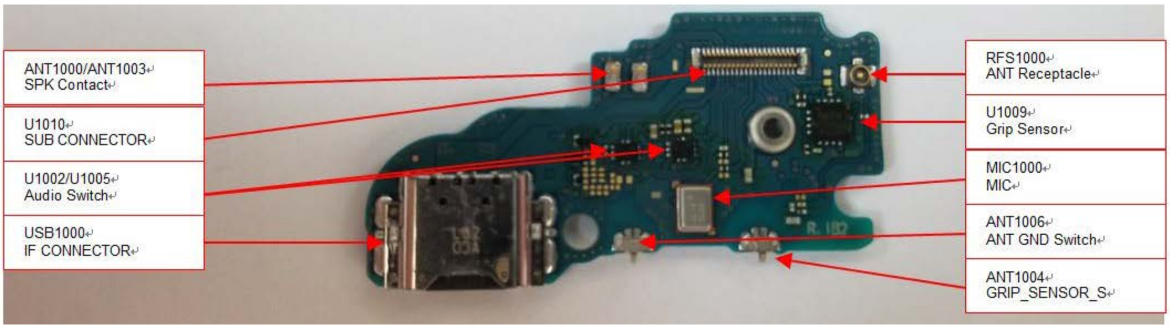
8. Level 3 Repair

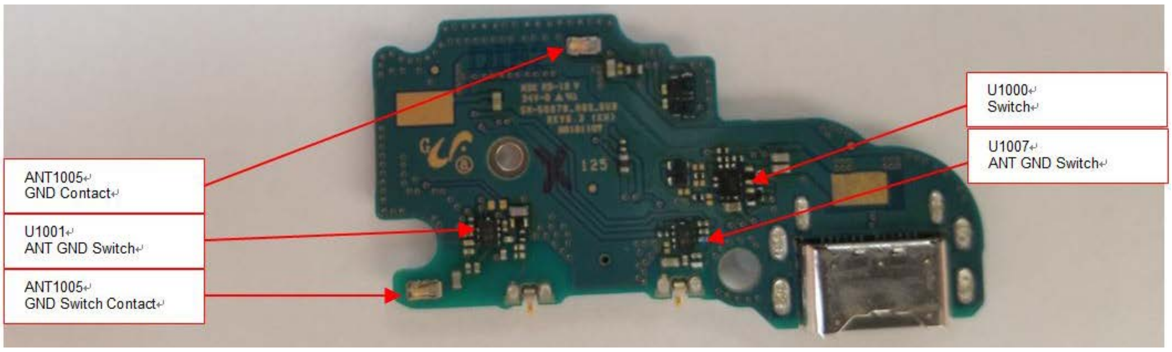
8-1. Components Layout





8. Level 3 Repair

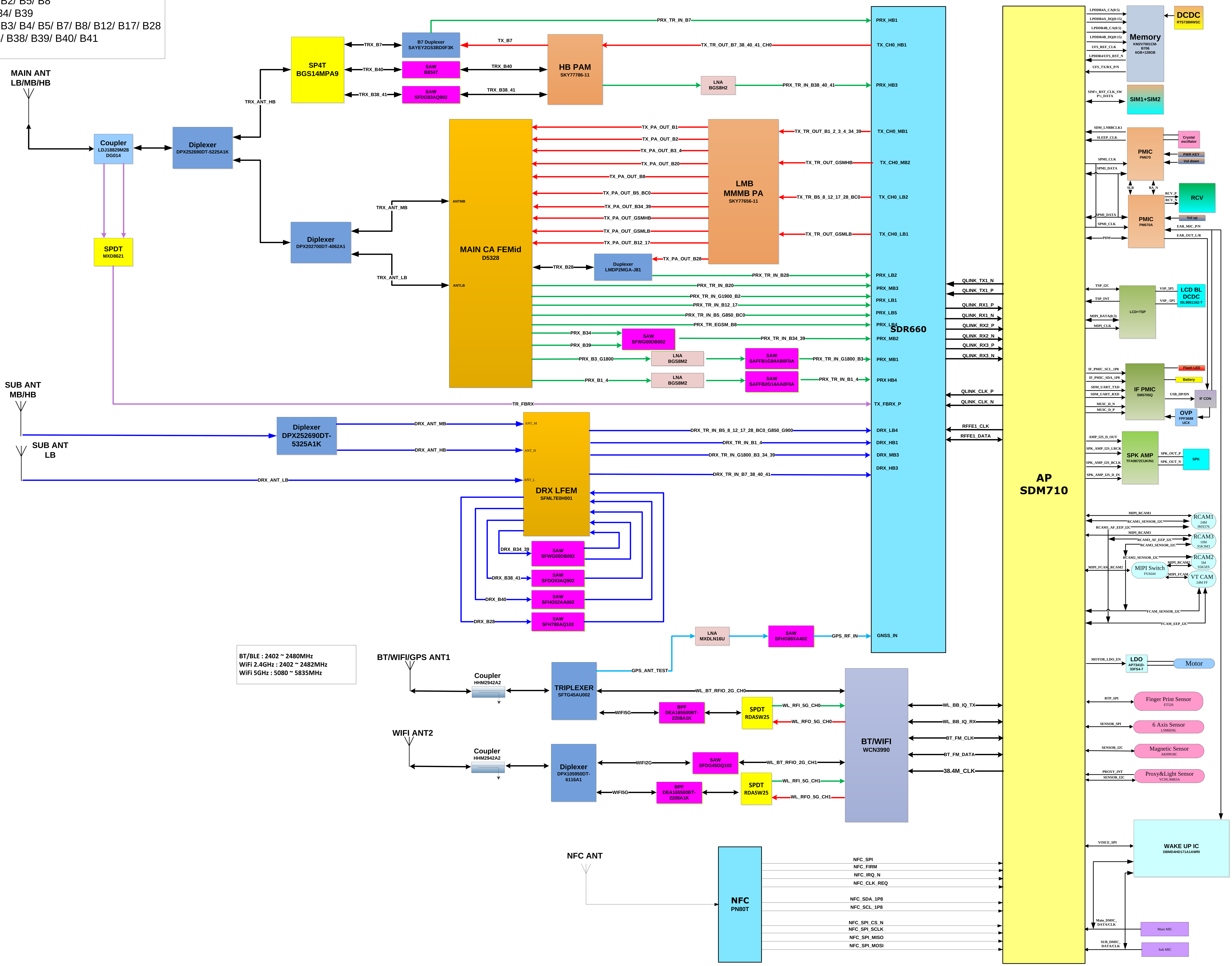




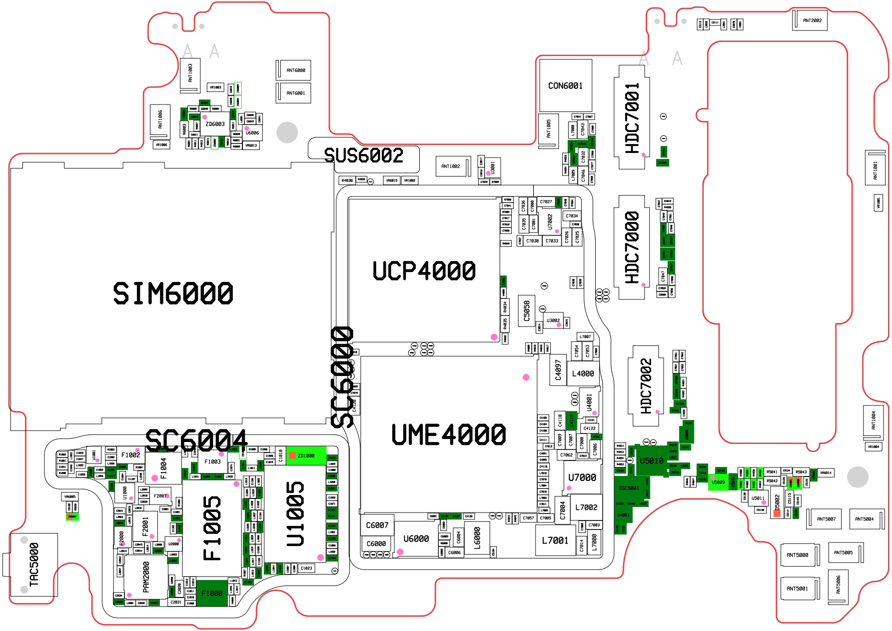
SM-G8870 SVC Manual Block Diagram

RF Band Information

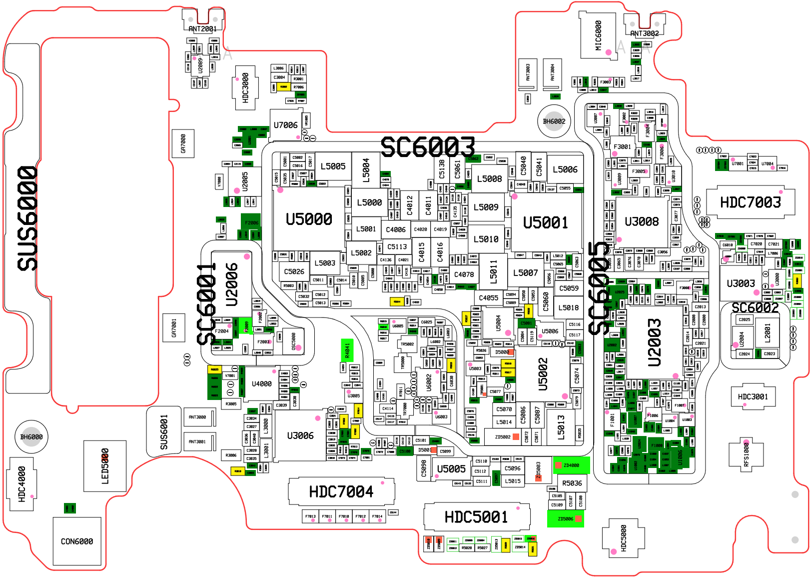
GSM: 850/ 900/ 1800/ 1900
CDMA: BC0
WCDMA: B1/ B2/ B5/ B8
TDSCDMA: B34/ B39
LTE FDD: B1/ B3/ B4/ B5/ B7/ B8/ B12/ B17/ B28
LTE TDD: B34/ B38/ B39/ B40/ B41



BT/BLE : 2402 ~ 2480MHz
WiFi 2.4GHz : 2402 ~ 2482MHz
WiFi 5GHz : 5080 ~ 5835MHz



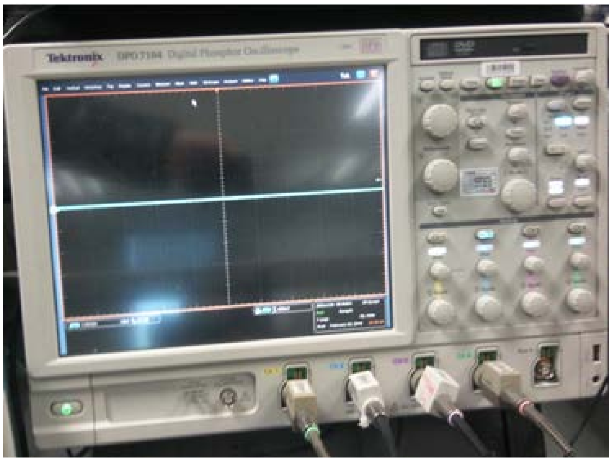
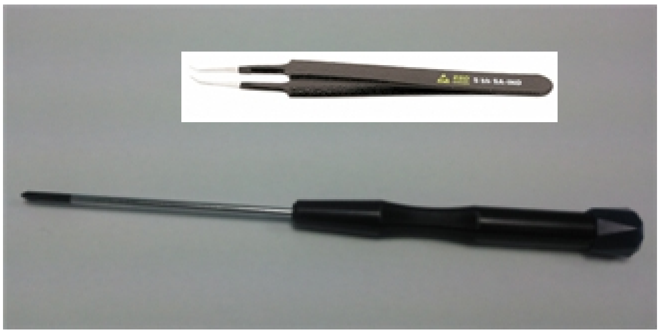
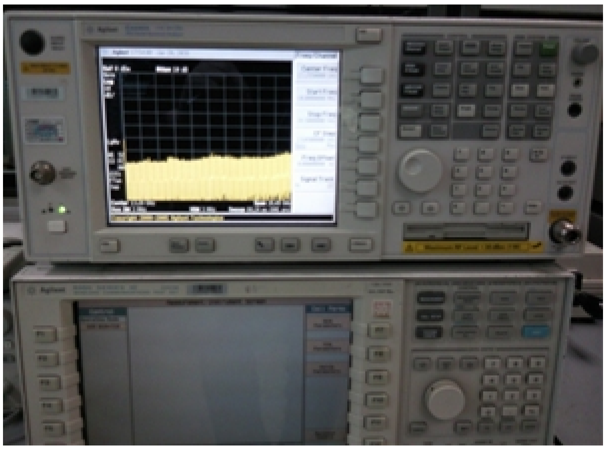
SIM6000
SUS6002
UCP4000
UME4000
SC6000
SC6004
CON6001
HDC7001
HDC7000
HDC7002
F1005
U1005
F1000
F1004
F1002
F1003
F2001
F2007
PAM2000
TAC5000
ZD6003
U6006
ZD1000
U6000
C6007
C6000
L6000
L7001
L7002
U7000
L4000
C4097
U4001
U5010
OSC5001
U5009
U5011
U5002
ANT6000
ANT6001
ANT1003
ANT1006
ANT1005
ANT1002
ANT1001
ANT1004
ANT2002
ANT5000
ANT5001
ANT5004
ANT5005
ANT5006
ANT5007
C5058
U3001
U3002
R5041
R5042
R5043



SUS6000
ANT2001
HDC3000
U7006
GA7000
U2005
SC6003
L5005
L5004
U5000
L5000
L5001
C4006
C4012
C4011
L5008
L5009
U5001
L5010
L5006
BH6002
MIC6000
ANT3002
ANT3003
ANT3004
F3001
U3008
HDC7003
SC6001
U2006
L5003
L5002
C5026
C5113
C4015
C4016
C4078
L5011
L5007
C5059
C4055
C5060
L5018
SC6005
U2003
U3003
SC6002
L2001
GA7001
OSC5000
U4000
R4041
U5002
C5074
C5070
L5014
C5086
C5087
L5013
U3006
L3000
L3001
SUS6001
ANT3000
ANT3001
BH6000
HDC4000
LED5000
CON6000
HDC7004
U5005
C5096
L5015
R5036
ZD4000
ZD5006
HDC5001
HDC5000
HDC3001
RFS1000

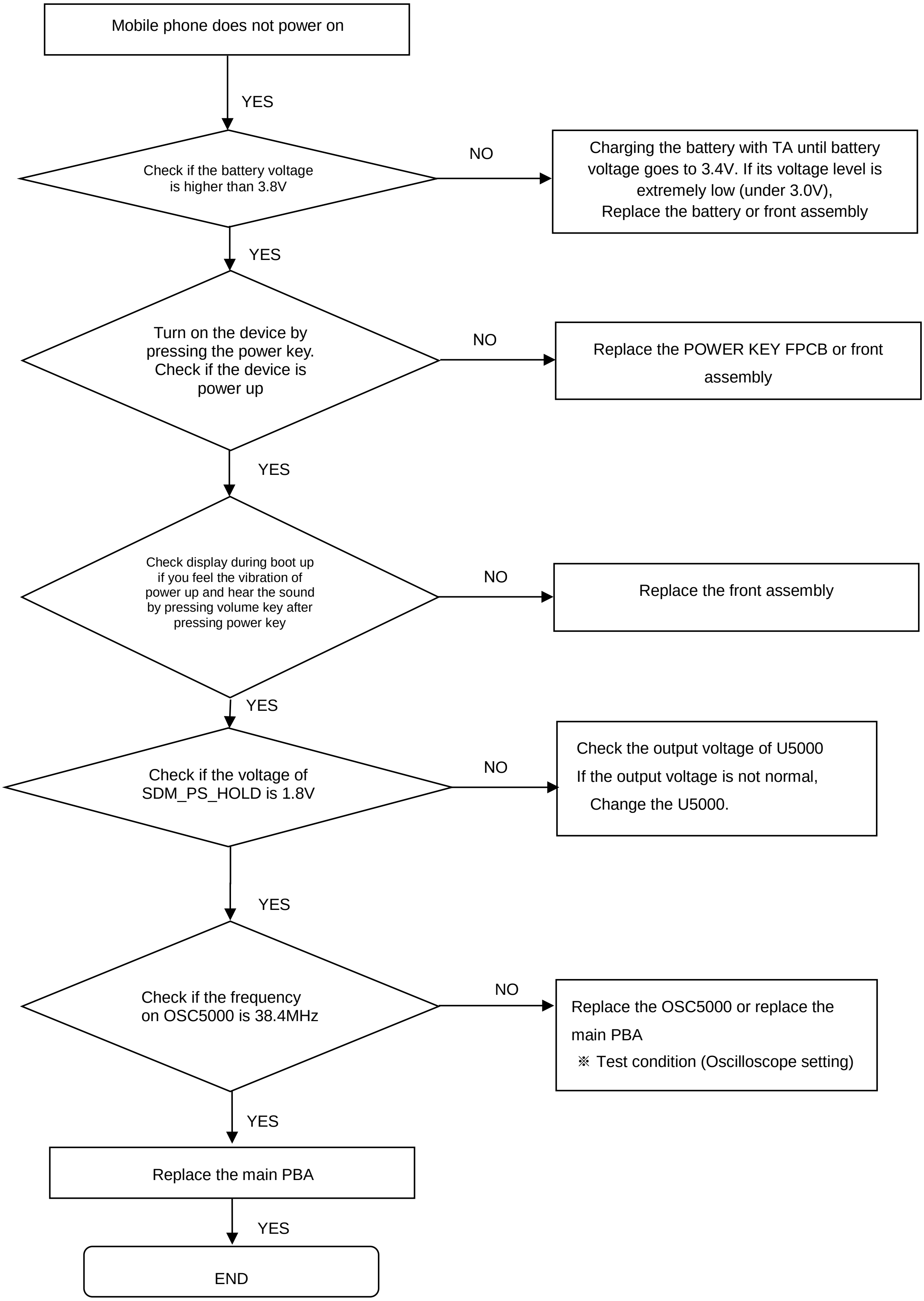
8. Level 3 Repair

8-2. Flow chart of Troubleshooting.

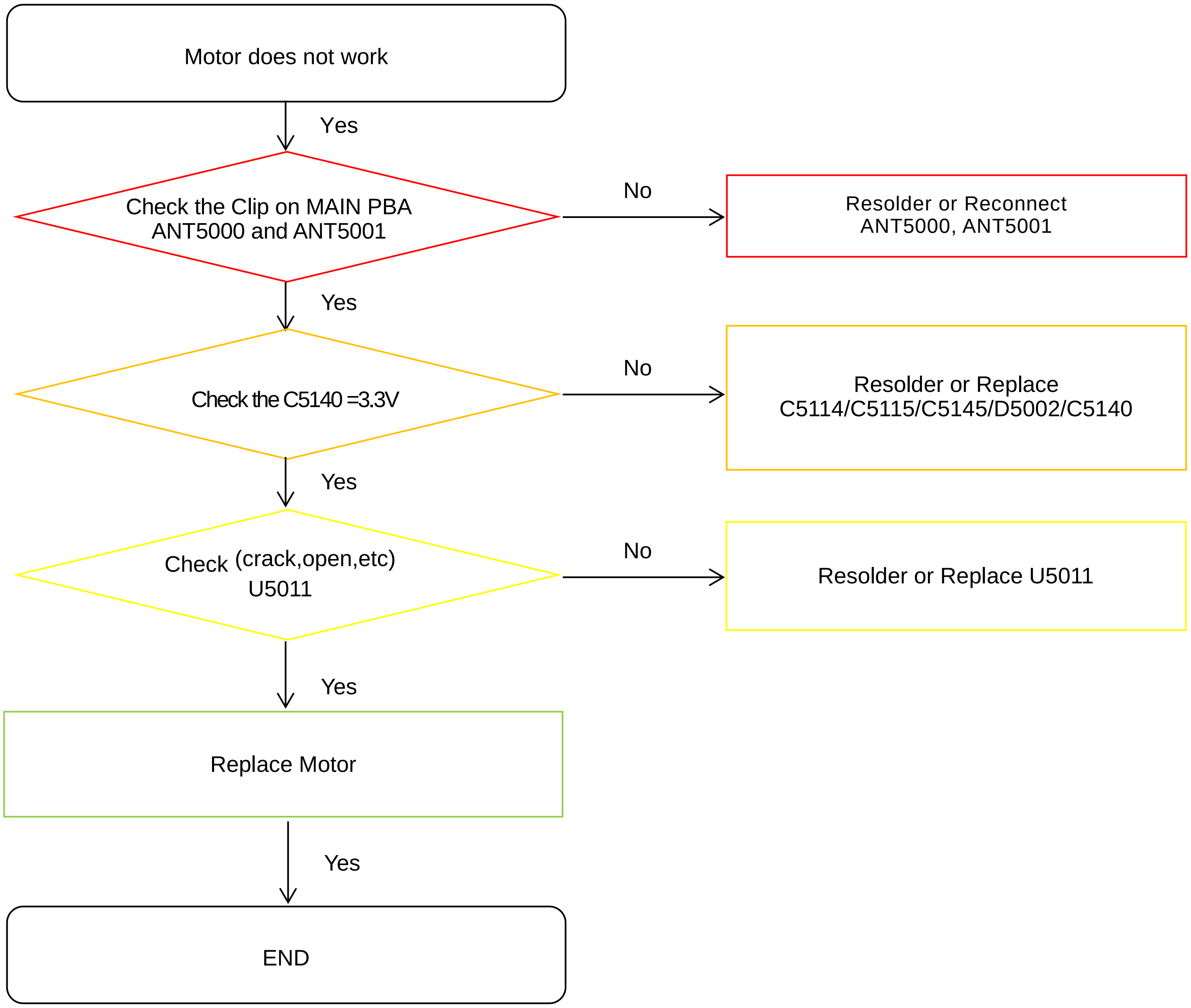
	
Oscilloscope	**Digital Multimeter**
	
Power Supply	**+ driver, ESD Safe Tweezer**
	
8960 & Spectrum Analyzer	**Soldering iron**

8. Level 3 Repair

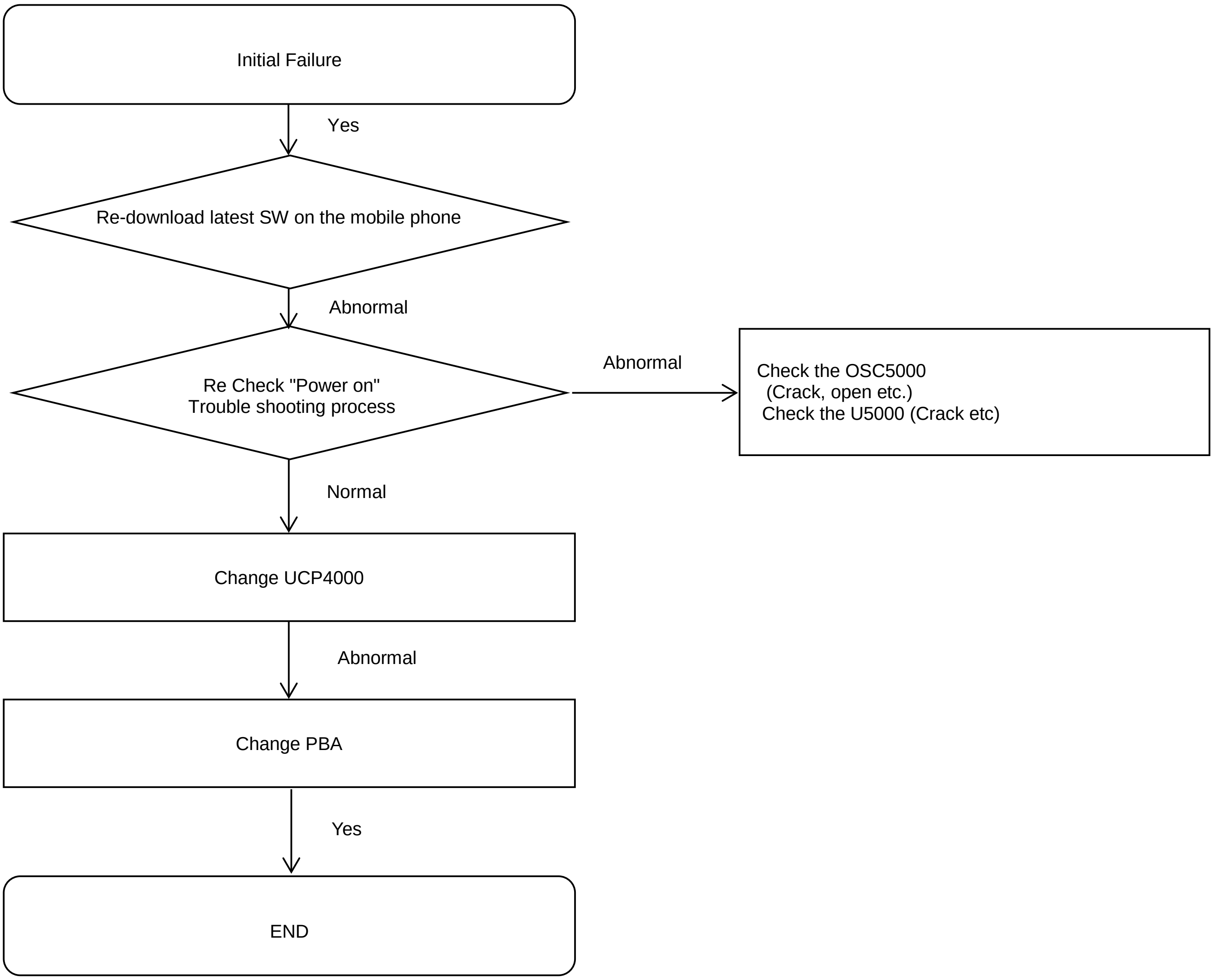
8-3-1. Power on



8-3-2. Motor

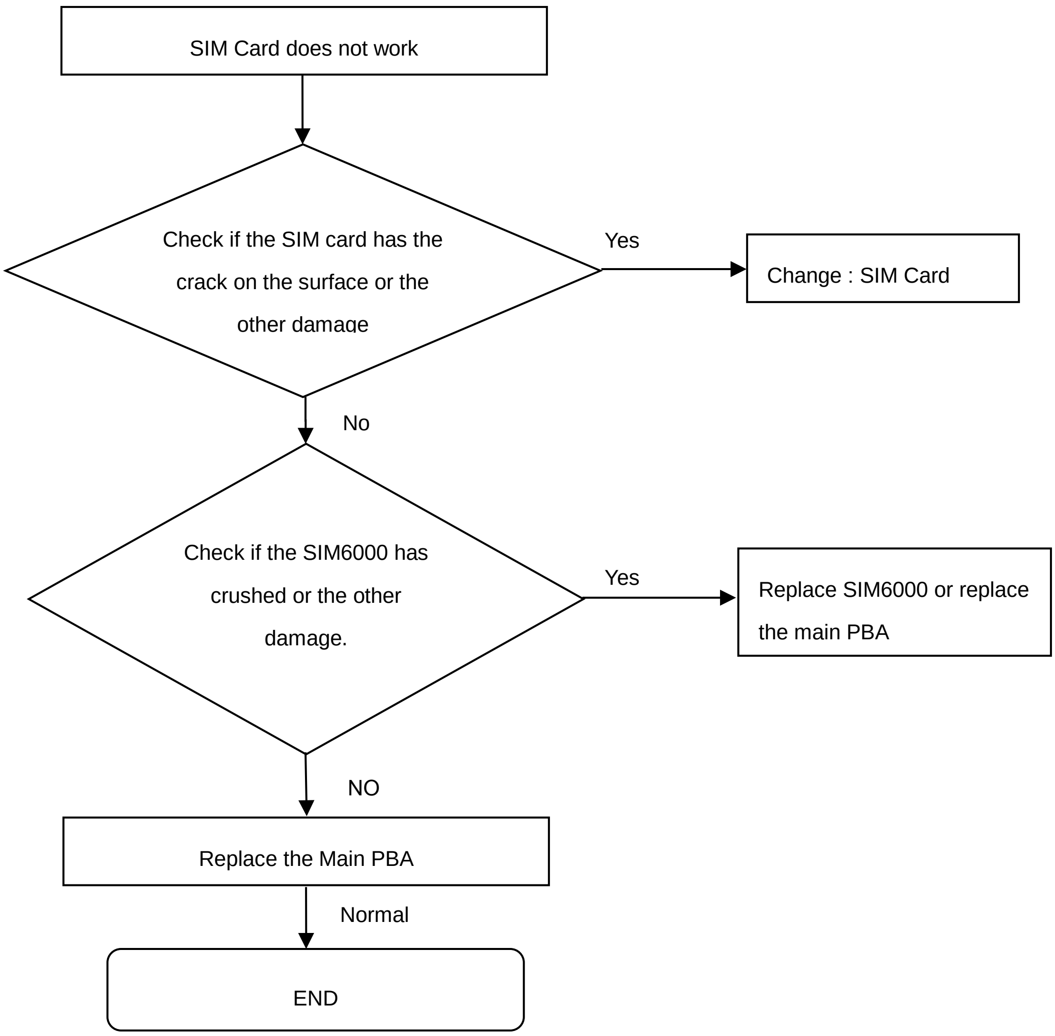

Motor does not work
Yes
Check the Clip on MAIN PBA
ANT5000 and ANT5001
No
Resolder or Reconnect
ANT5000, ANT5001
Yes
Check the C5140 =3.3V
No
Resolder or Replace
C5114/C5115/C5145/D5002/C5140
Yes
Check (crack,open,etc)
U5011
No
Resolder or Replace U5011
Yes
Replace Motor
Yes
END

8-3-3. initial

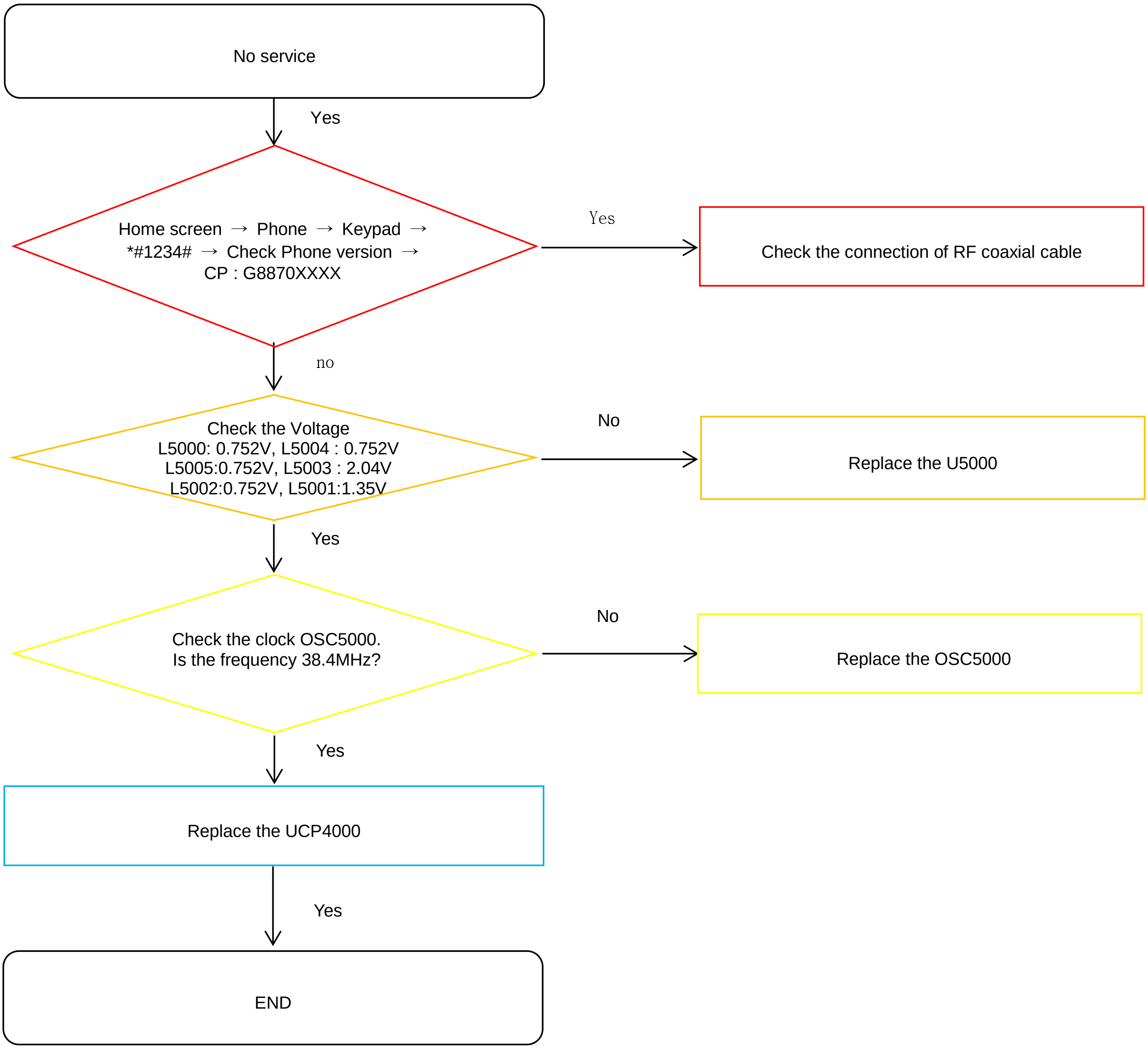


Initial Failure
Yes
Re-download latest SW on the mobile phone
Abnormal
Re Check "Power on"
Trouble shooting process
Abnormal
Check the OSC5000
(Crack, open etc.)
Check the U5000 (Crack etc)
Normal
Change UCP4000
Abnormal
Change PBA
Yes
END

8-3-4. SIM PART

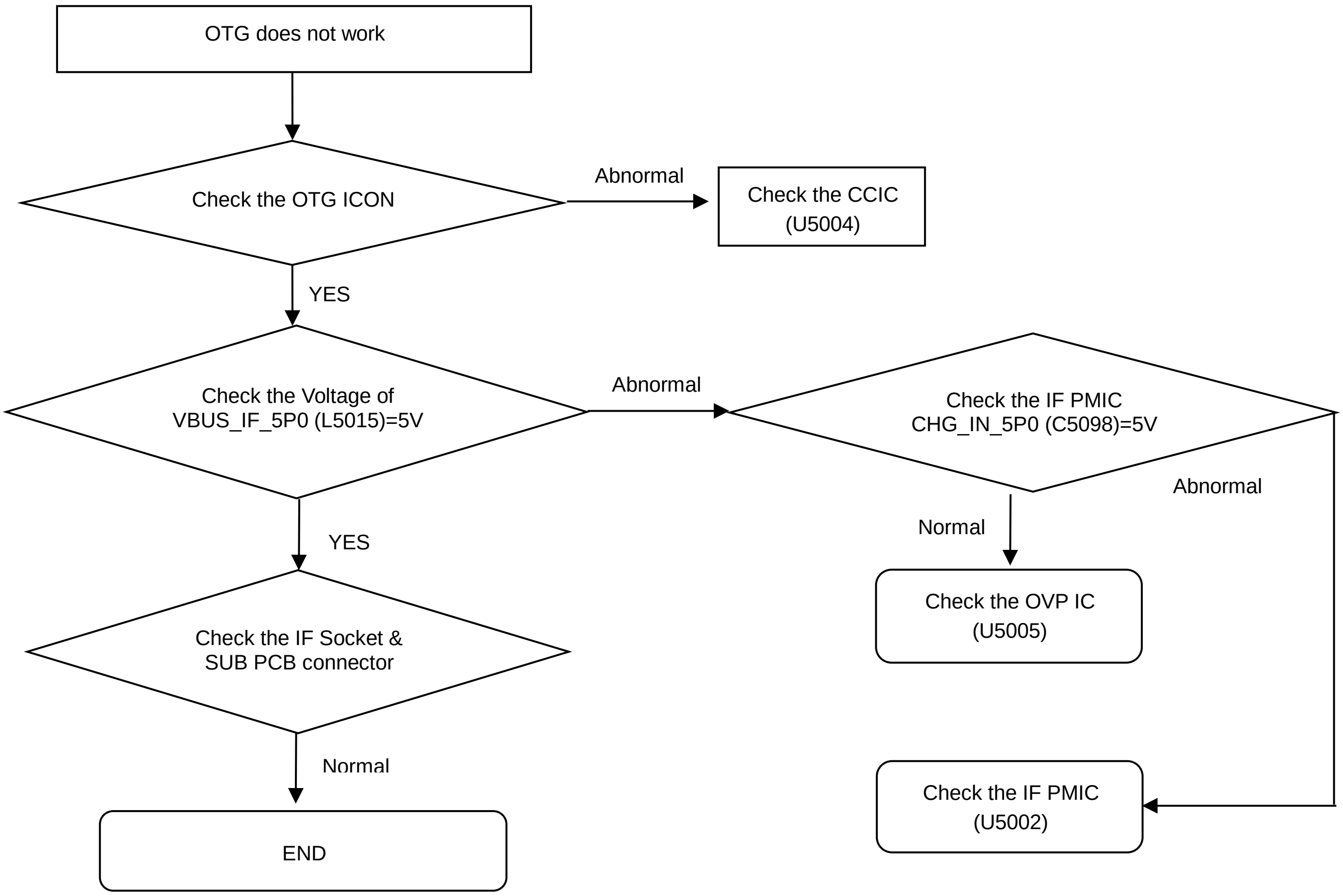

SIM Card does not work
Check if the SIM card has the
crack on the surface or the
other damage
Yes
Change : SIM Card
No
Check if the SIM6000 has
crushed or the other
damage.
Yes
Replace SIM6000 or replace
the main PBA
NO
Replace the Main PBA
Normal
END

8-3-5. No Service

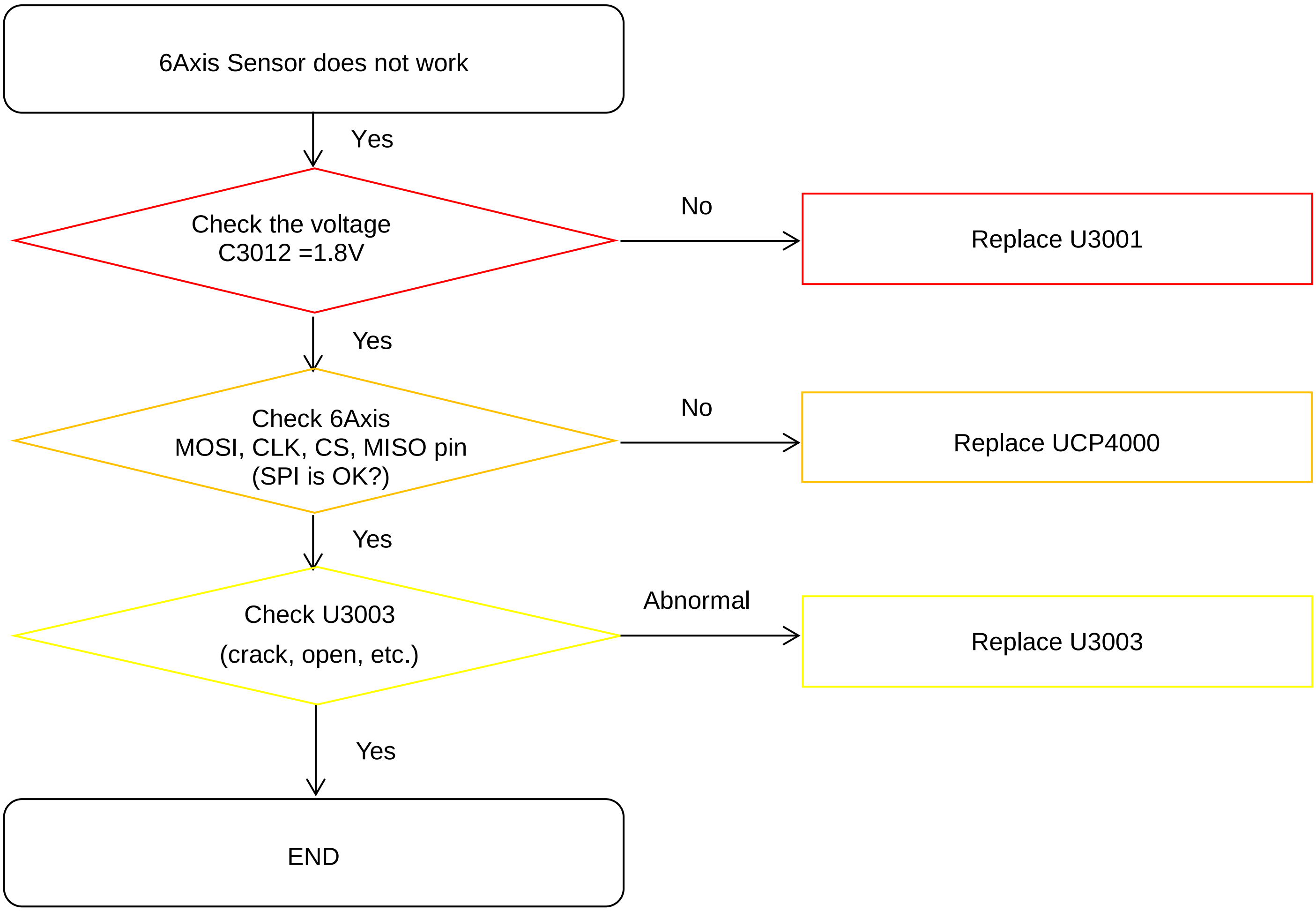


8. Level 3 Repair

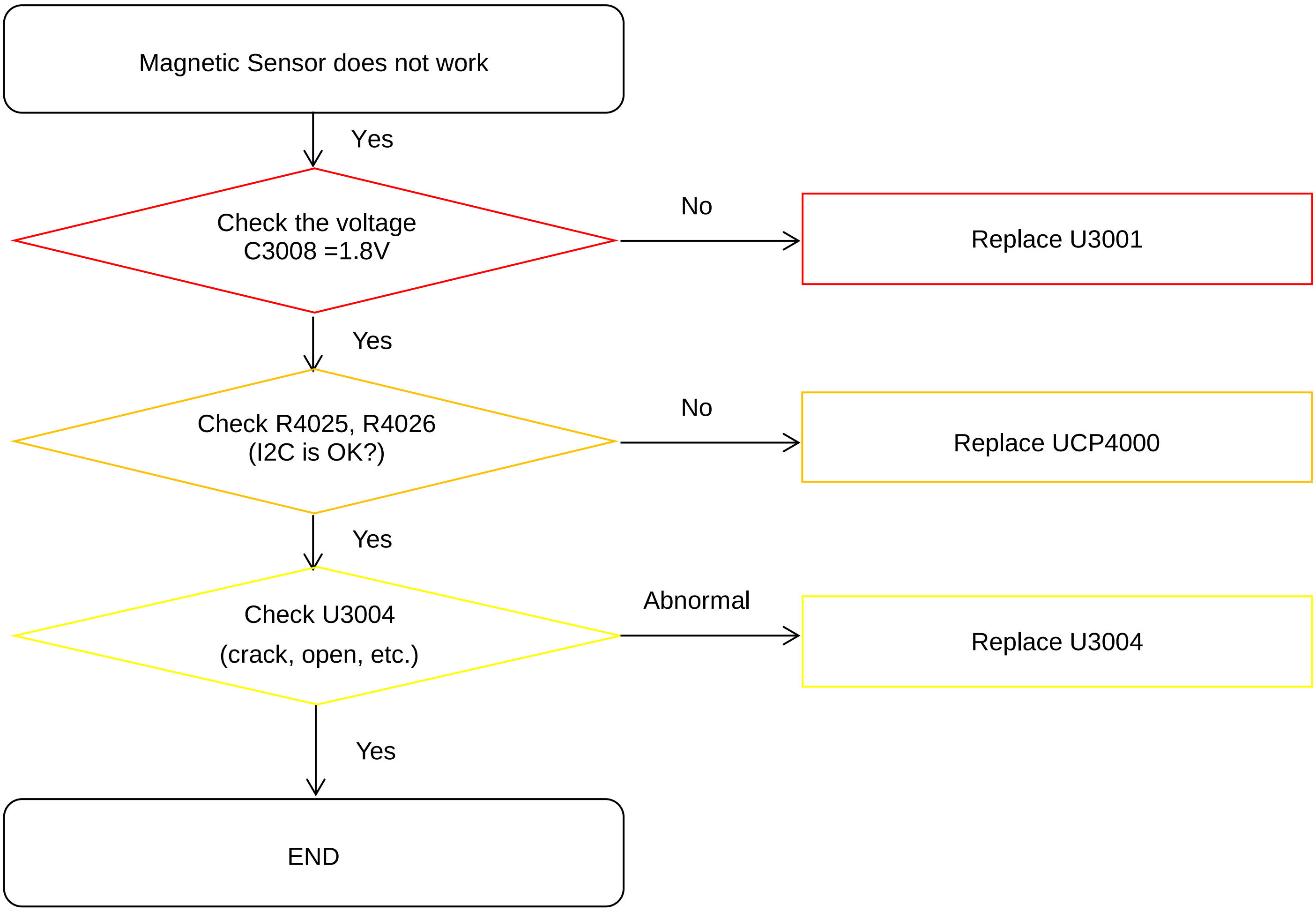
8-3-6. OTG



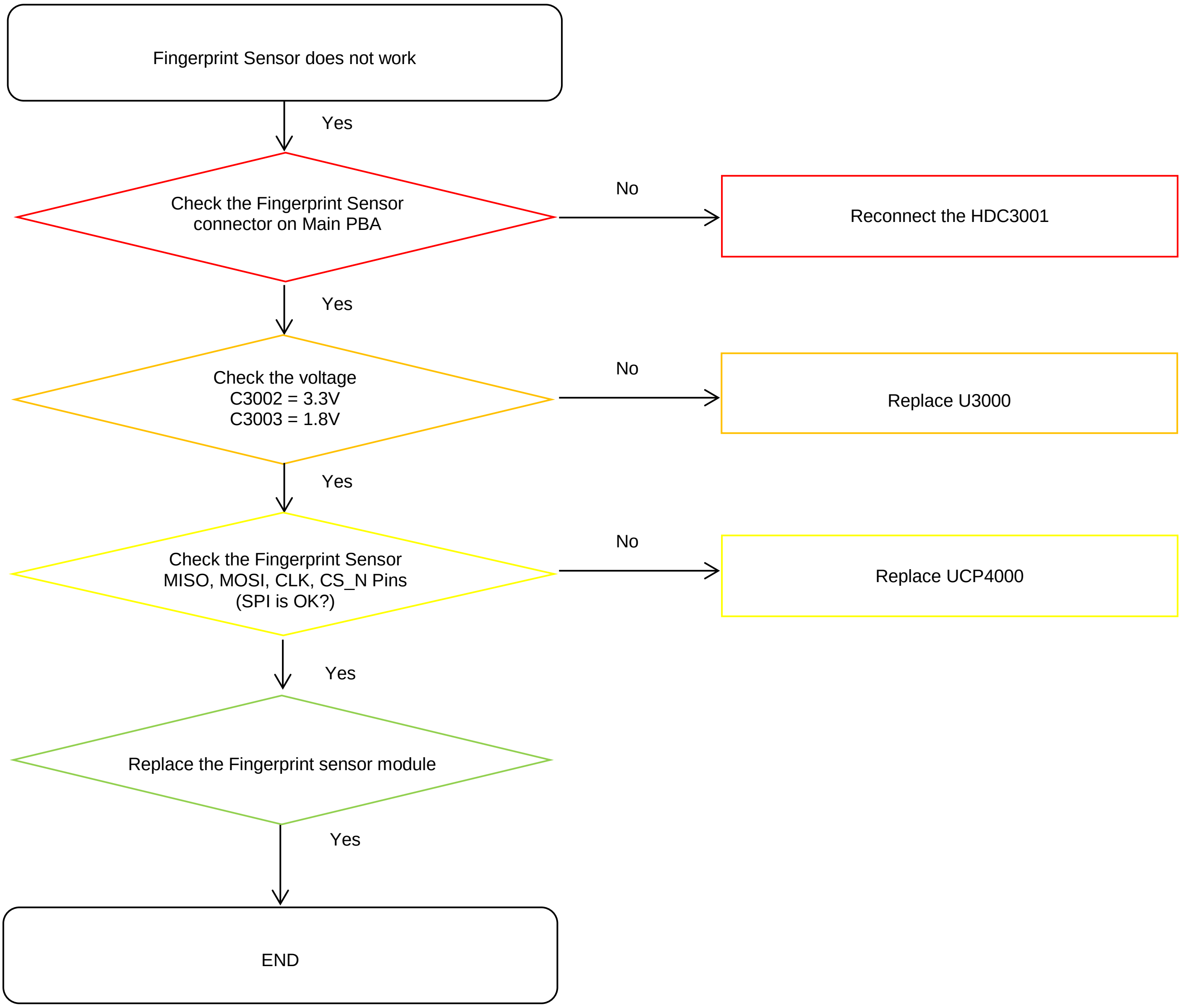
8-3-7. 6Axis sensor



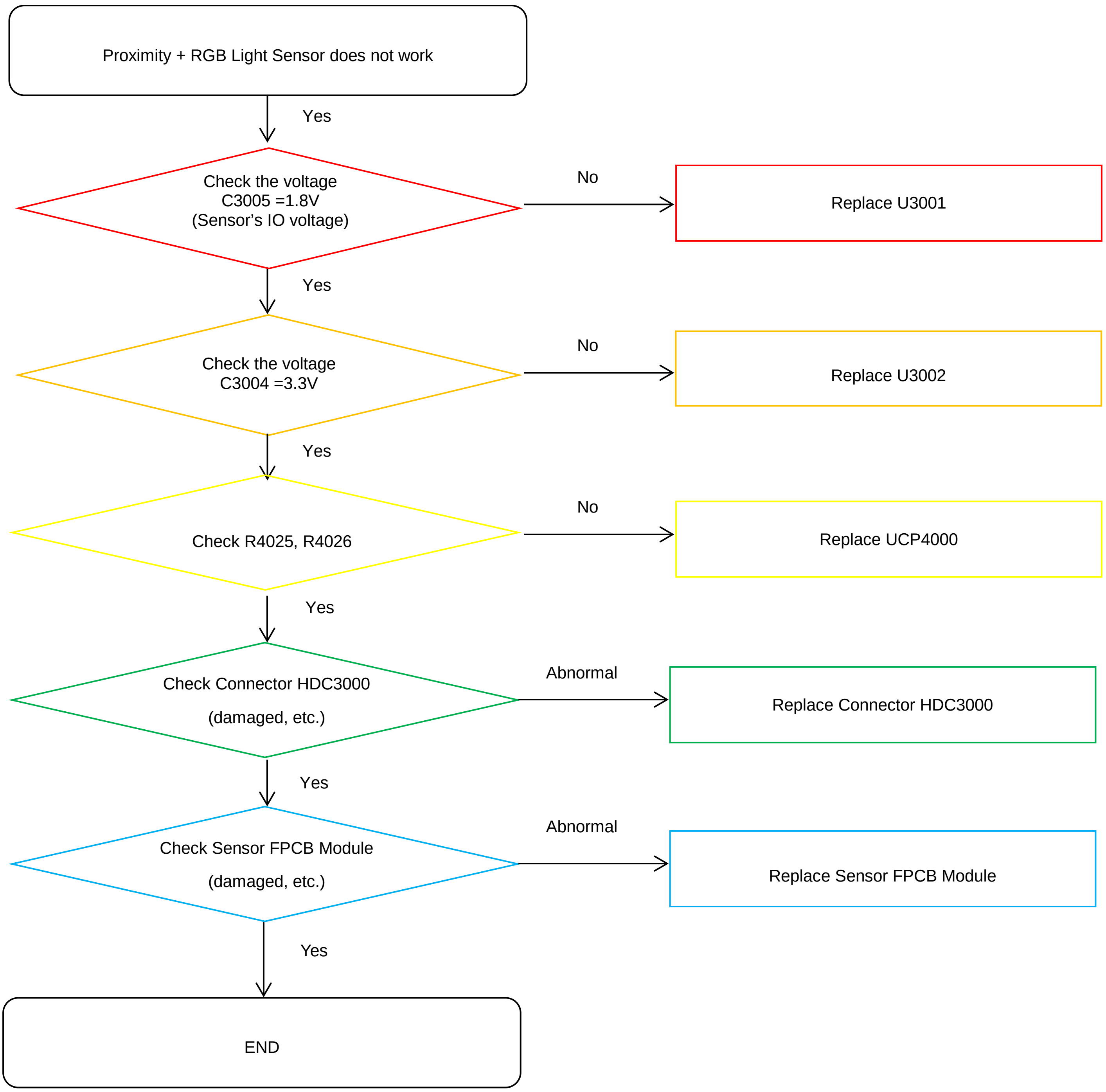
8-3-8. Magnetic sensor



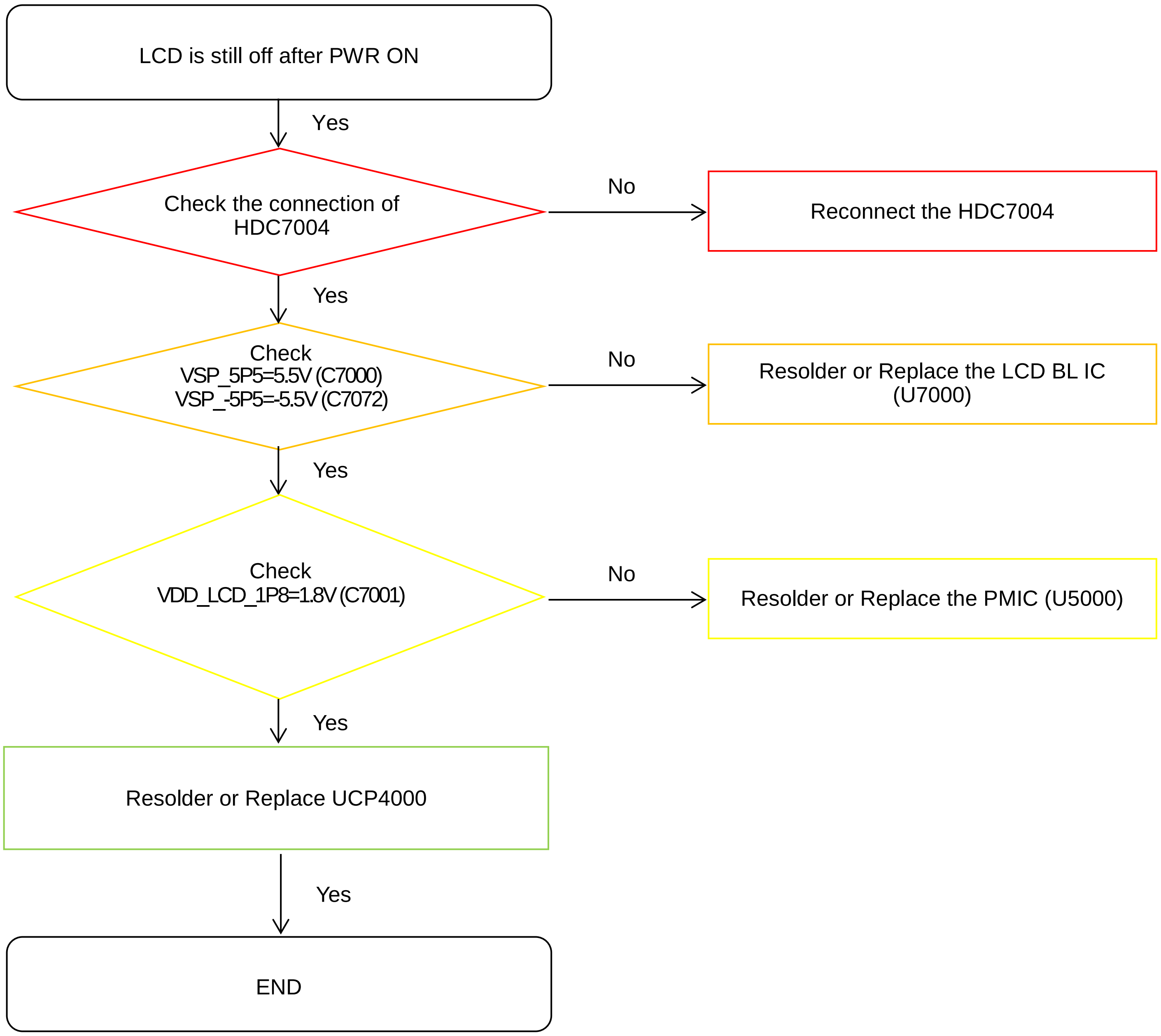
8-3-9. Fingerprint Sensor



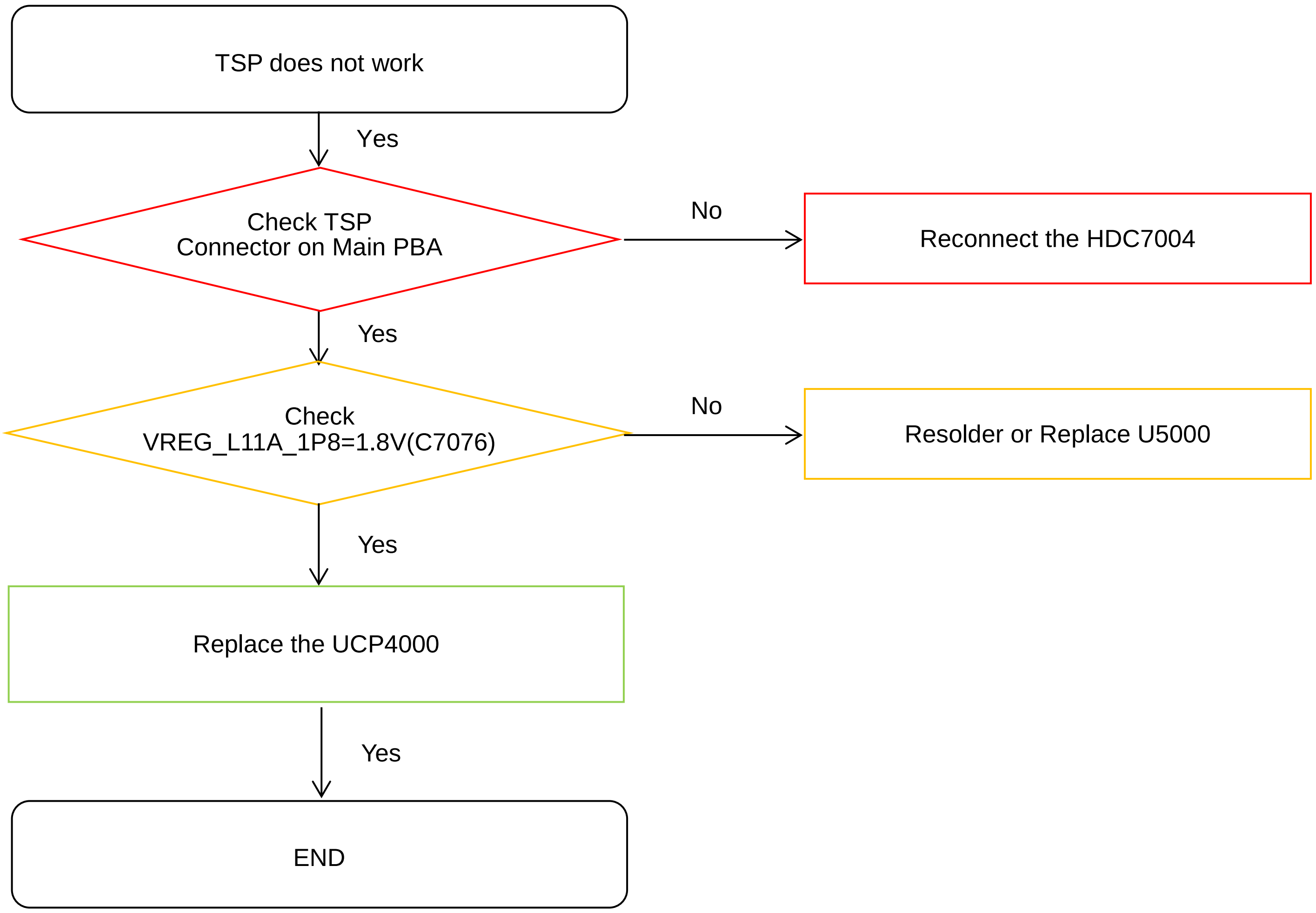
8-3-10. Proximity + RGB sensor



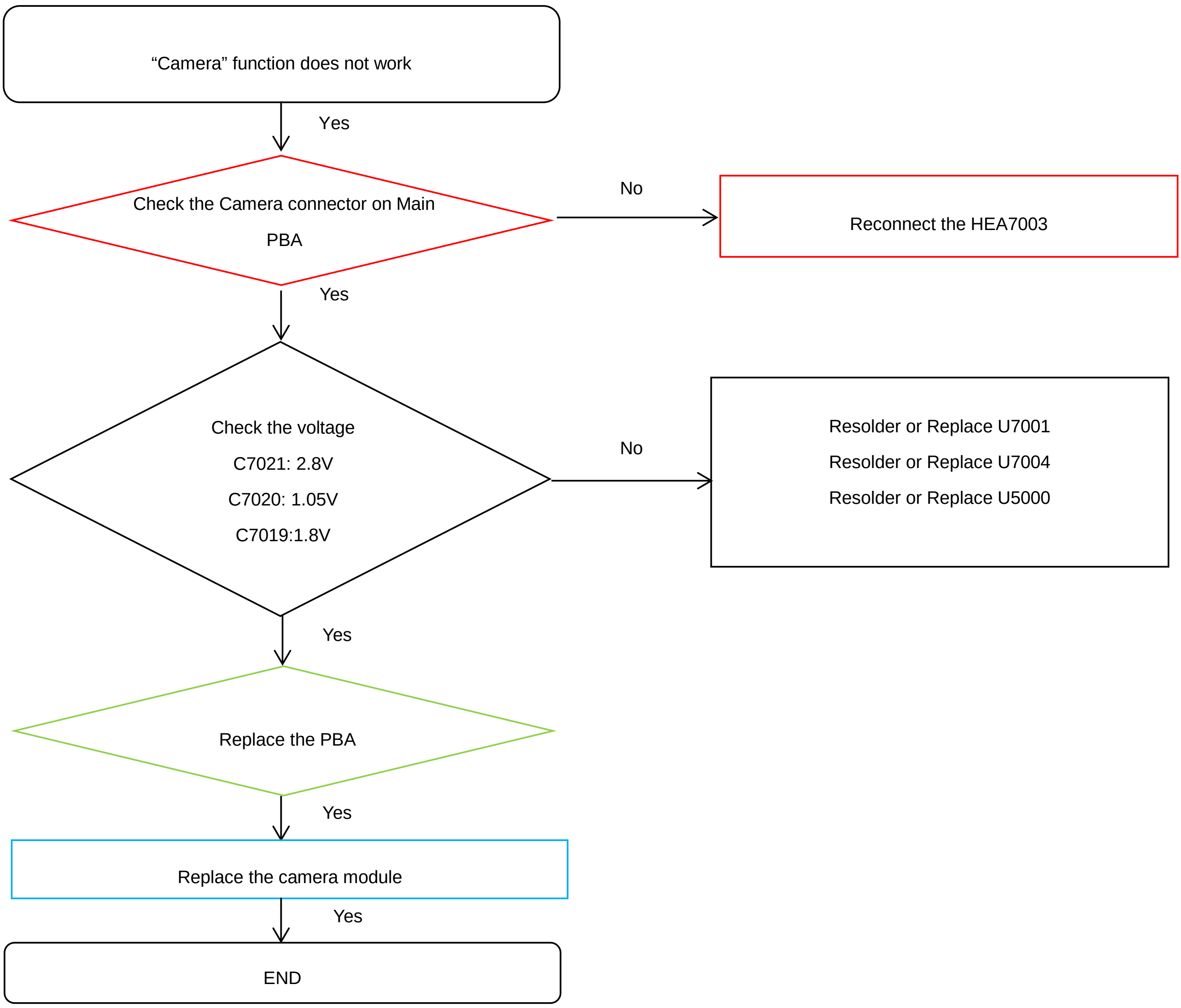
8-3-11. Display



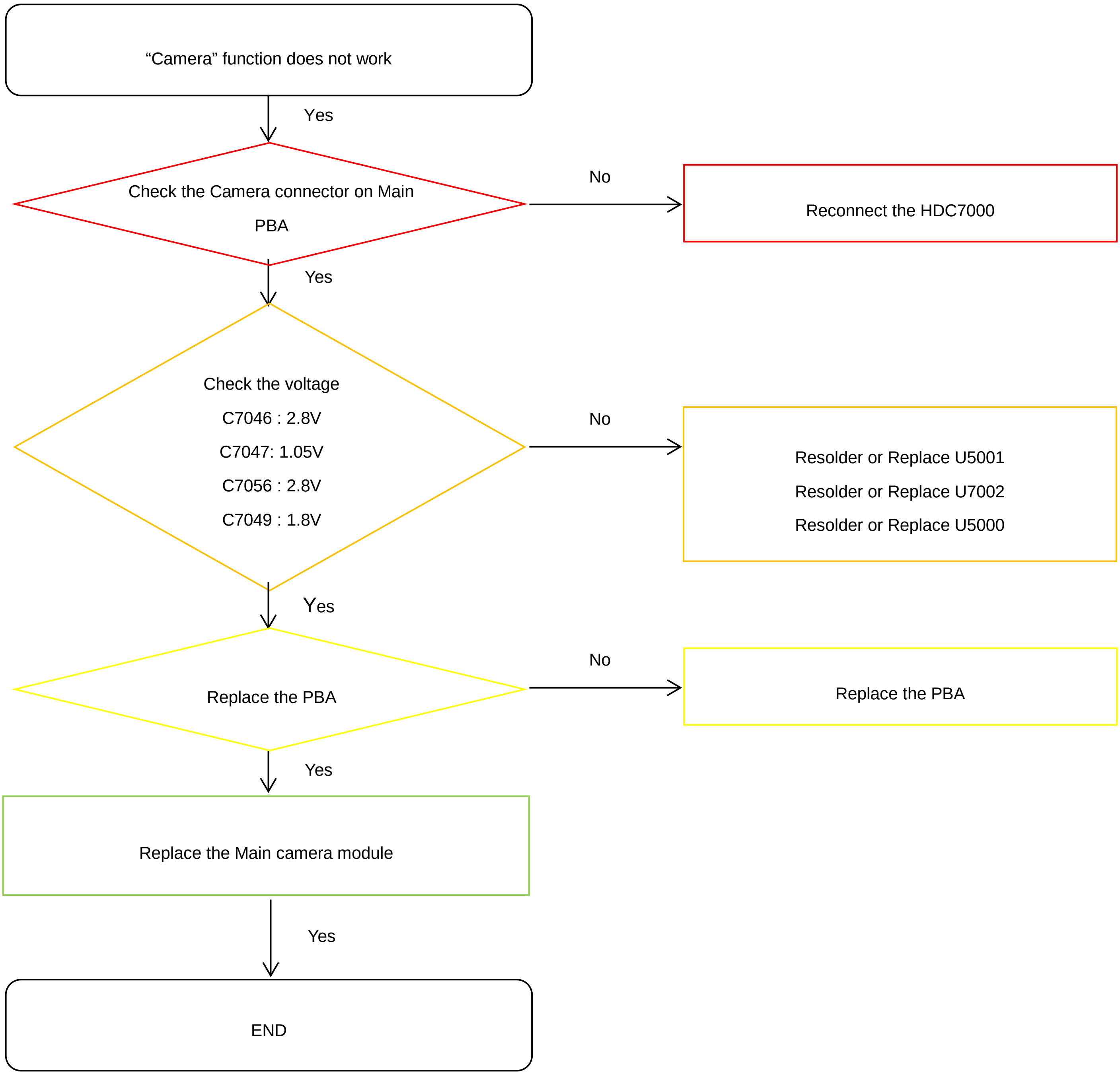
8-3-12. TSP


TSP does not work
Yes
Check TSP
Connector on Main PBA
No
Reconnect the HDC7004
Yes
Check
VREG_L11A_1P8=1.8V(C7076)
No
Resolder or Replace U5000
Yes
Replace the UCP4000
Yes
END

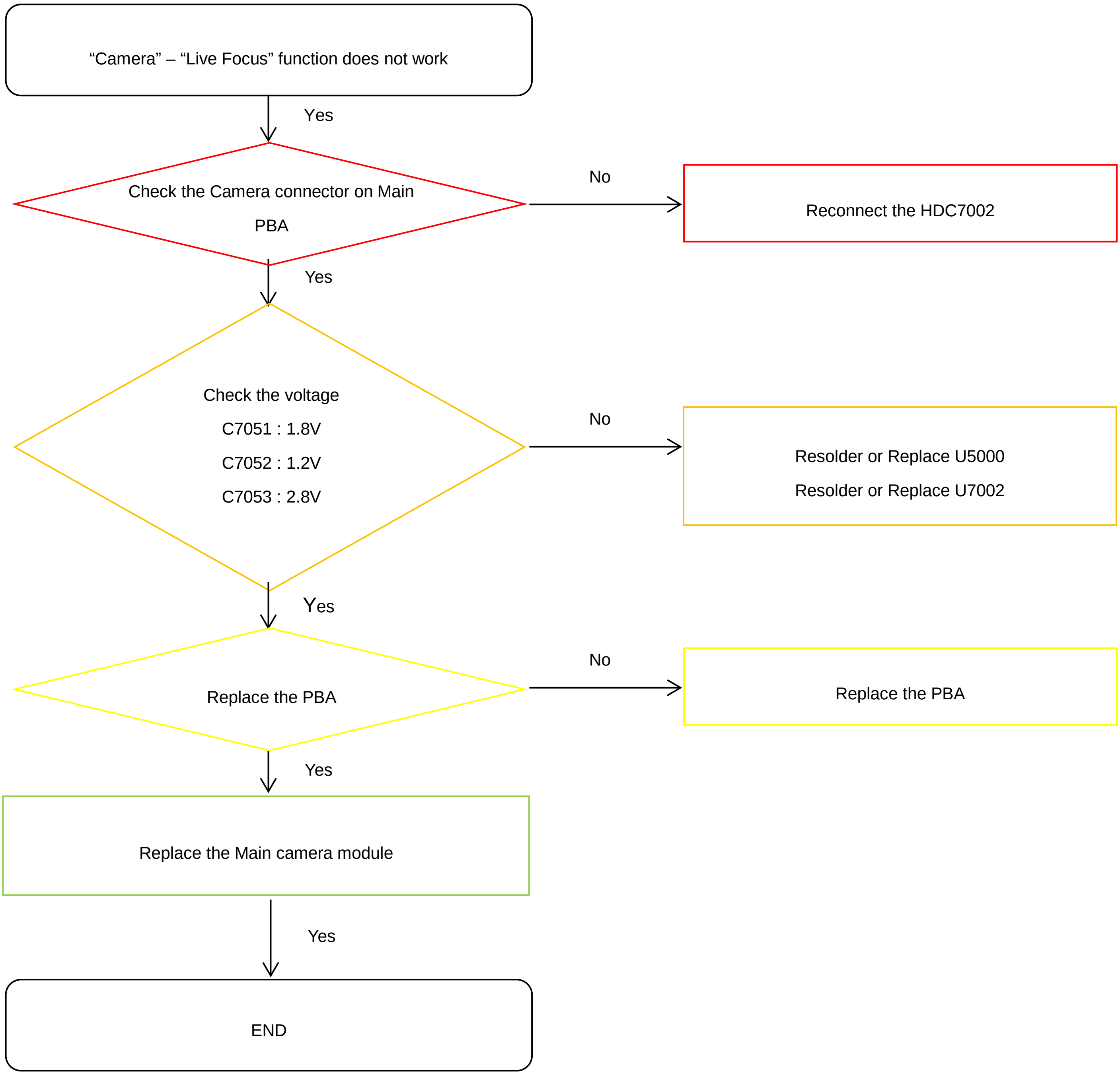
8-3-13. VT CAMERA



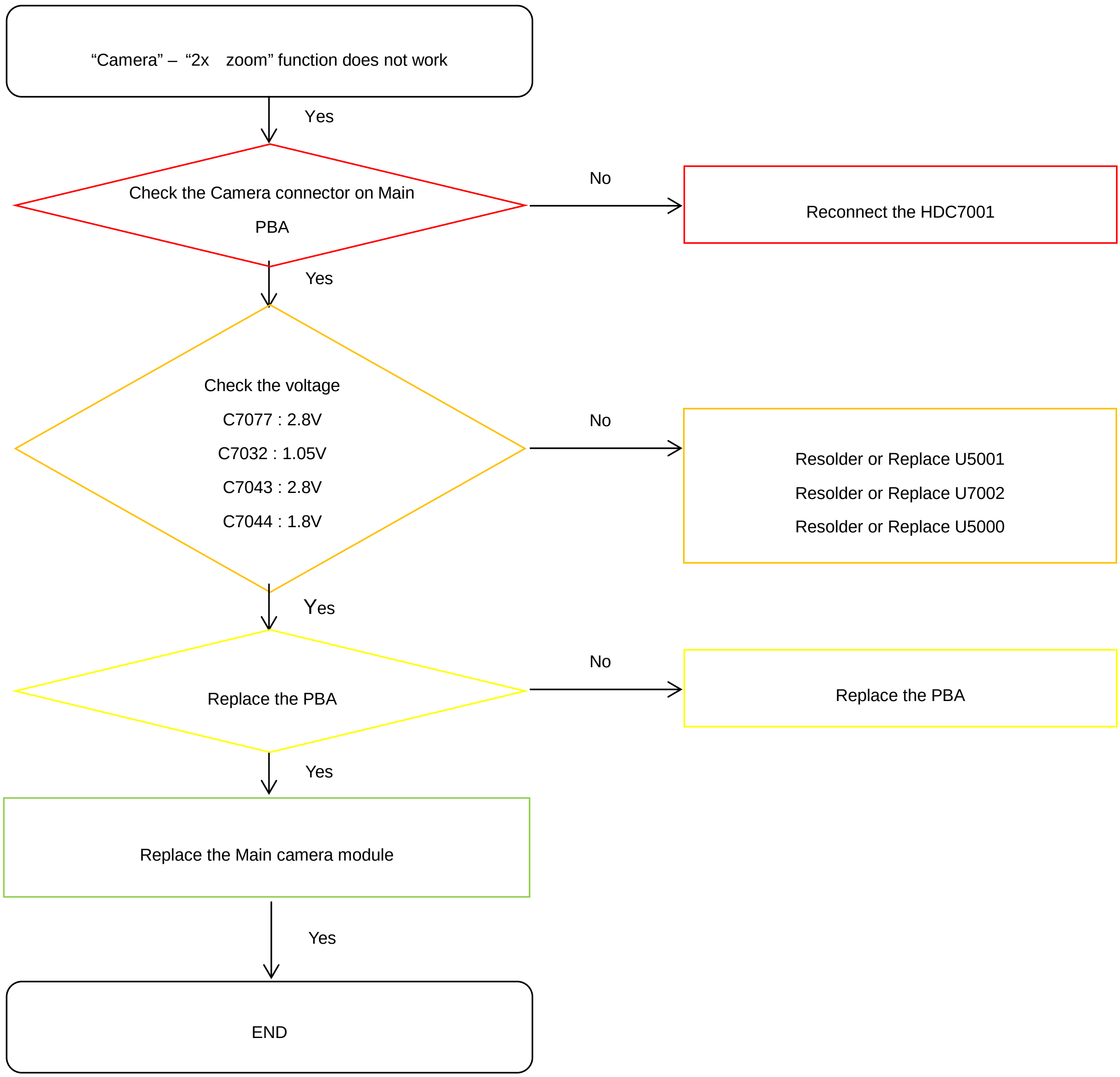
8-3-14. 24M REAR CAMERA



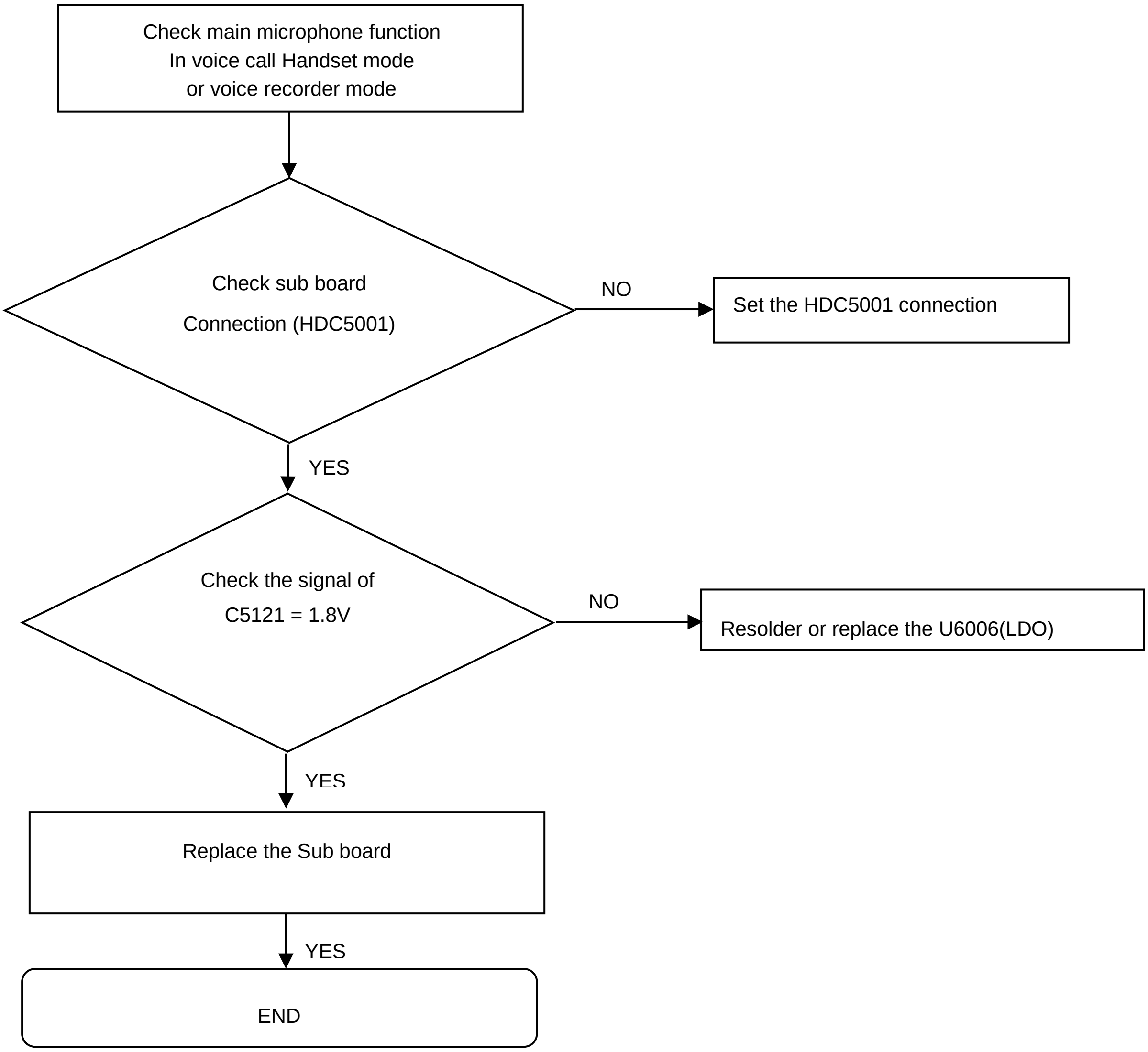
8-3-15. 5M REAR CAMERA



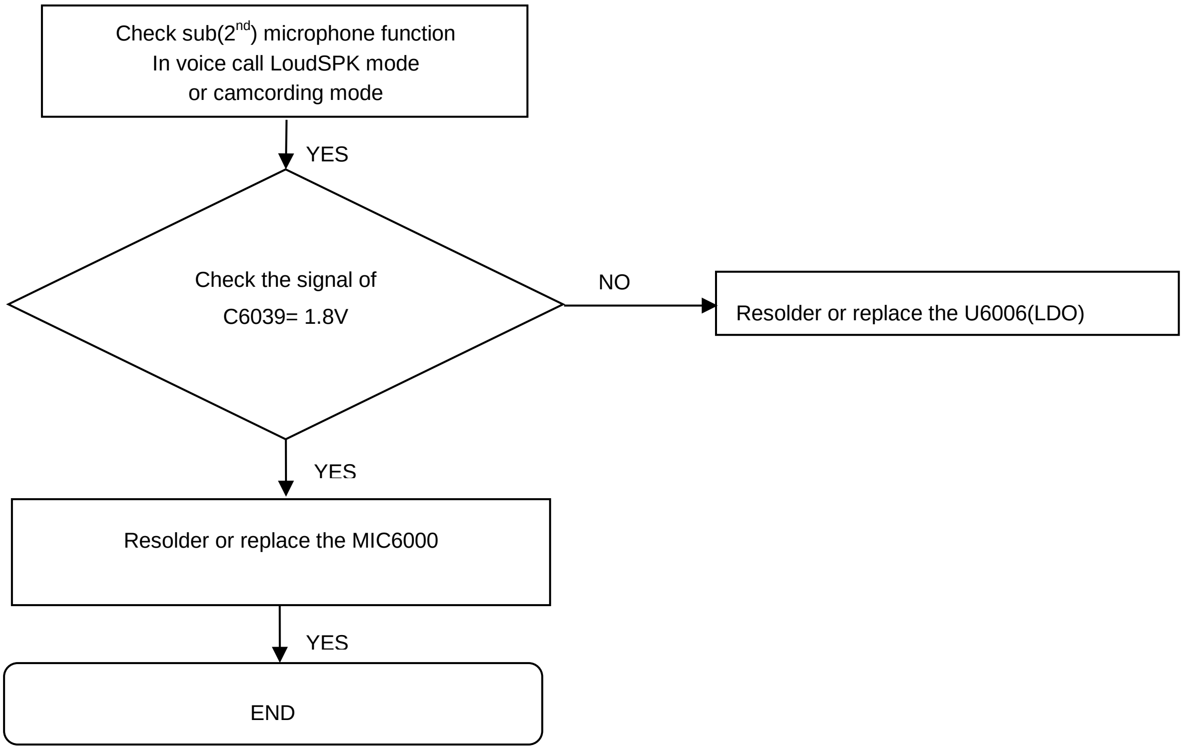
8-3-16. 10M REAR CAMERA



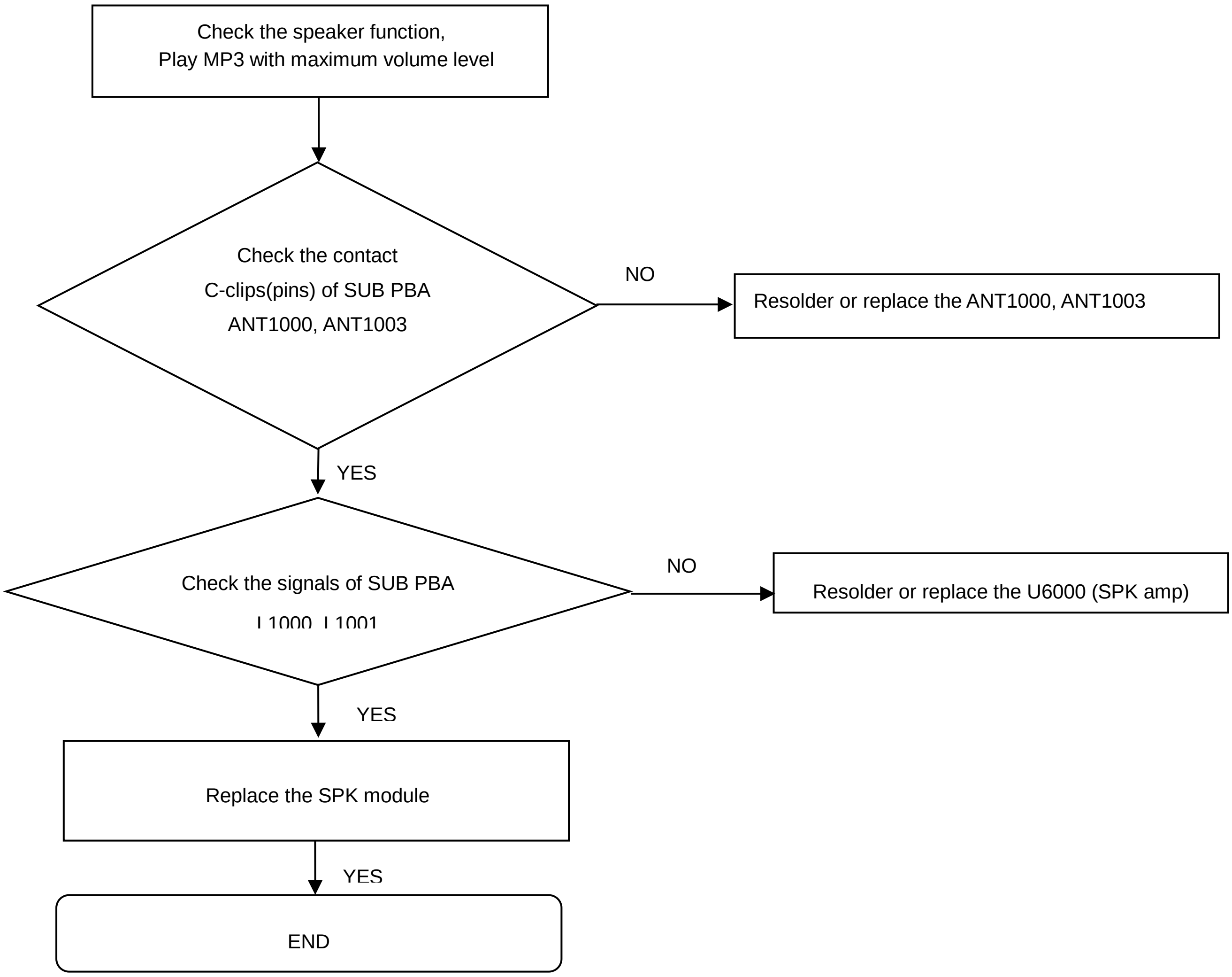
8-3-17-1. Main MIC



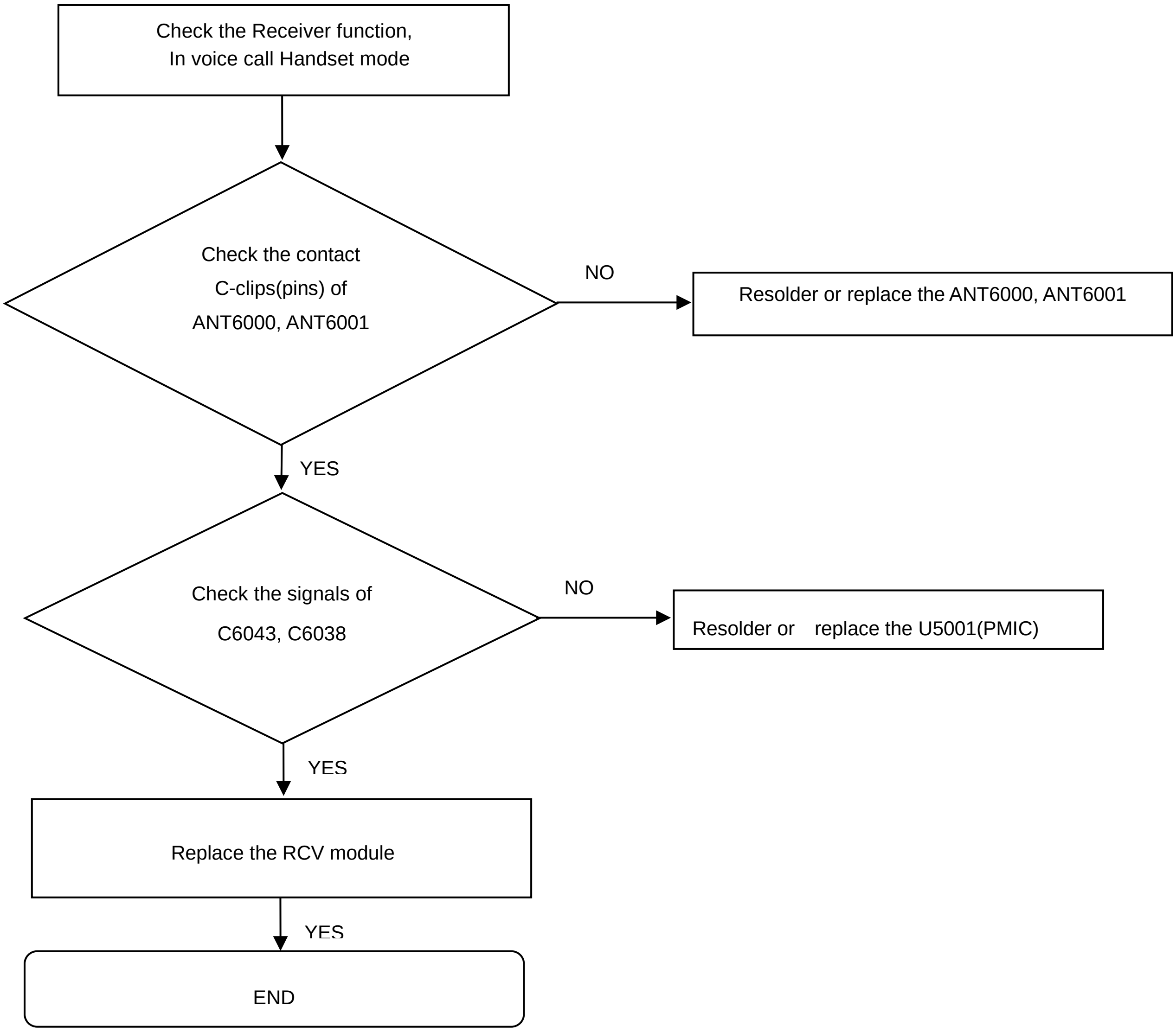
8-3-17-2. Sub(2nd) MIC



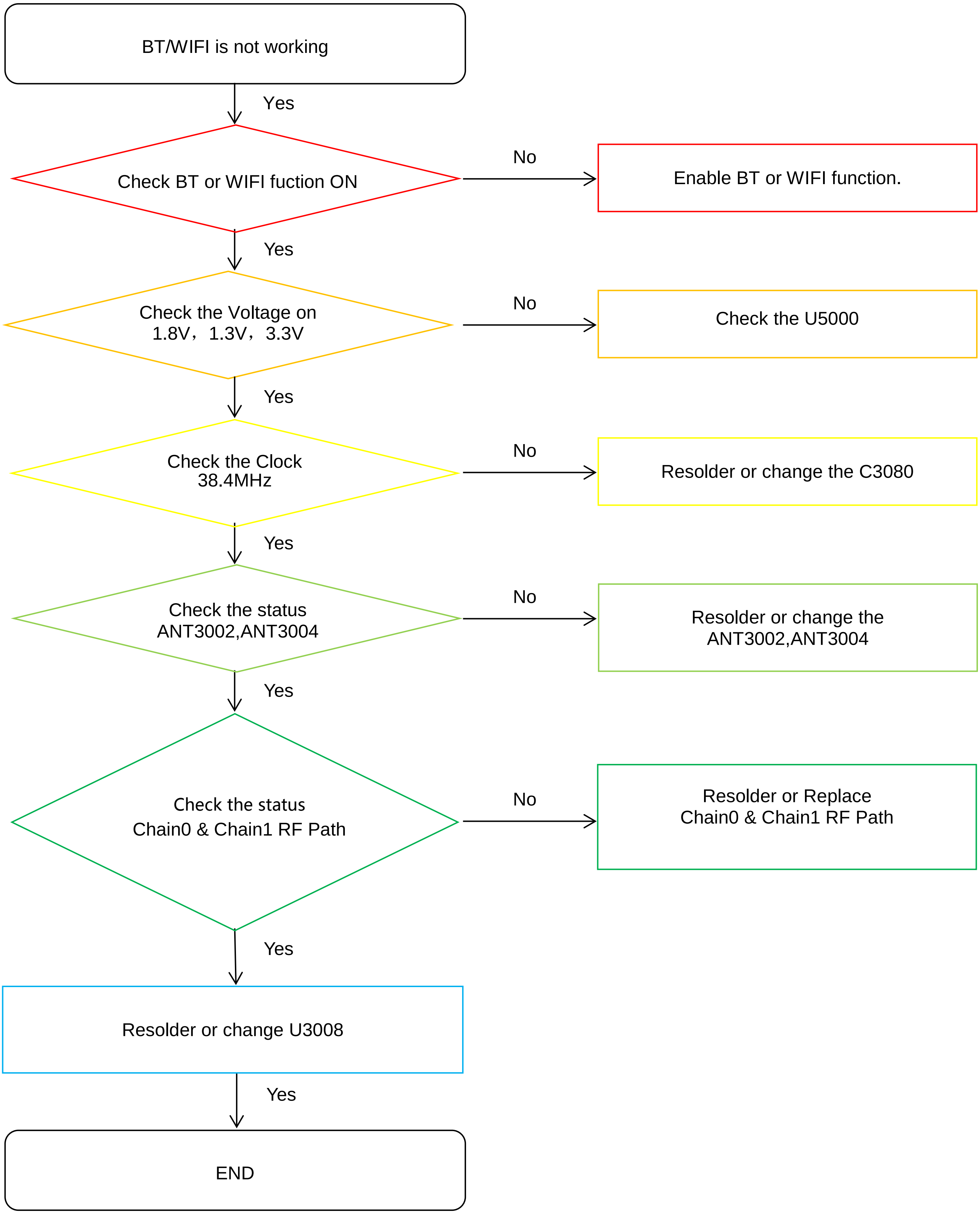
8-3-18. Speaker part



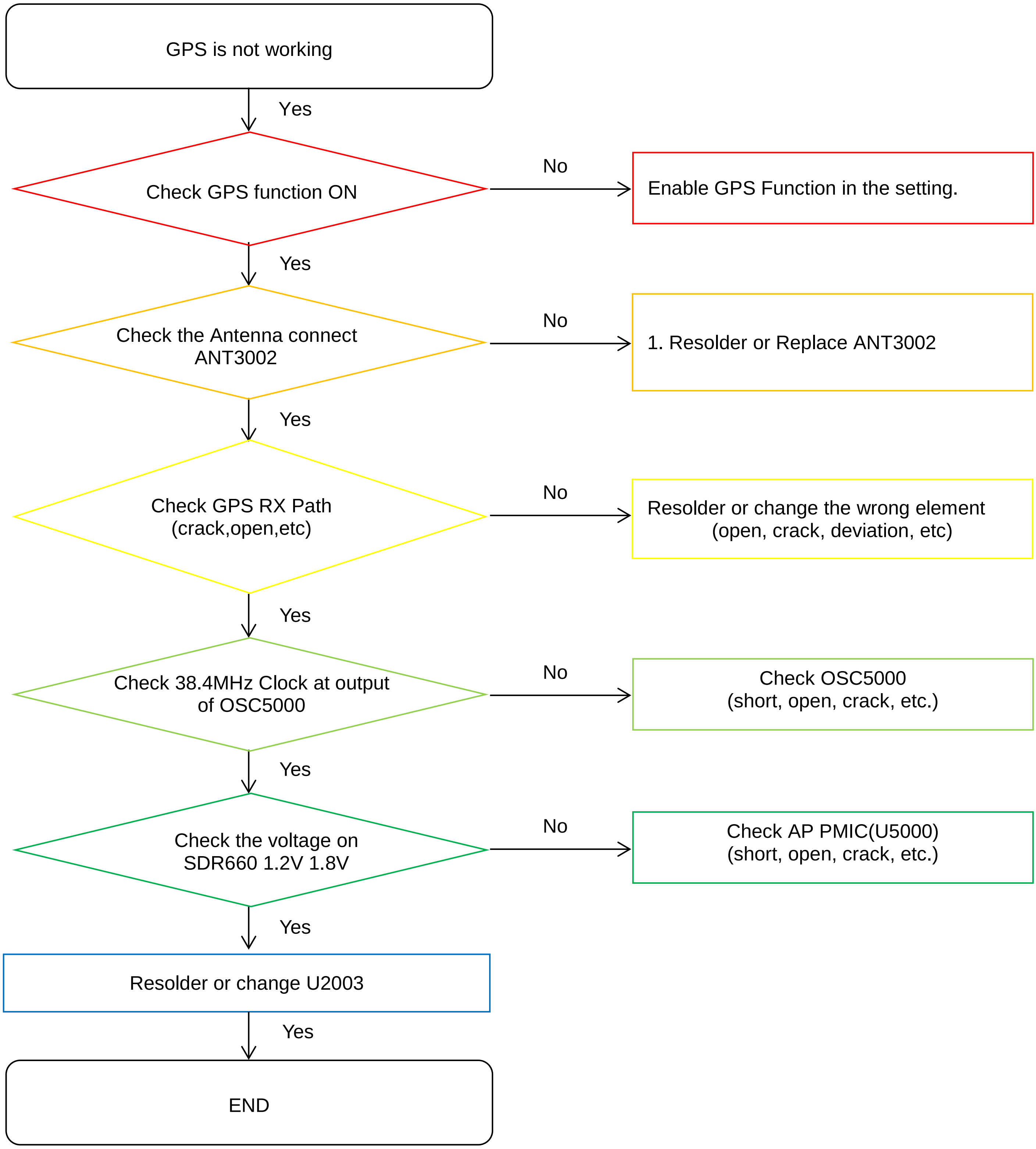
8-3-19. Receiver part



8-3-20. BT/WIFI

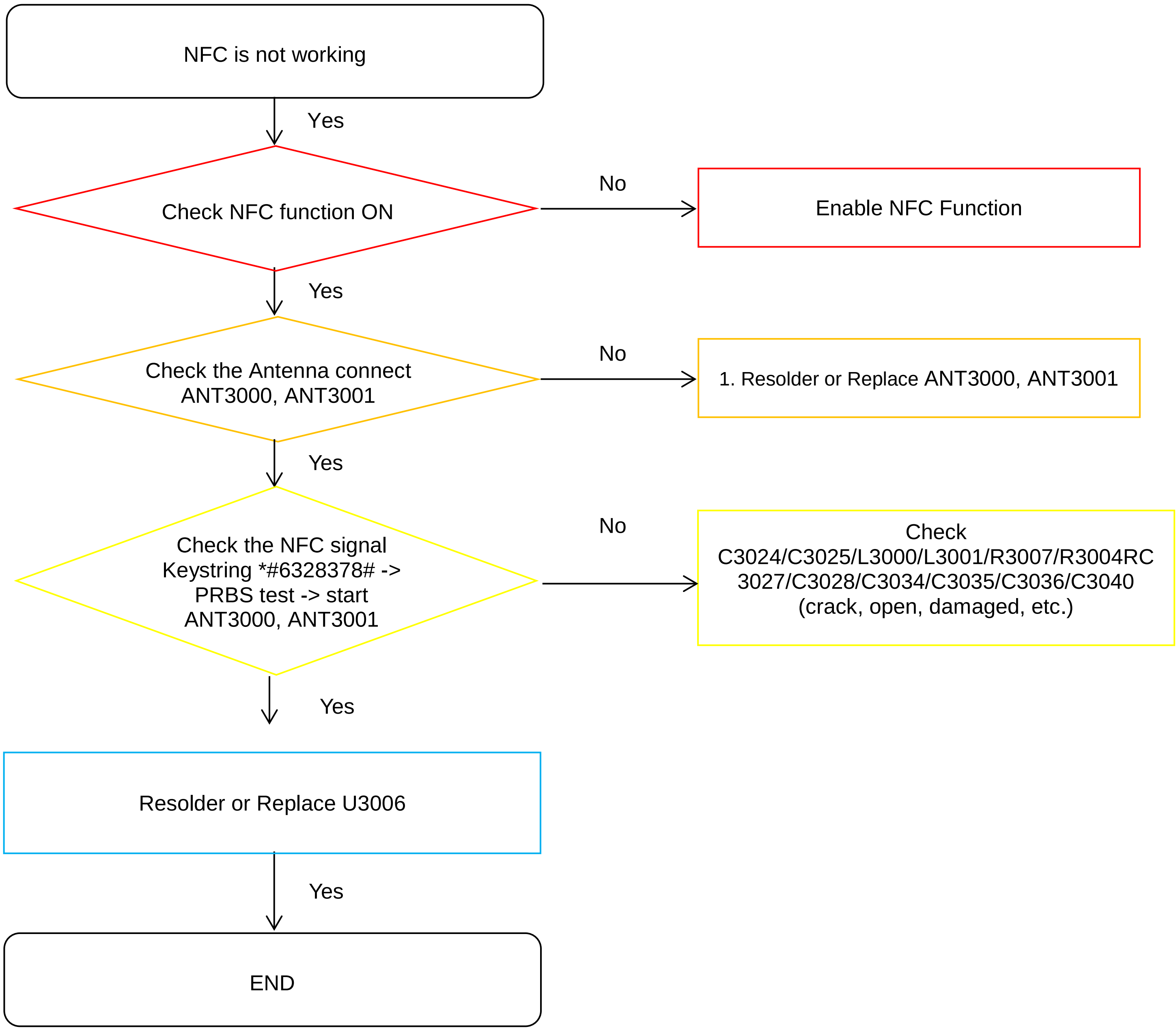

BT/WIFI is not working
Yes
Check BT or WIFI fuction ON
No
Enable BT or WIFI function.
Yes
Check the Voltage on
1.8V，1.3V，3.3V
No
Check the U5000
Yes
Check the Clock
38.4MHz
No
Resolder or change the C3080
Yes
Check the status
ANT3002,ANT3004
No
Resolder or change the
ANT3002,ANT3004
Yes
Check the status
Chain0 & Chain1 RF Path
No
Resolder or Replace
Chain0 & Chain1 RF Path
Yes
Resolder or change U3008
Yes
END

8-3-21. GPS



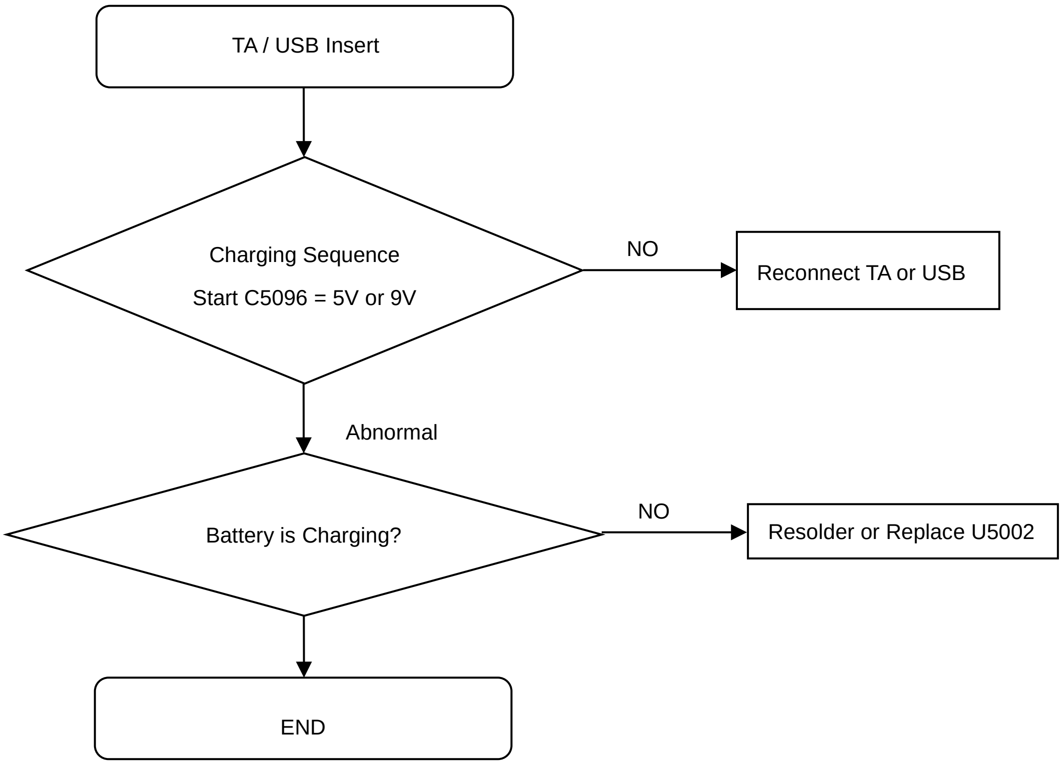
GPS is not working
Yes
Check GPS function ON
No
Enable GPS Function in the setting.
Yes
Check the Antenna connect ANT3002
No
1. Resolder or Replace ANT3002
Yes
Check GPS RX Path (crack,open,etc)
No
Resolder or change the wrong element (open, crack, deviation, etc)
Yes
Check 38.4MHz Clock at output of OSC5000
No
Check OSC5000 (short, open, crack, etc.)
Yes
Check the voltage on SDR660 1.2V 1.8V
No
Check AP PMIC(U5000) (short, open, crack, etc.)
Yes
Resolder or change U2003
Yes
END

8-3-22. NFC


NFC is not working
Yes
Check NFC function ON
No
Enable NFC Function
Yes
Check the Antenna connect
ANT3000, ANT3001
No
1. Resolder or Replace ANT3000, ANT3001
Yes
Check the NFC signal
Keystring *#6328378# ->
PRBS test -> start
ANT3000, ANT3001
No
Check
C3024/C3025/L3000/L3001/R3007/R3004RC
3027/C3028/C3034/C3035/C3036/C3040
(crack, open, damaged, etc.)
Yes
Resolder or Replace U3006
Yes
END

8-3-23. Charging Part

Wire Charging (Normal/Fast)



8. Level 3 Repair

8-4. Service Schematics

- NC Point(Top View)

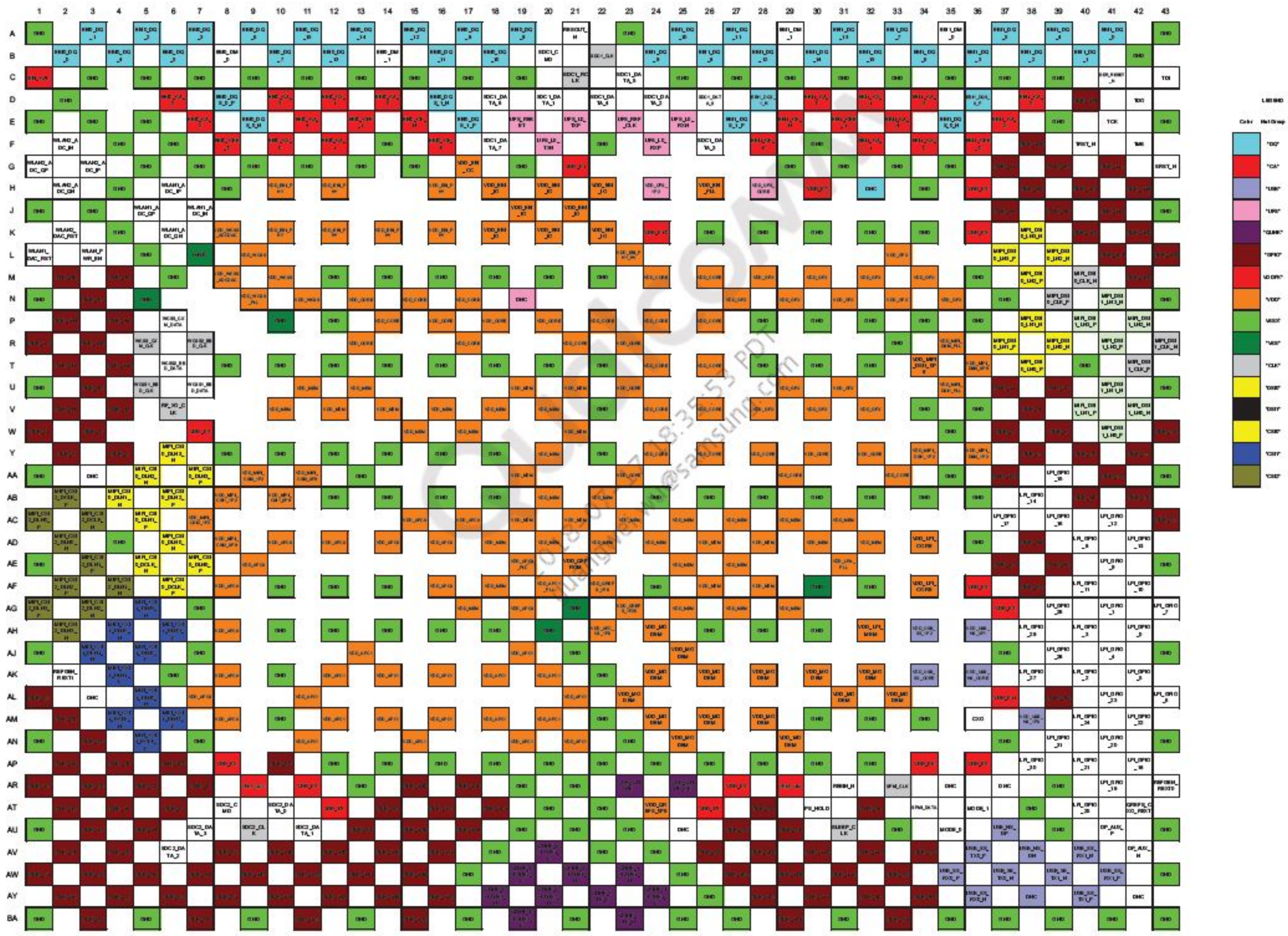


Figure 2-1 SDM710 pin assignments

9. Reference Abbreviation

Reference Abbreviation

— **AAC**: Advanced Audio Coding.
— **AVC** : Advanced Video Coding.
— **BER** : Bit Error Rate
— **BPSK**: Binary Phase Shift Keying
— **CA** : Conditional Access
— **CDM** : Code Division Multiplexing
— **C/I** : Carrier to Interference
— **DMB** : Digital Multimedia Broadcasting
— **EN :** European Standard
— **ES :** Elementary Stream
— **ETSI**: European Telecommunications Standards Institute
— **MPEG**: Moving Picture Experts Group
— **PN** : Pseudo-random Noise
— **PS** : Pilot Symbol
— **QPSK**: Quadrature Phase Shift Keying
— **RS** : Reed-Solomon
— **SI** : Service Information
— **TDM** : Time Division Multiplexing
— **TS :** Transport Stream

2018. 12. Rev 1.0